

FIGURES

Figure 1

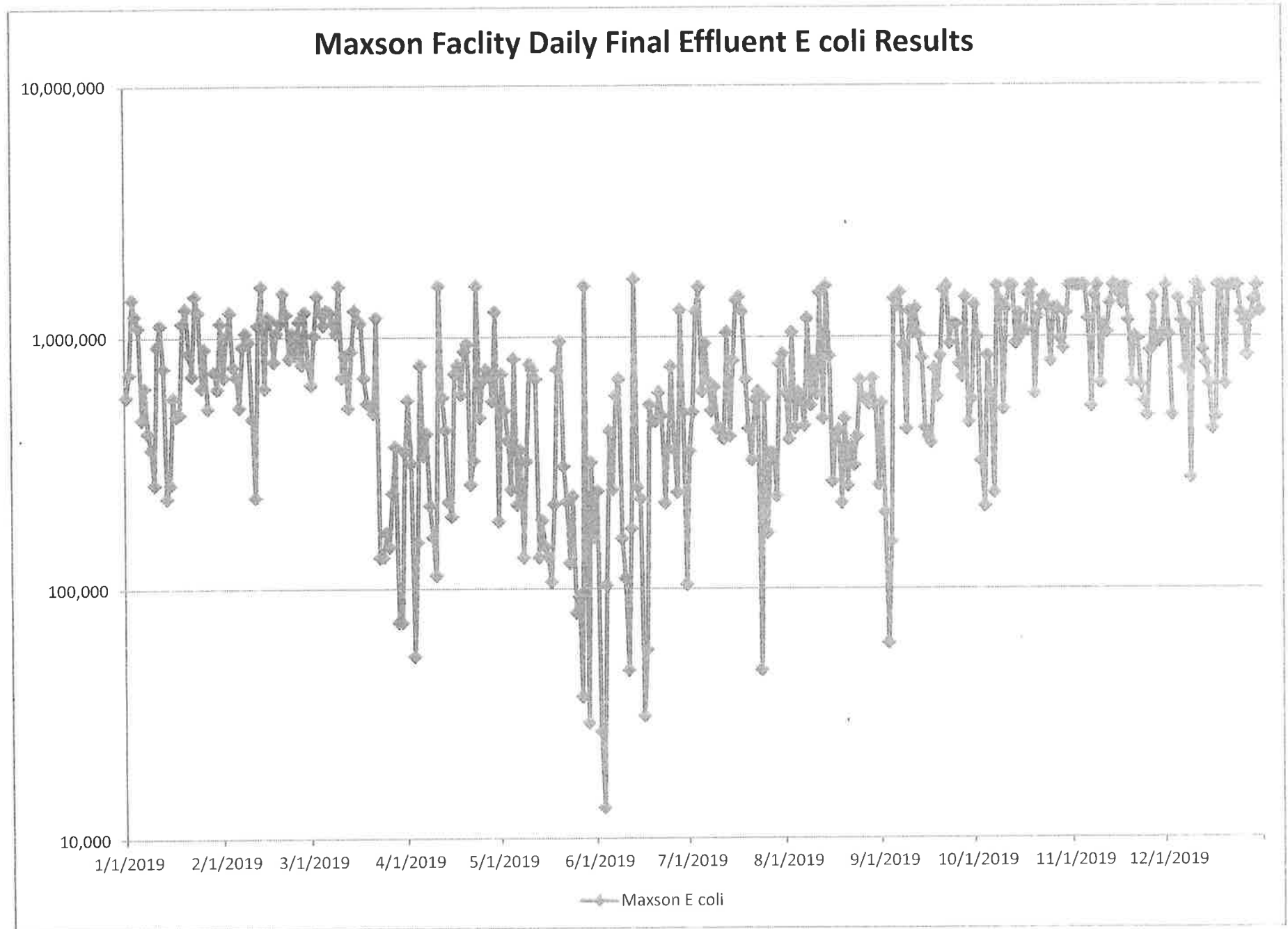


Figure 2

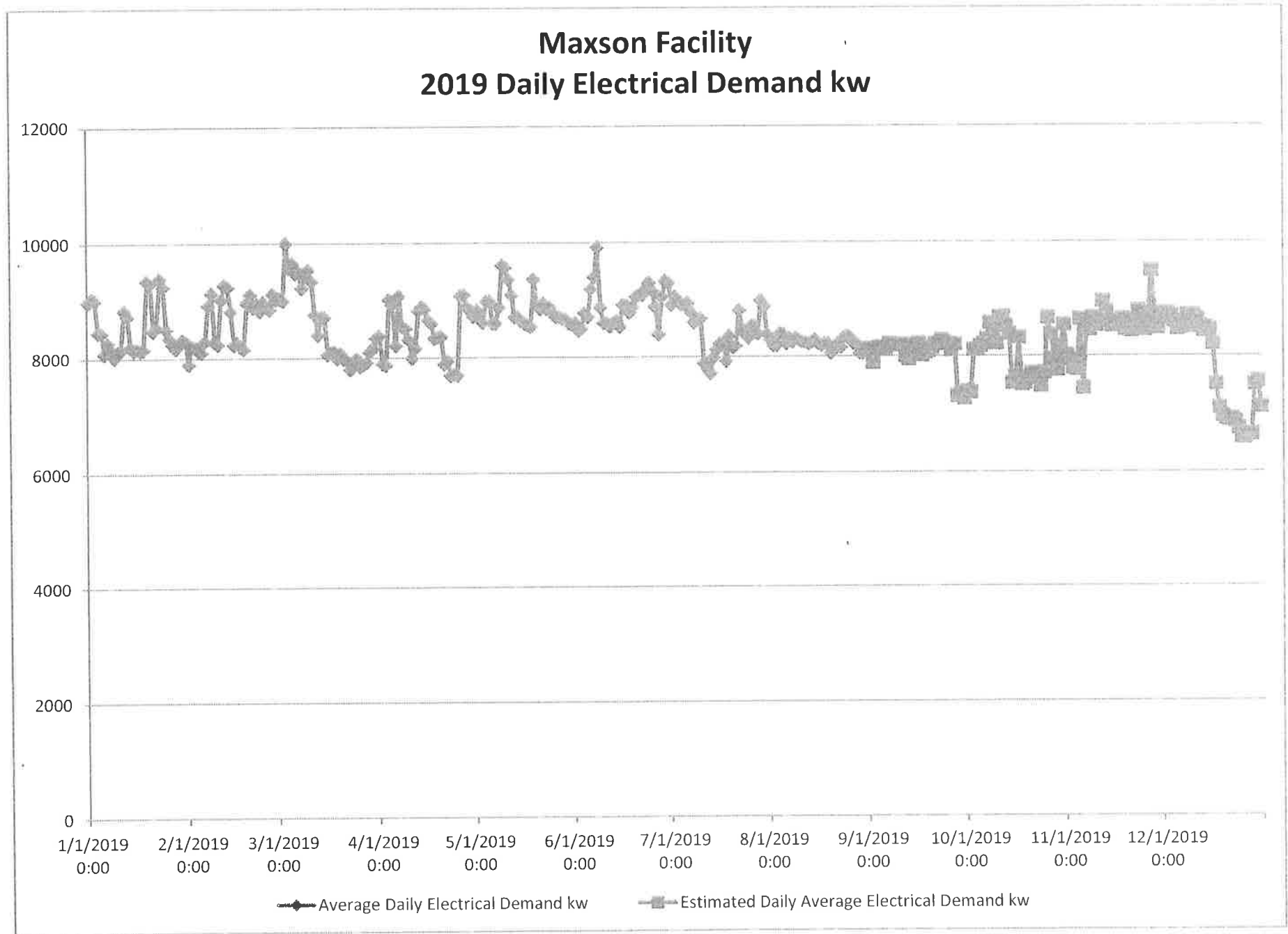


Figure 3

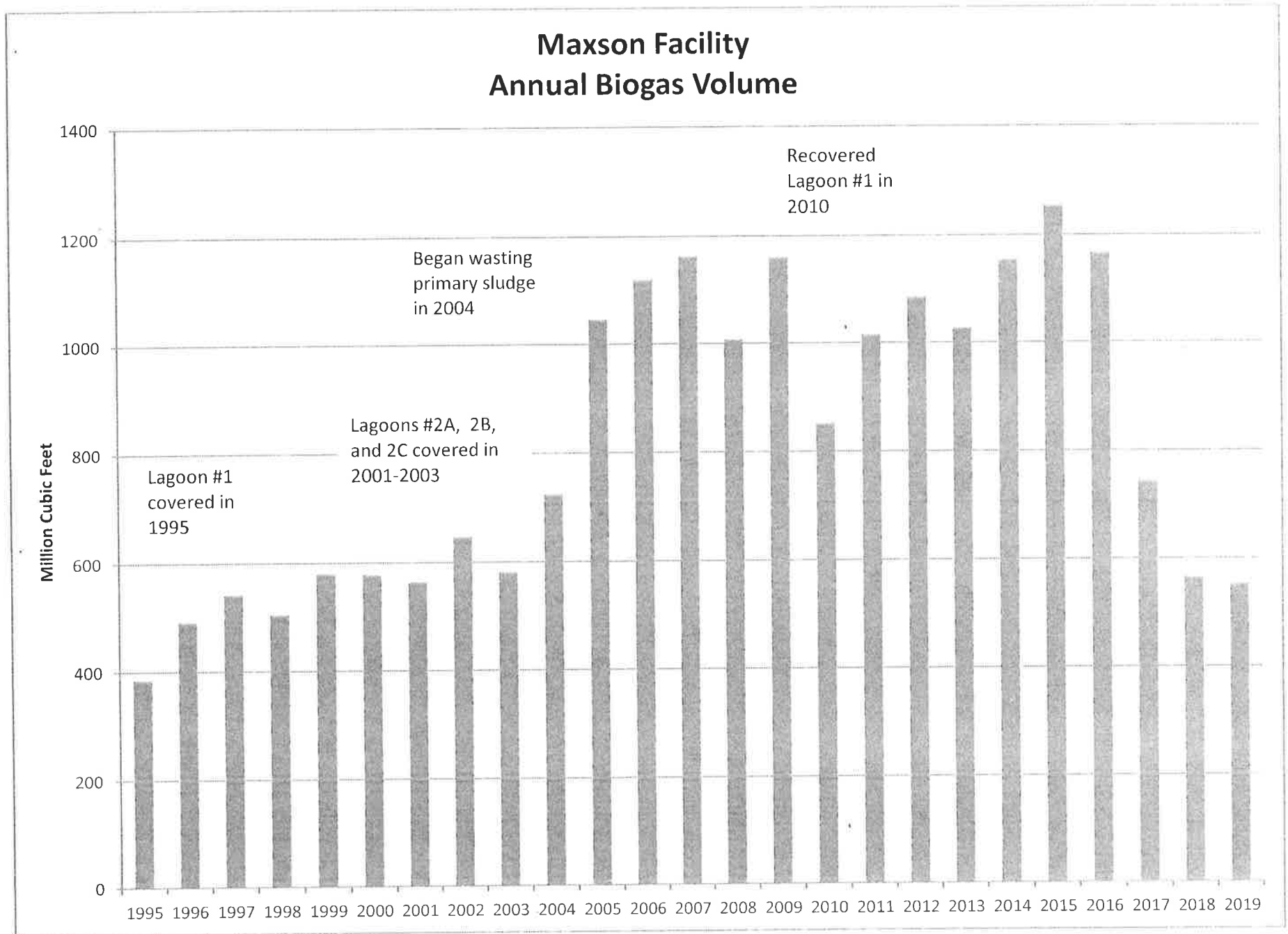


Figure 4

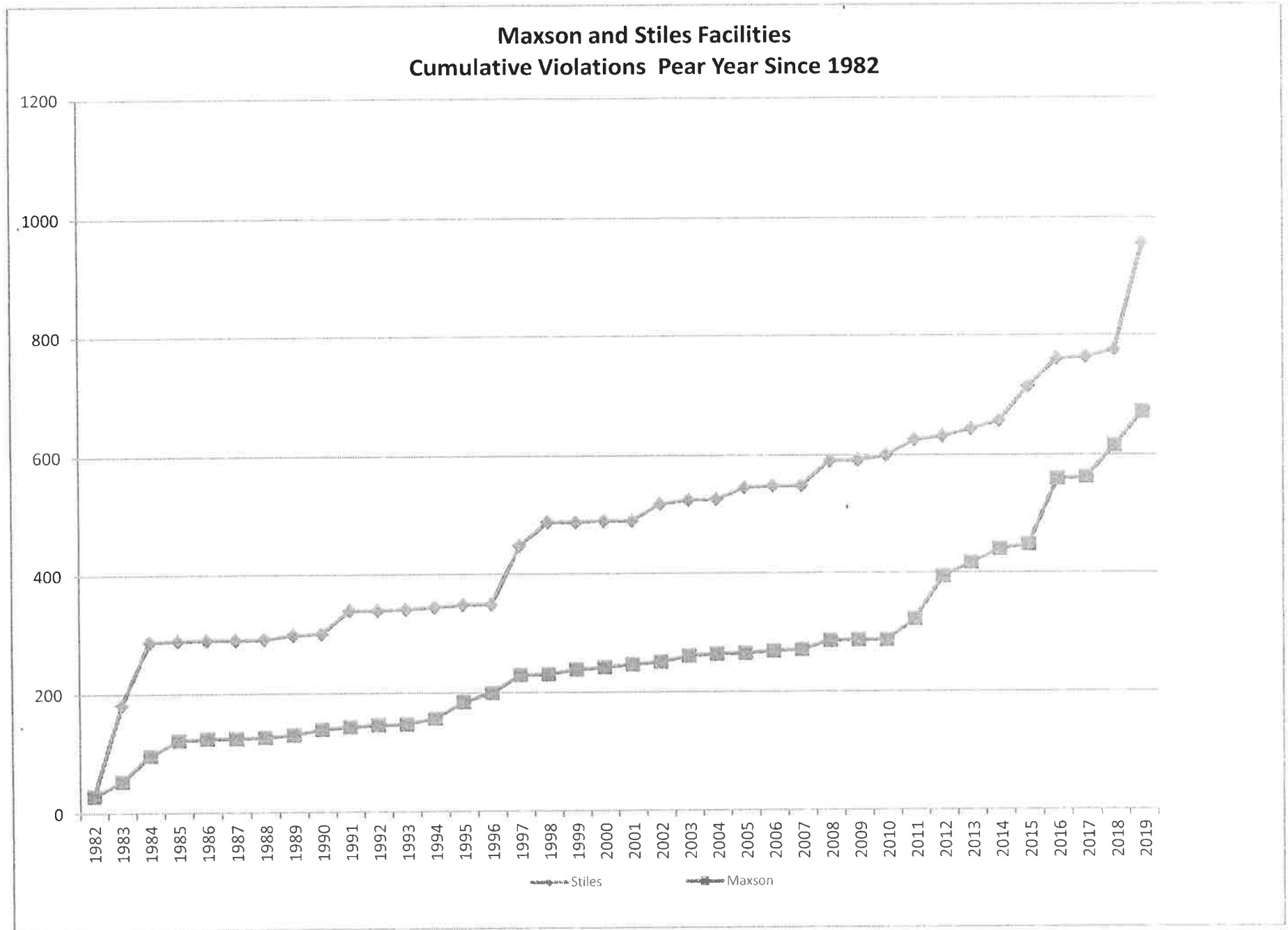


Figure 5

Maxson Facility Cumulative Lagooned Sludge and Surface Disposed Biosolids

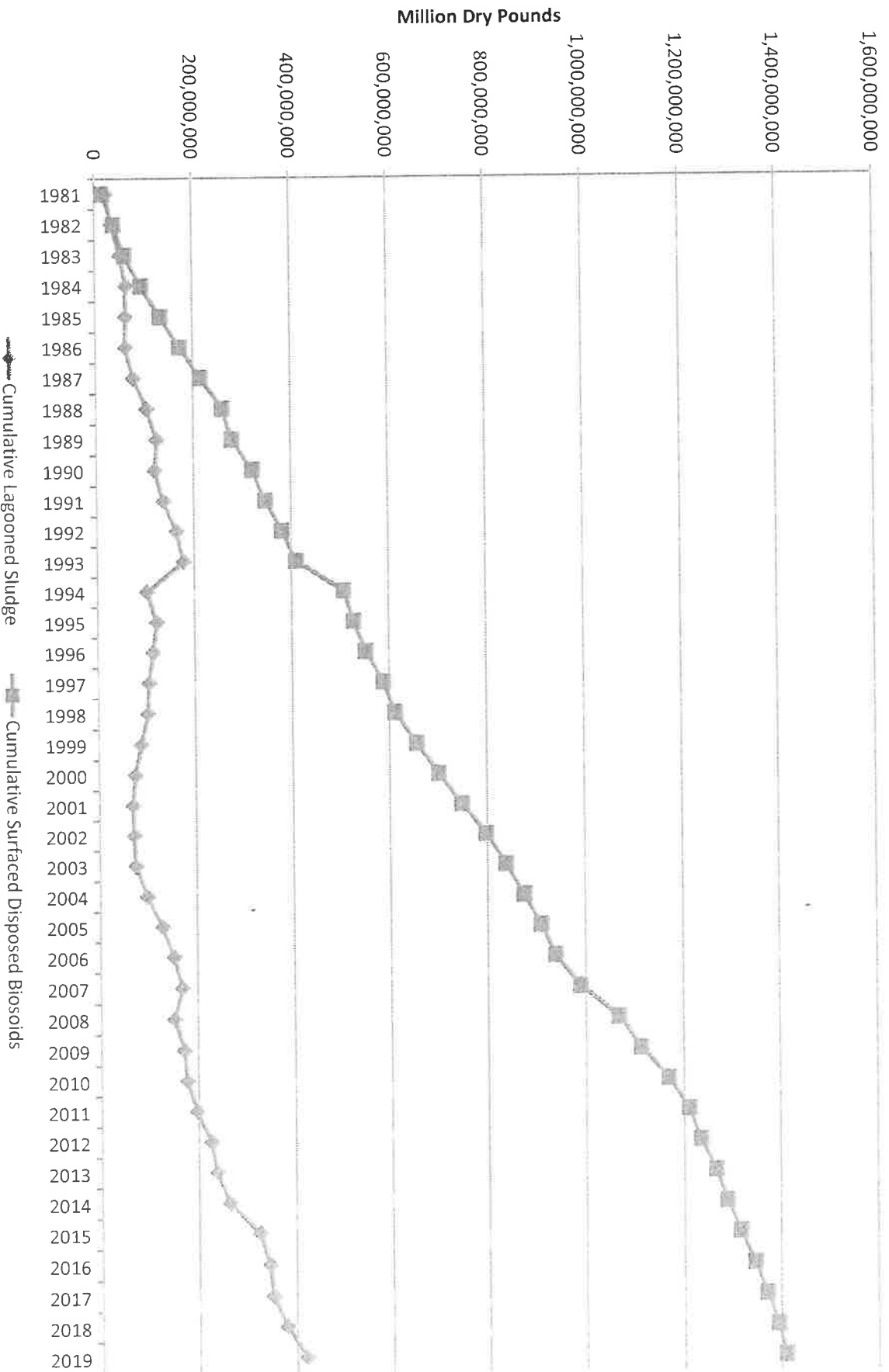


Figure 6

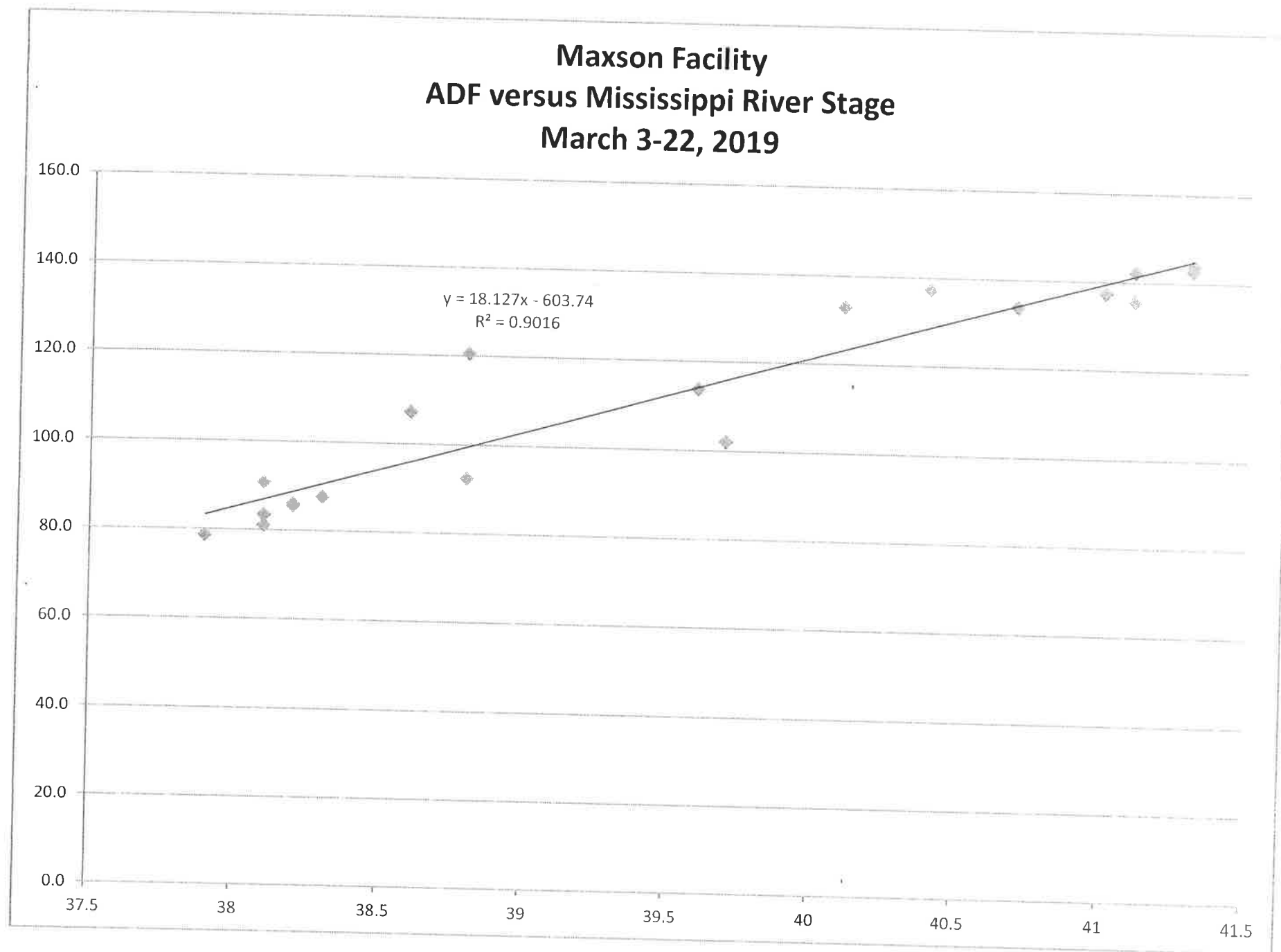


Figure 7

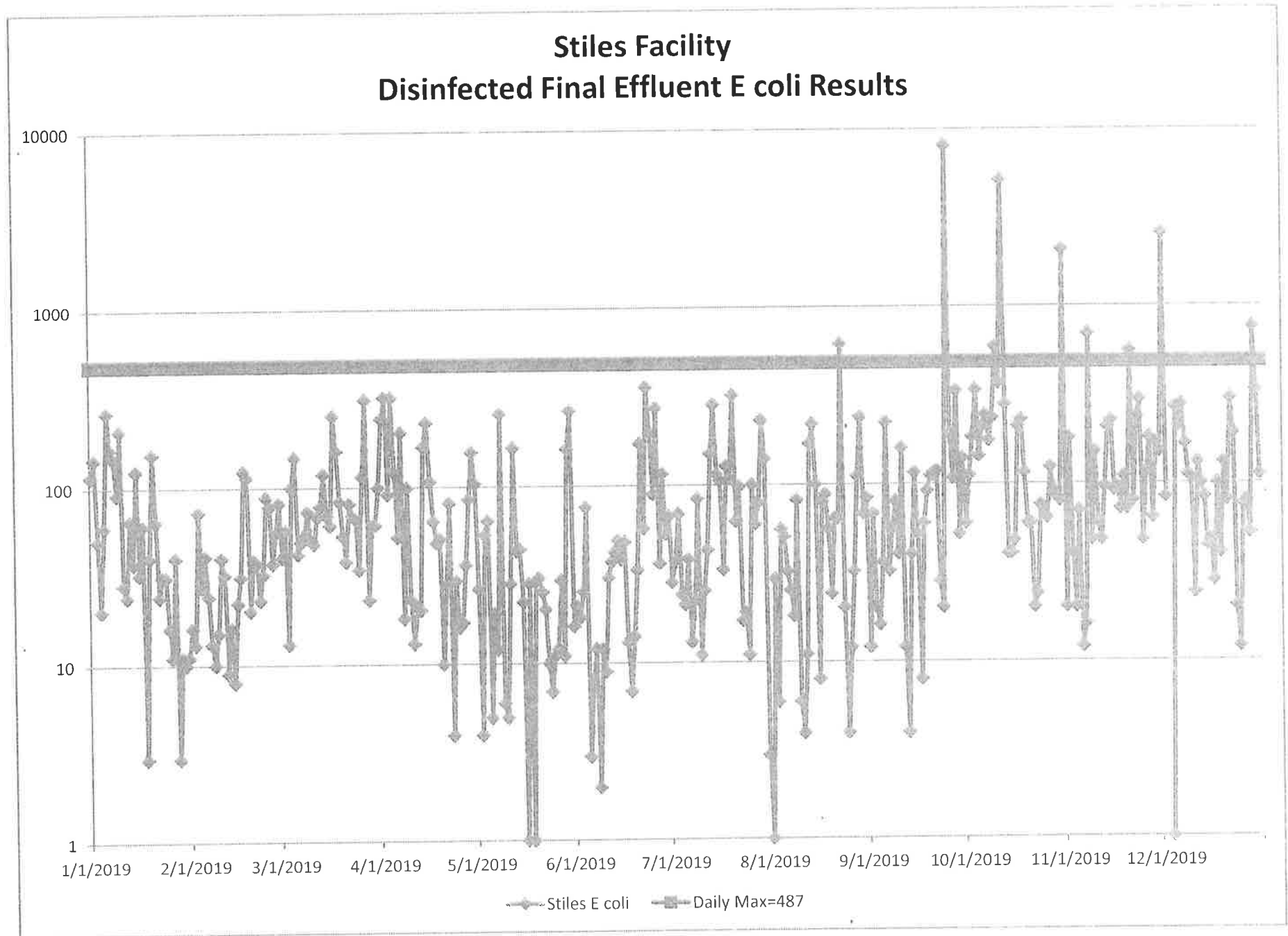


Figure 8

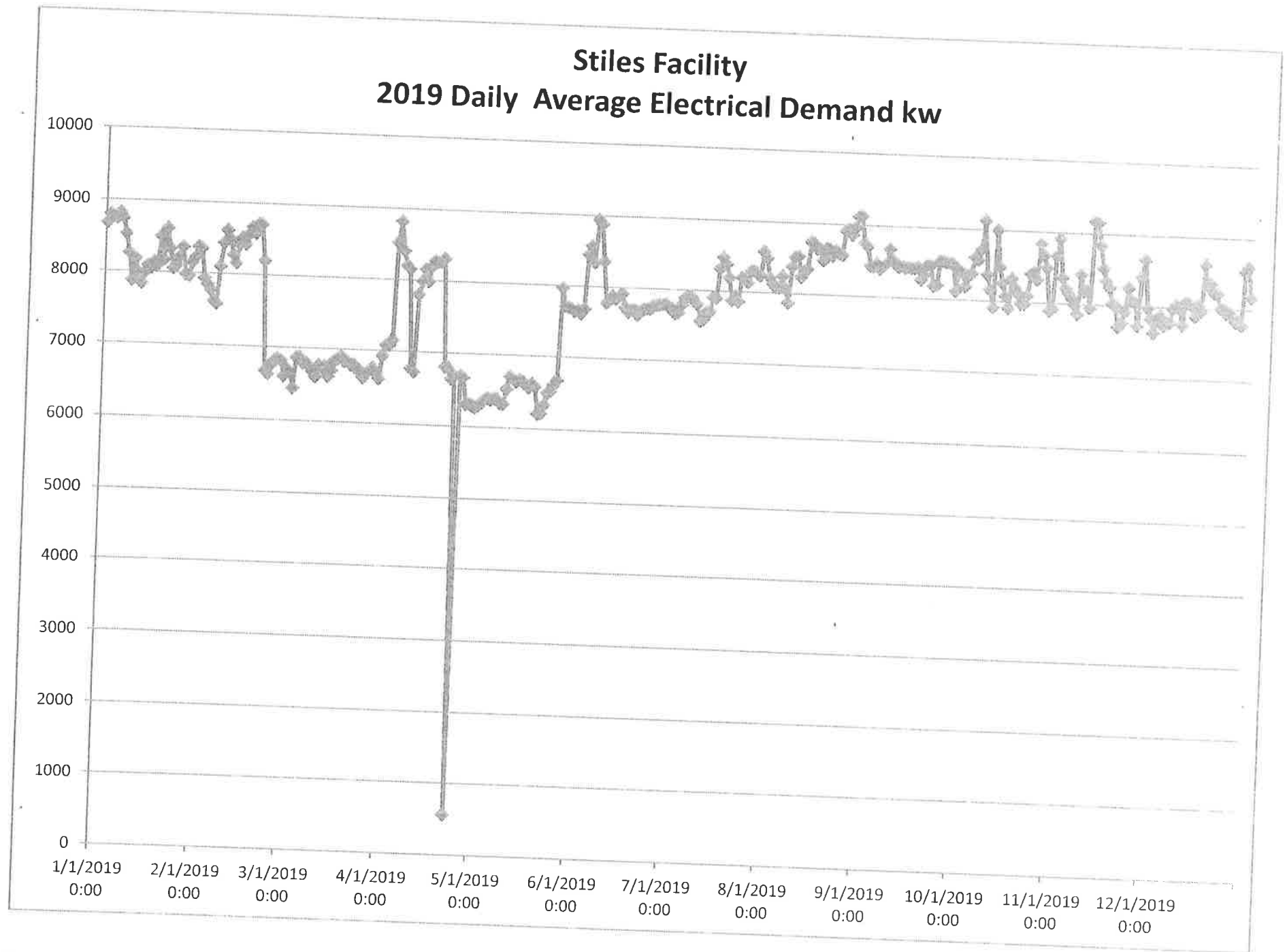


Figure 9

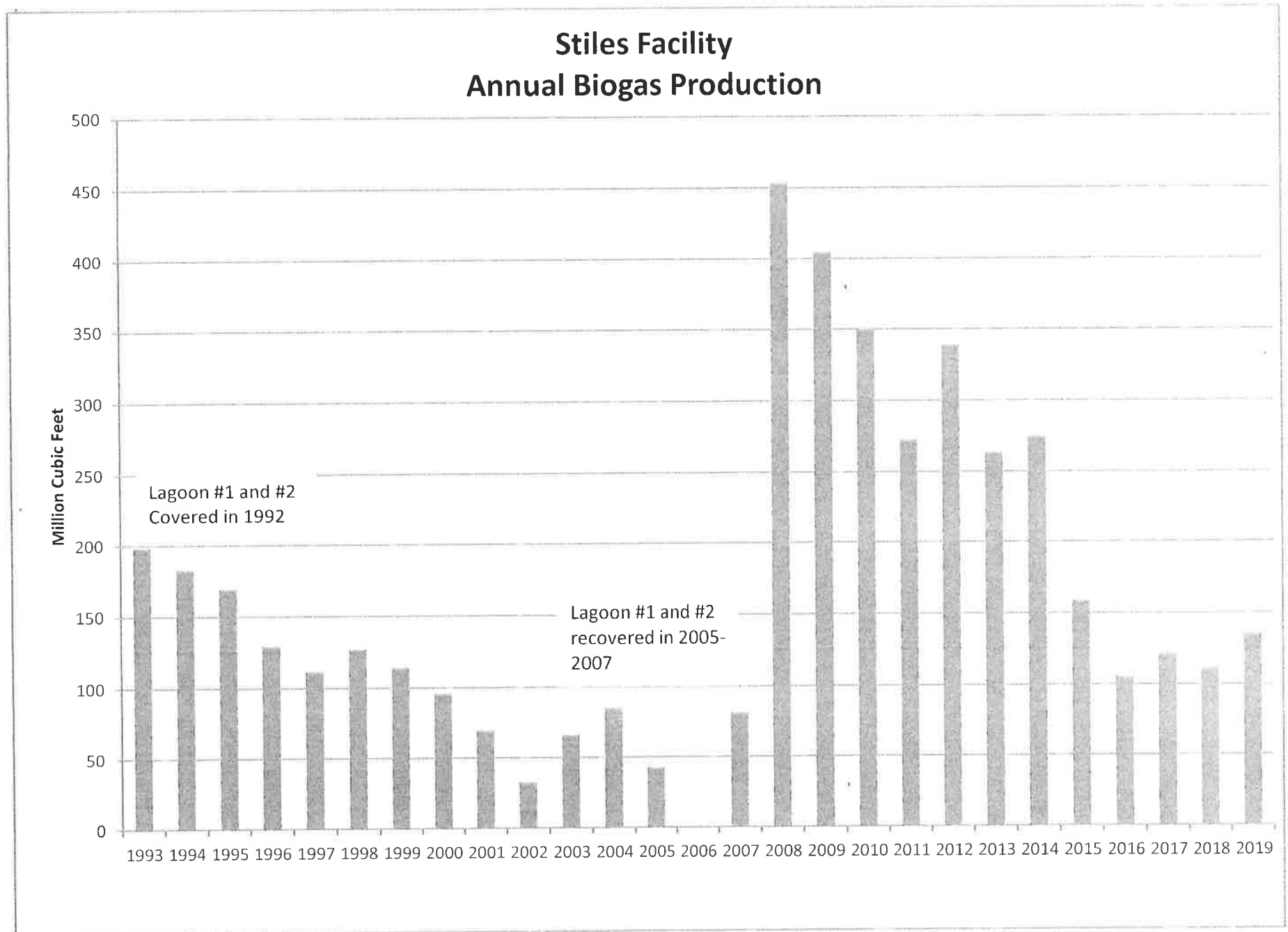


Figure 10

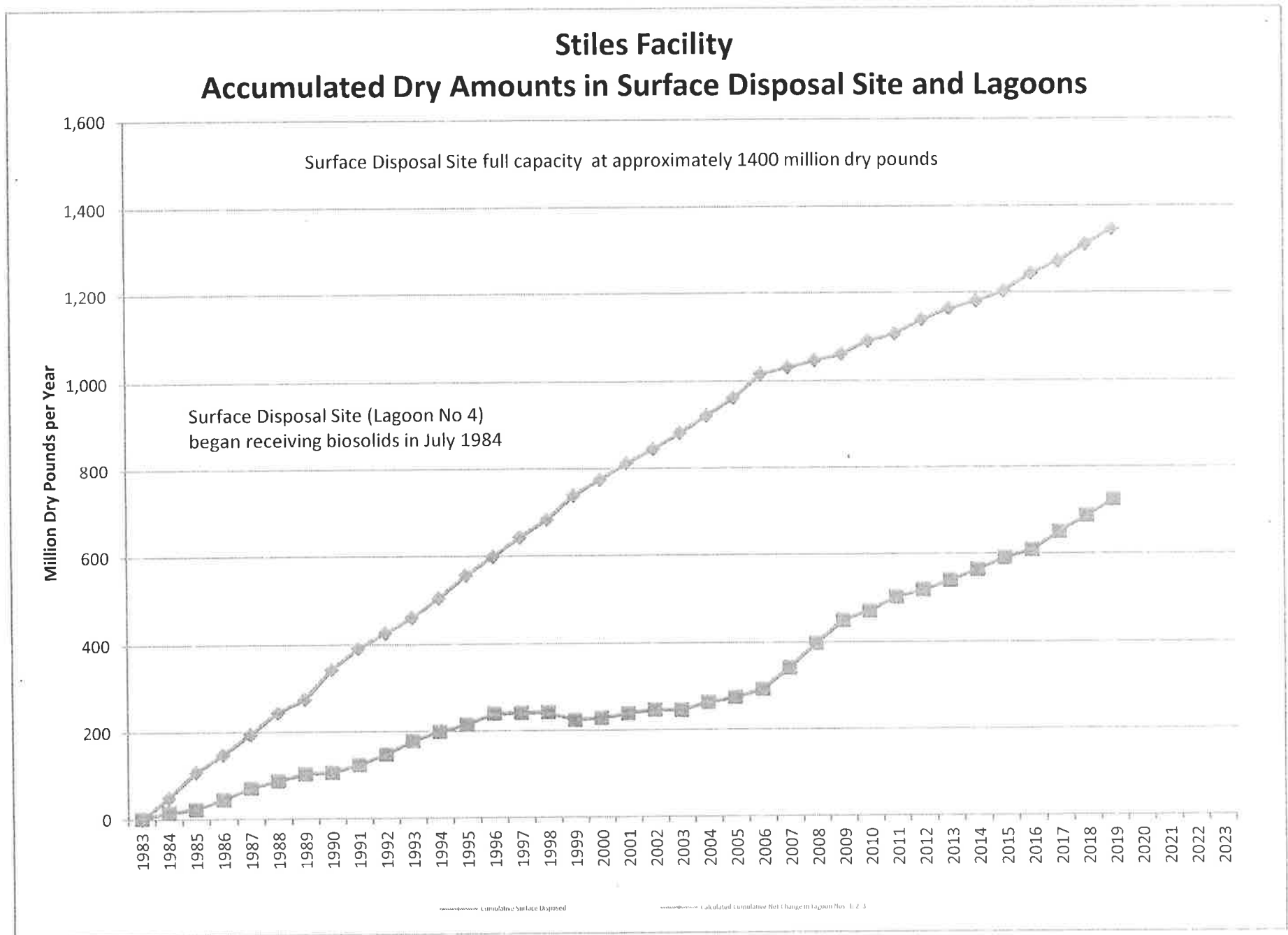


Figure 11

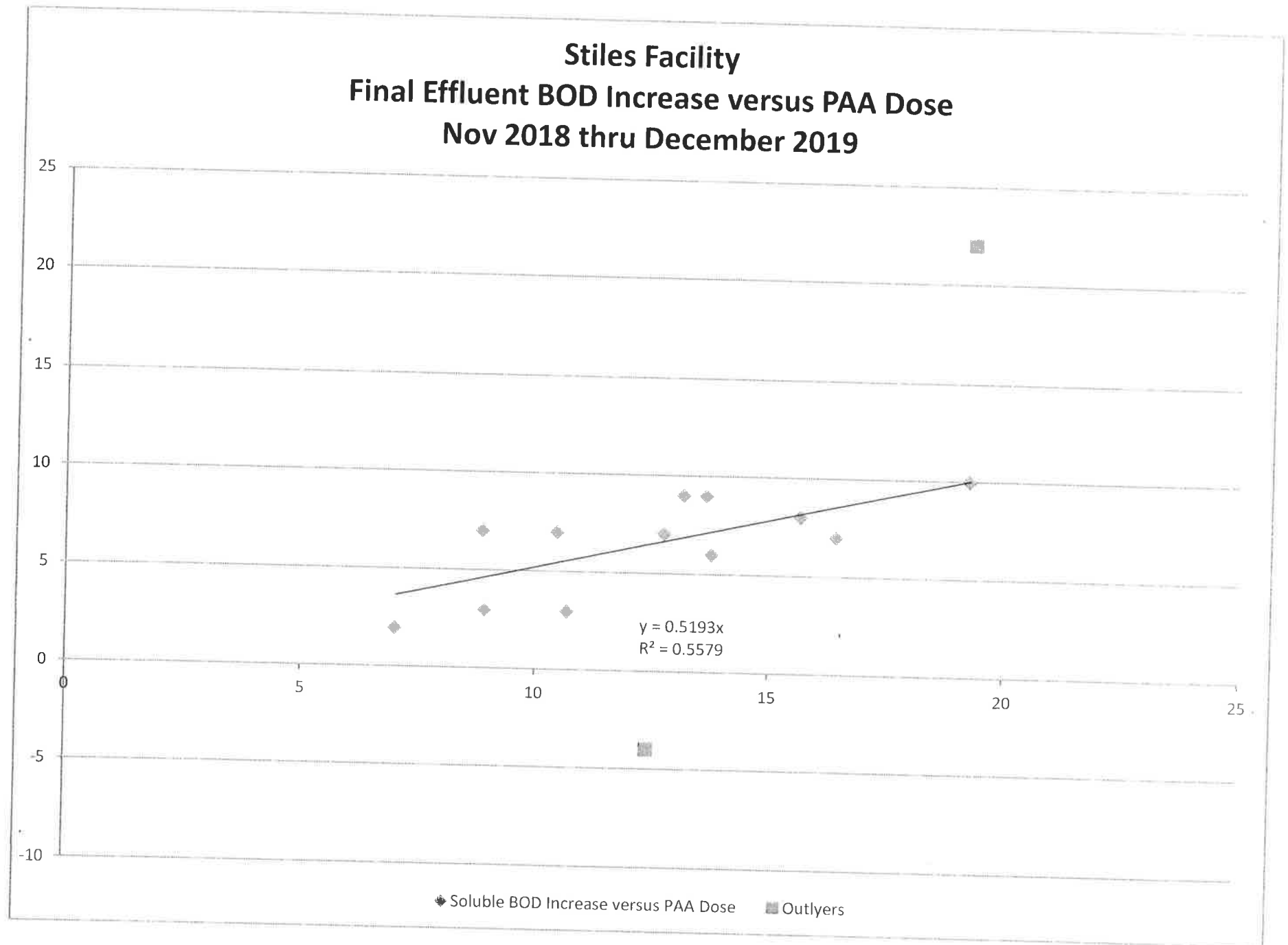
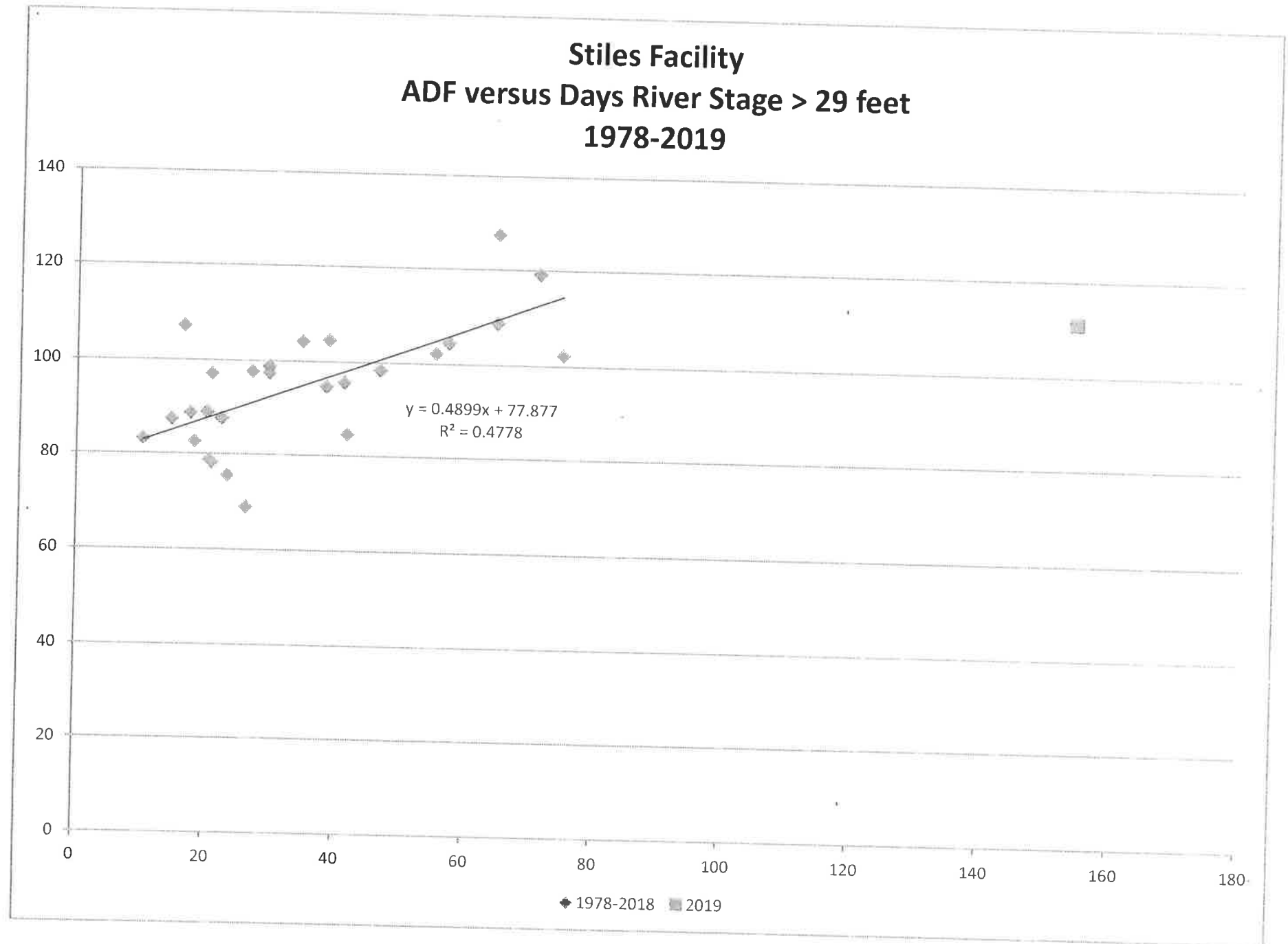


Figure 12



Historical Miss River Stage

Figure 13

Maxson Facility ADF and ADDWF mgd

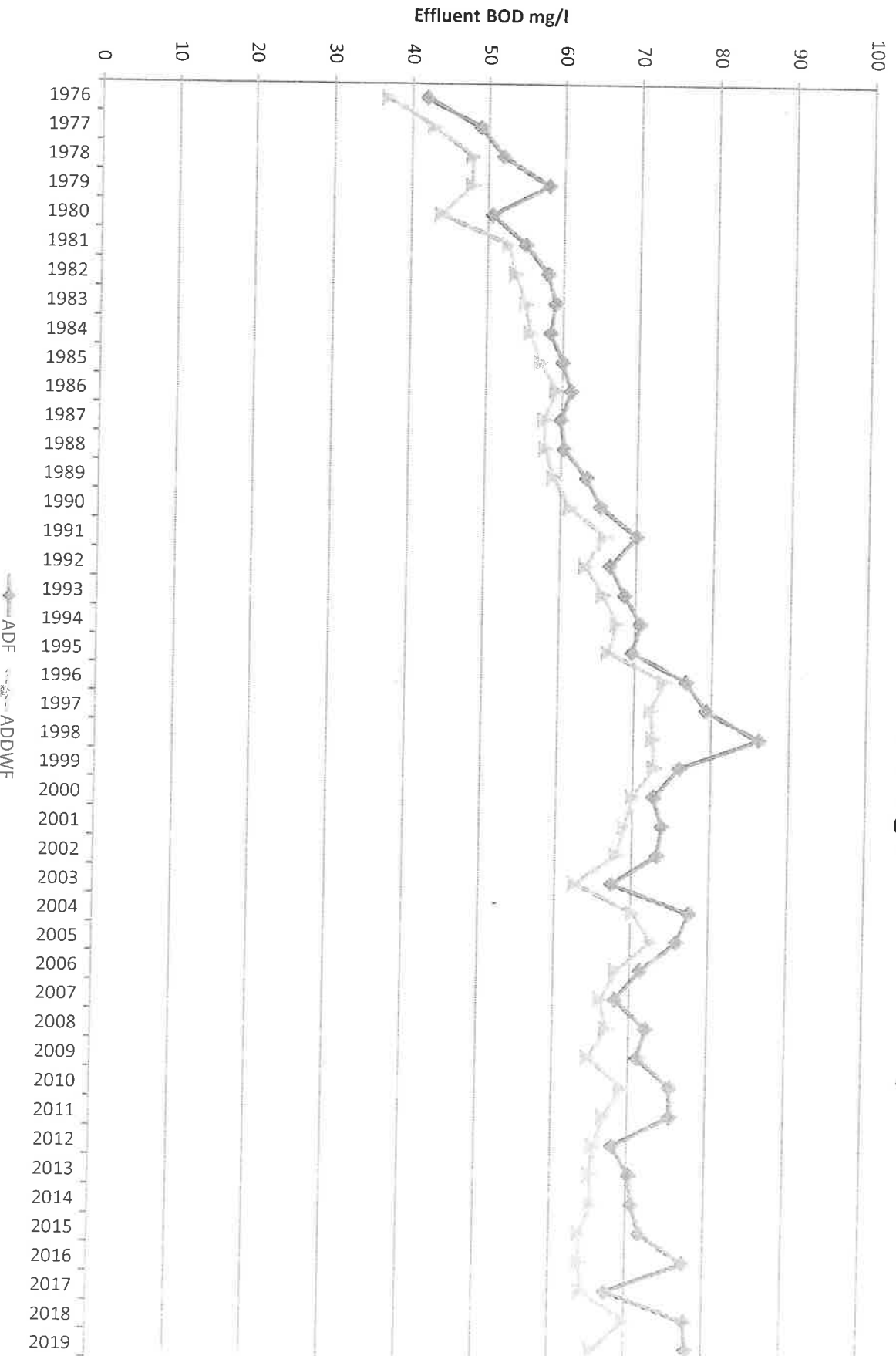


Figure 14

Maxson Facility Long Term BOD Loading

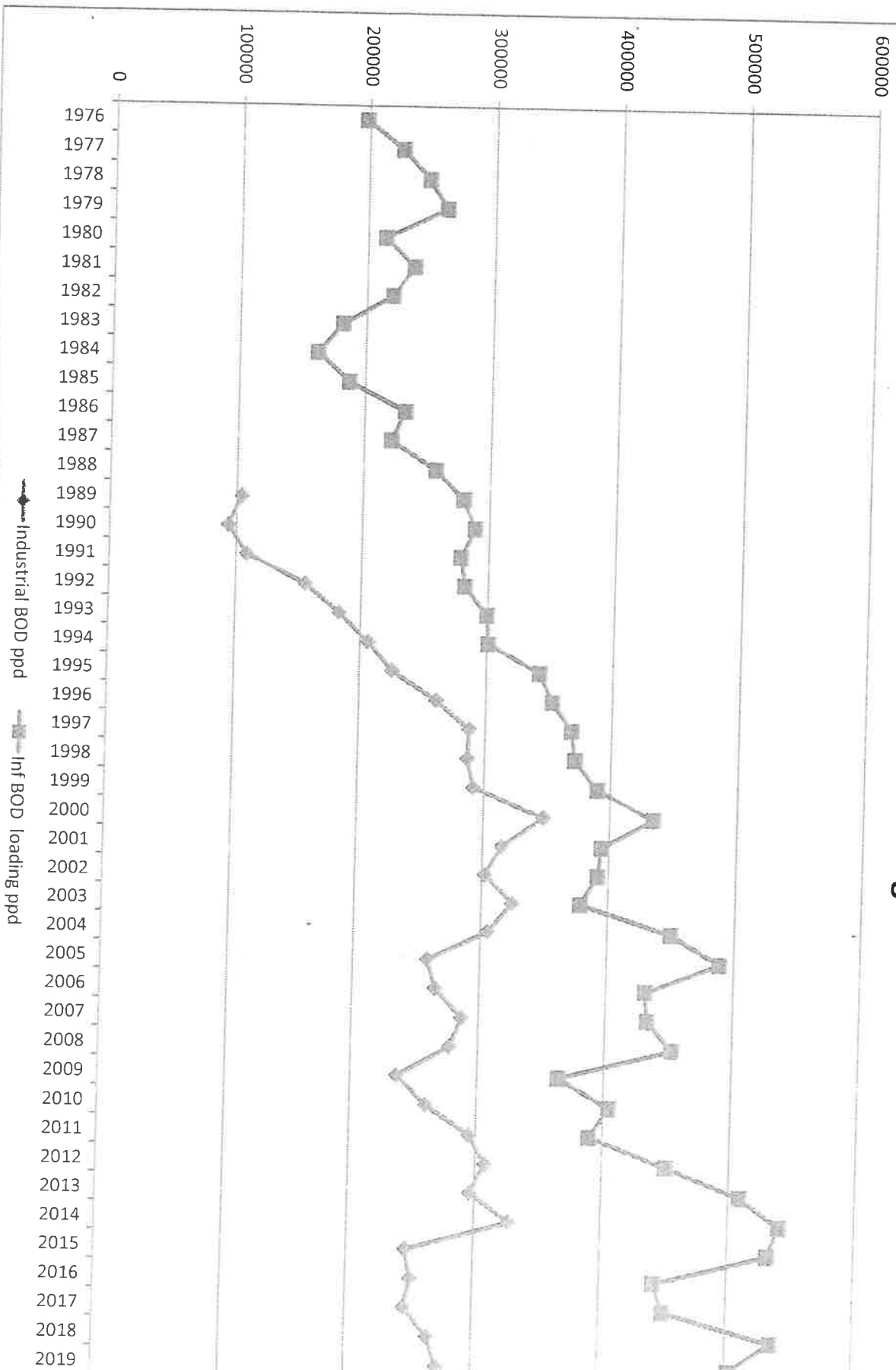


Figure 15

Maxson Facility Long Term TSS Loading

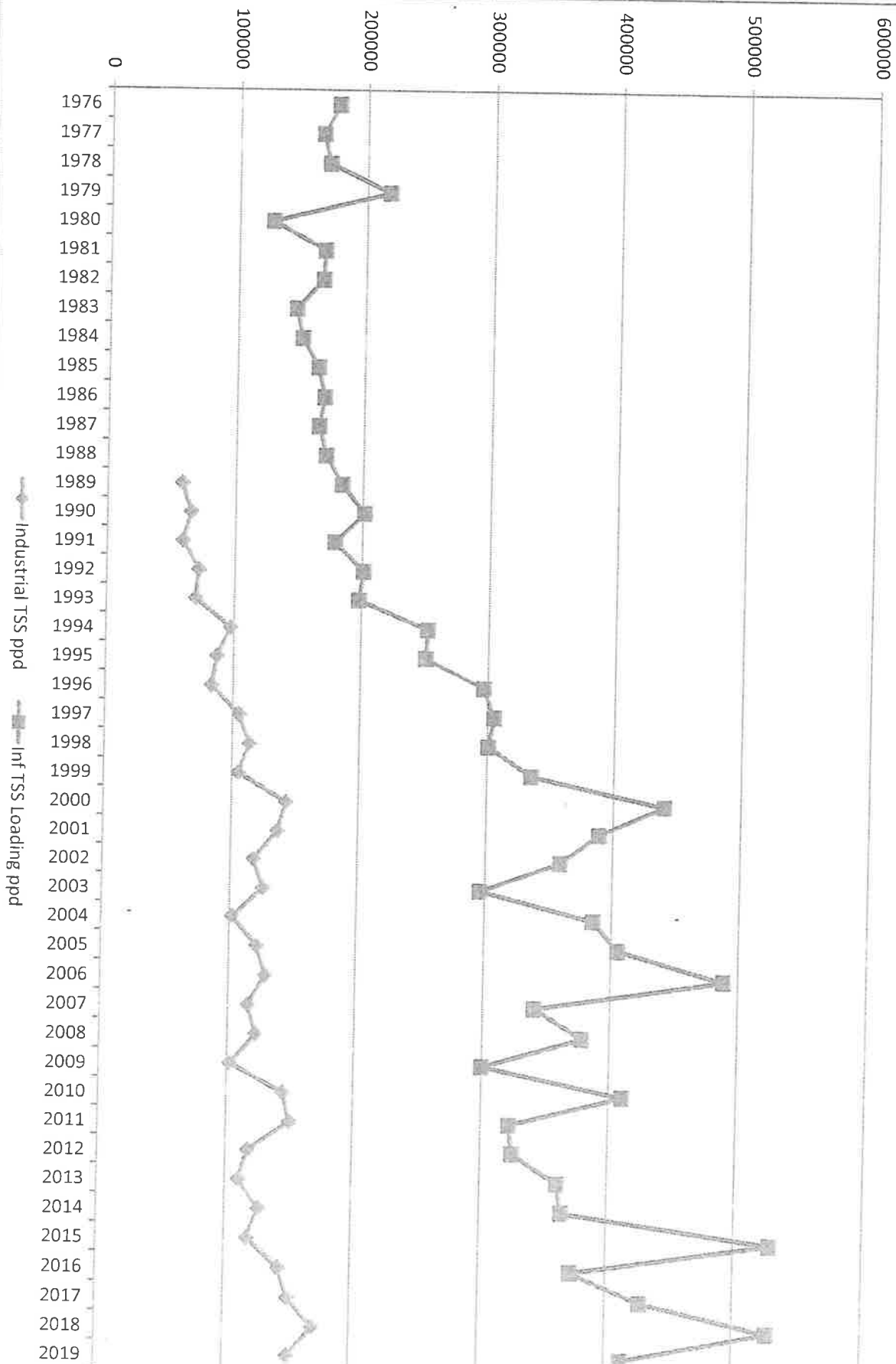


Figure 16

Stiles Facility ADF and ADDWF mgd

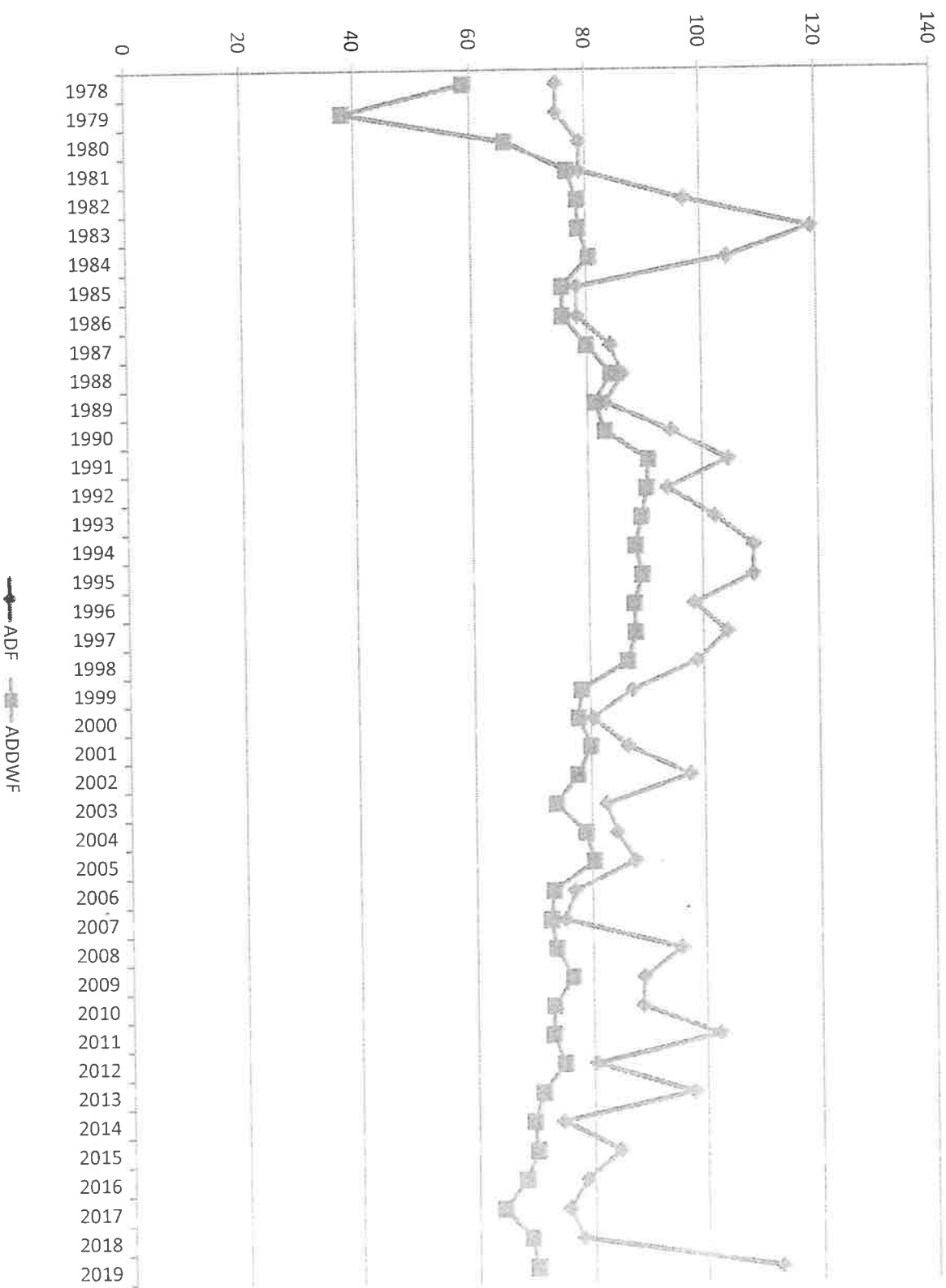


Figure 17

Stiles Facility Long Term BOD Loading ppd

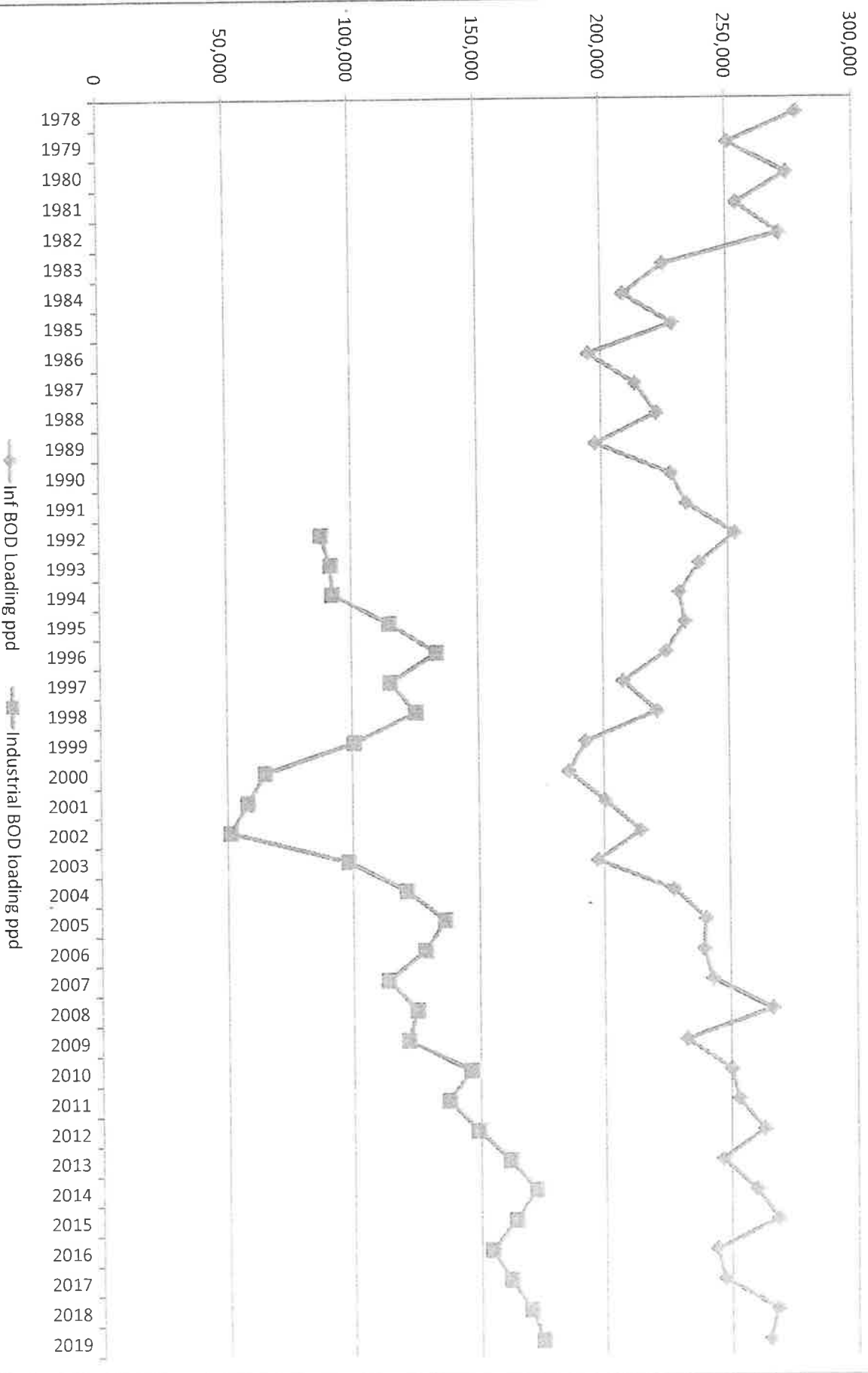


Figure 18

Stiles Facility Long Term TSS Loading ppd

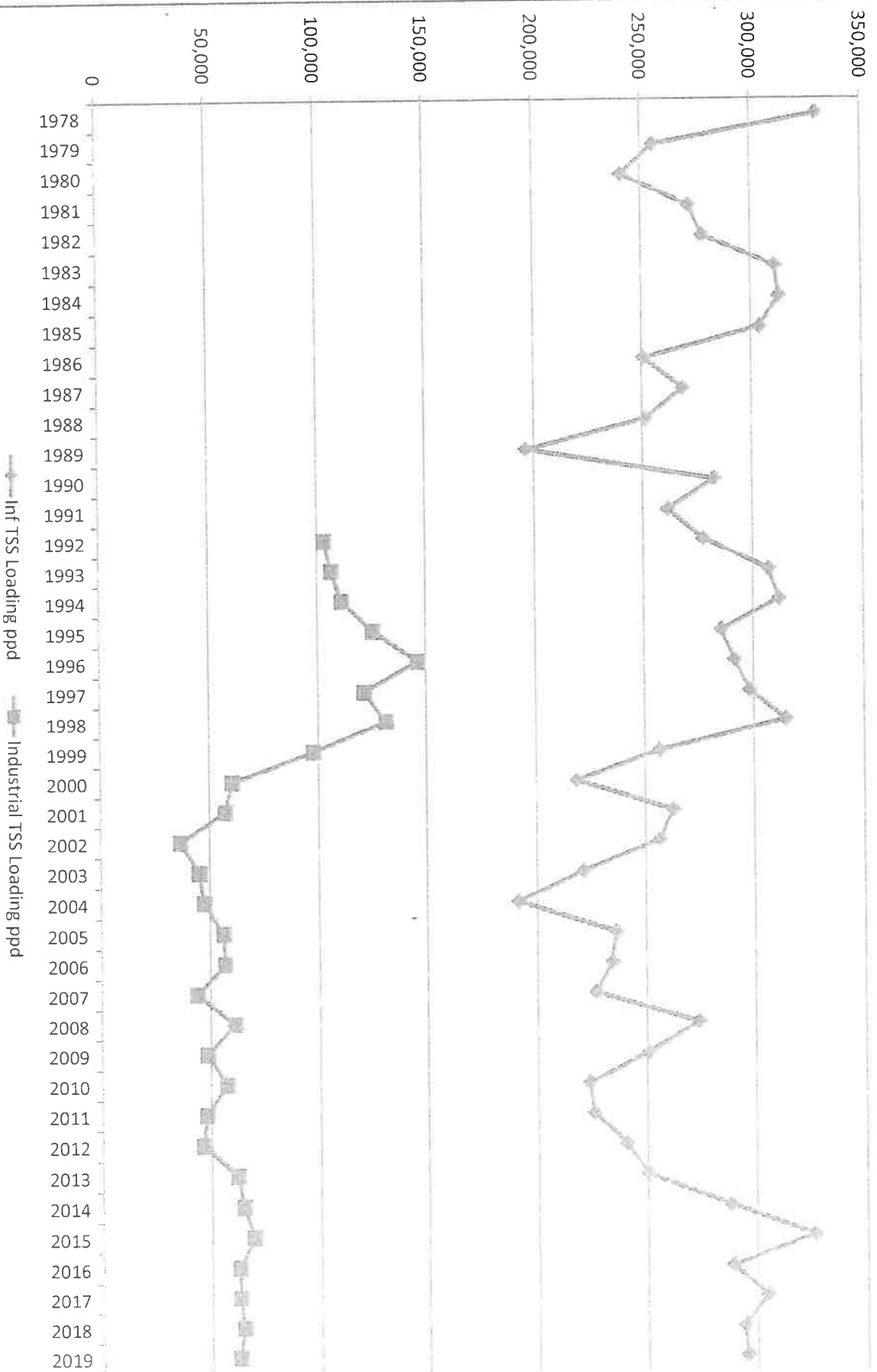


Figure 19

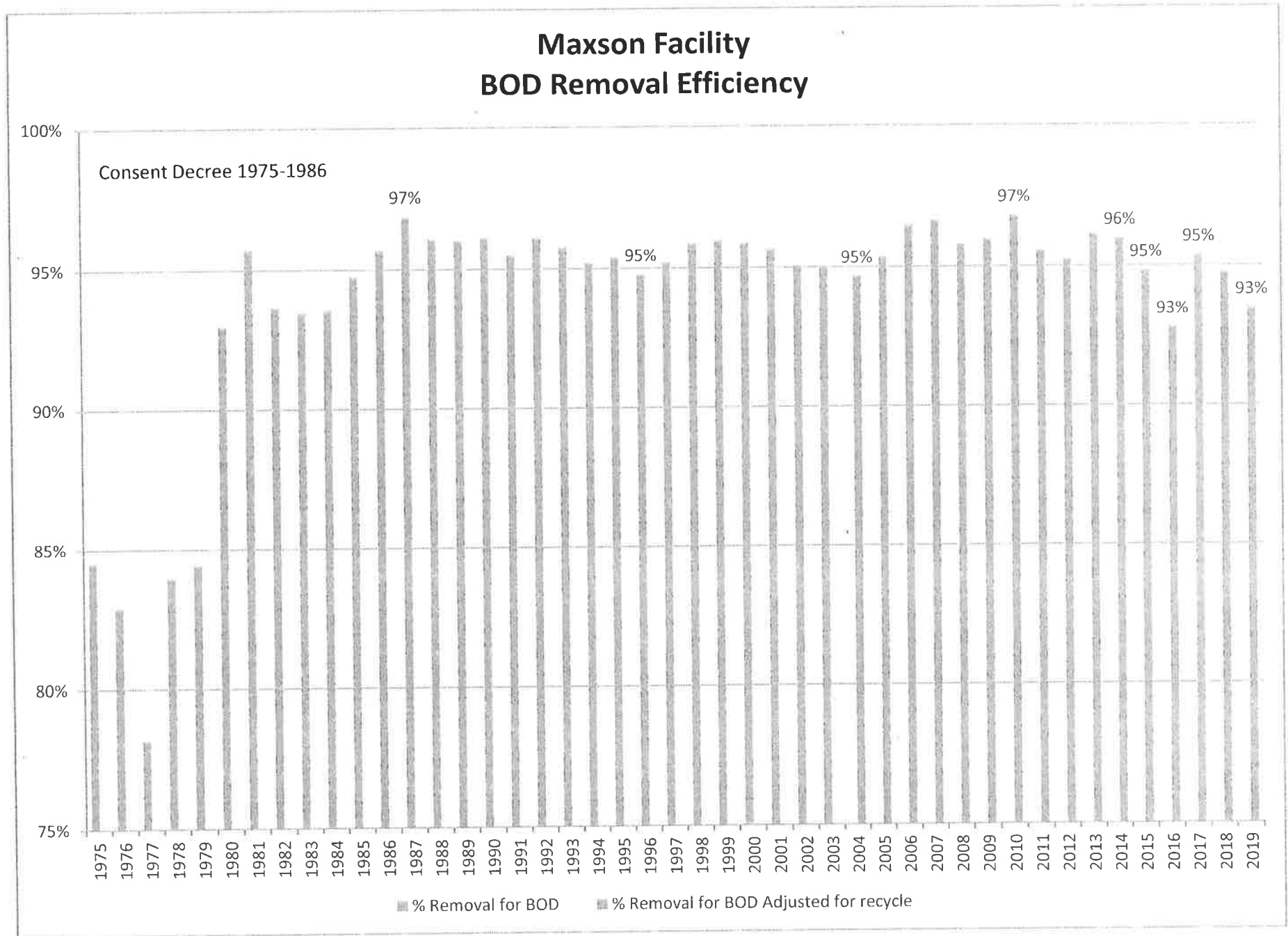


Figure 20

Maxson Facility TSS % Removal

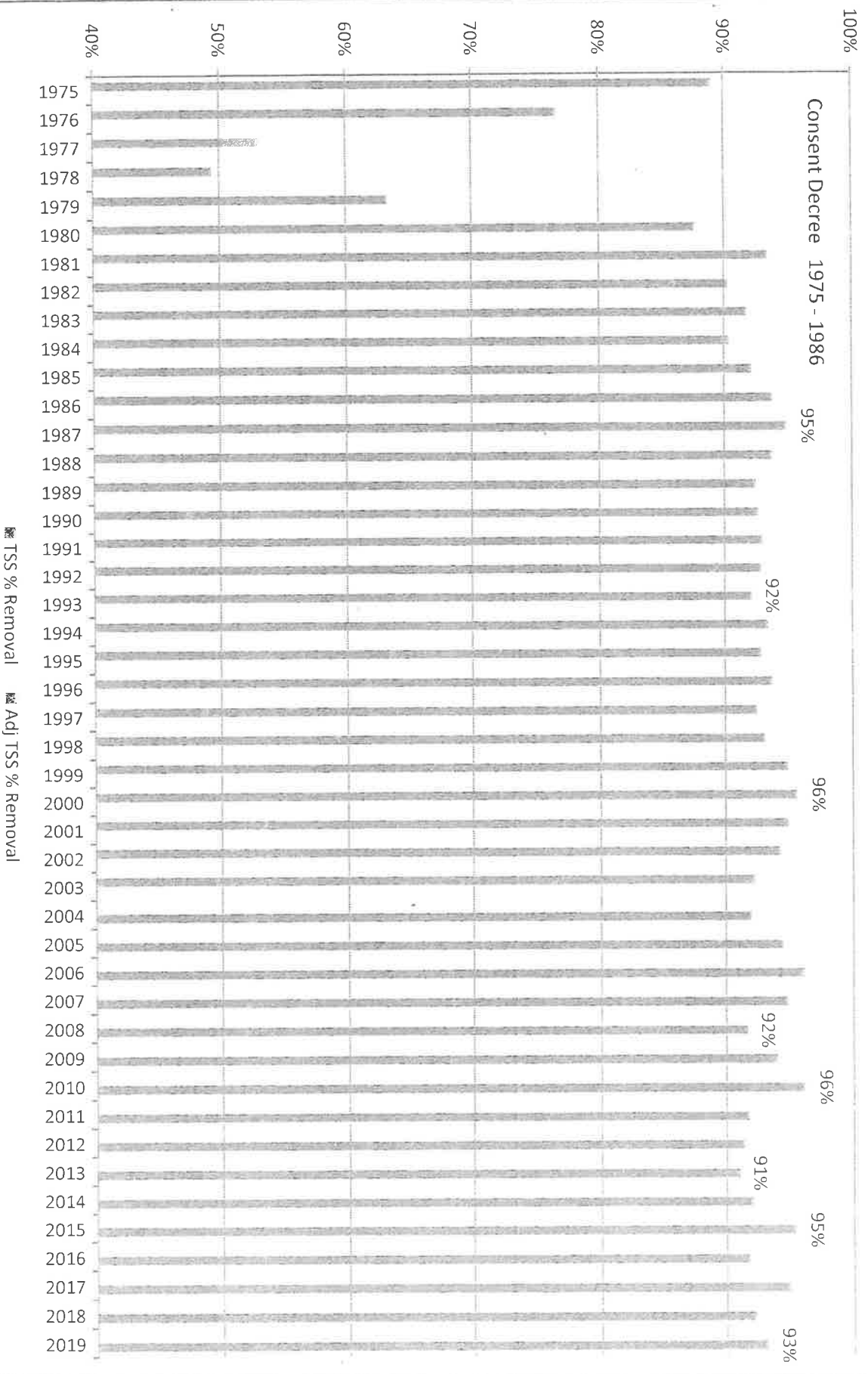


Figure 21

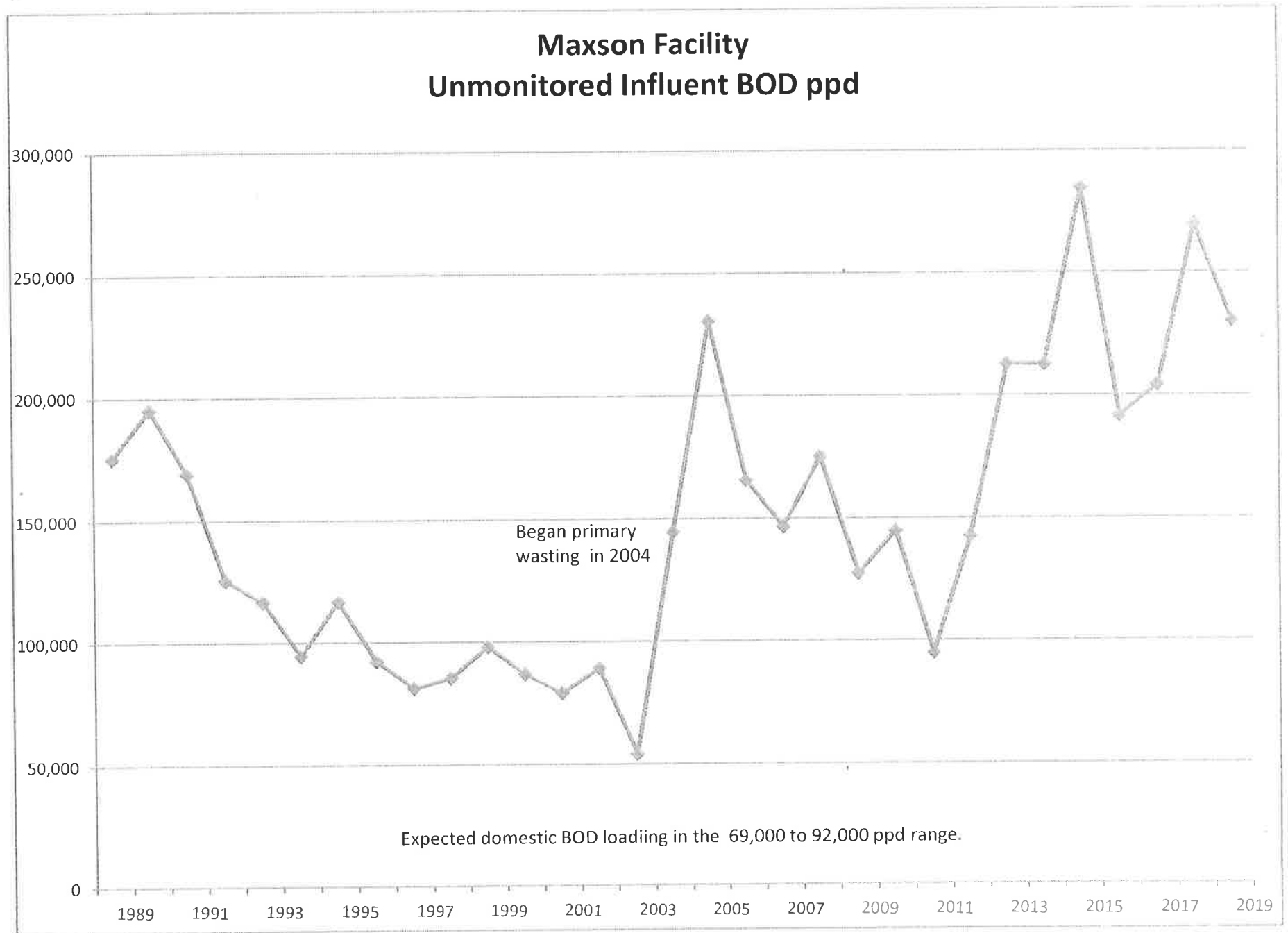


Figure 22

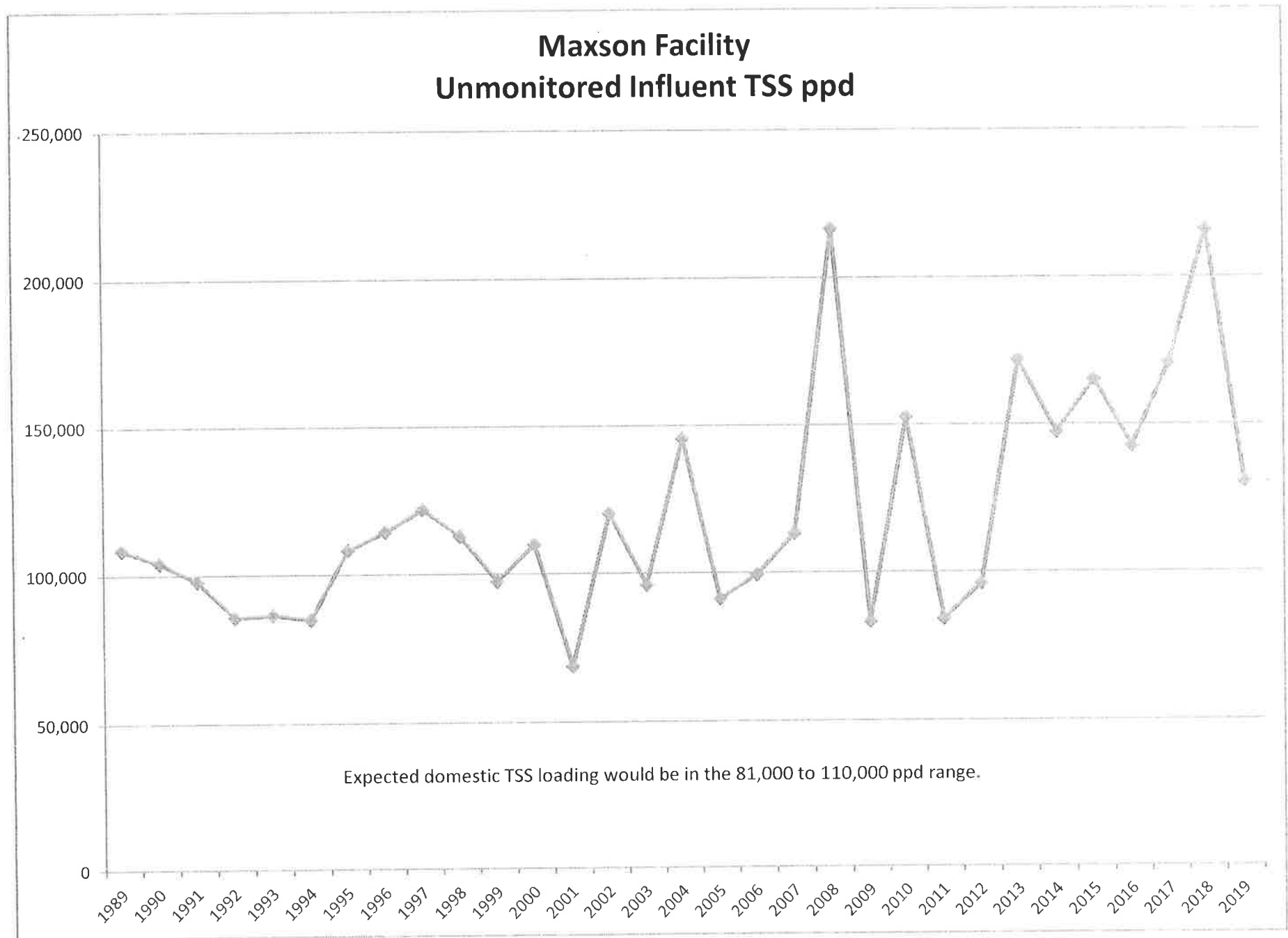


Figure 23

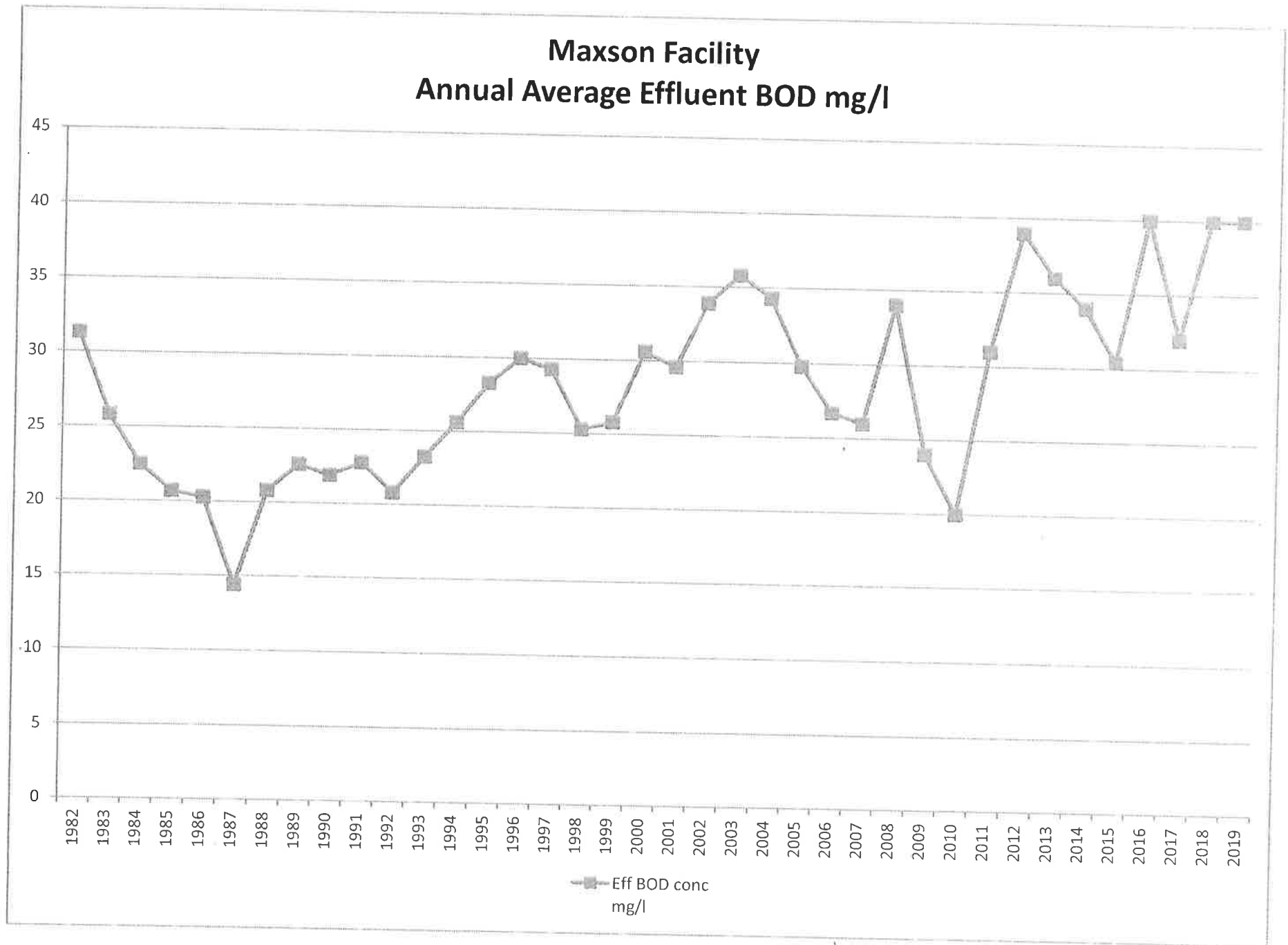


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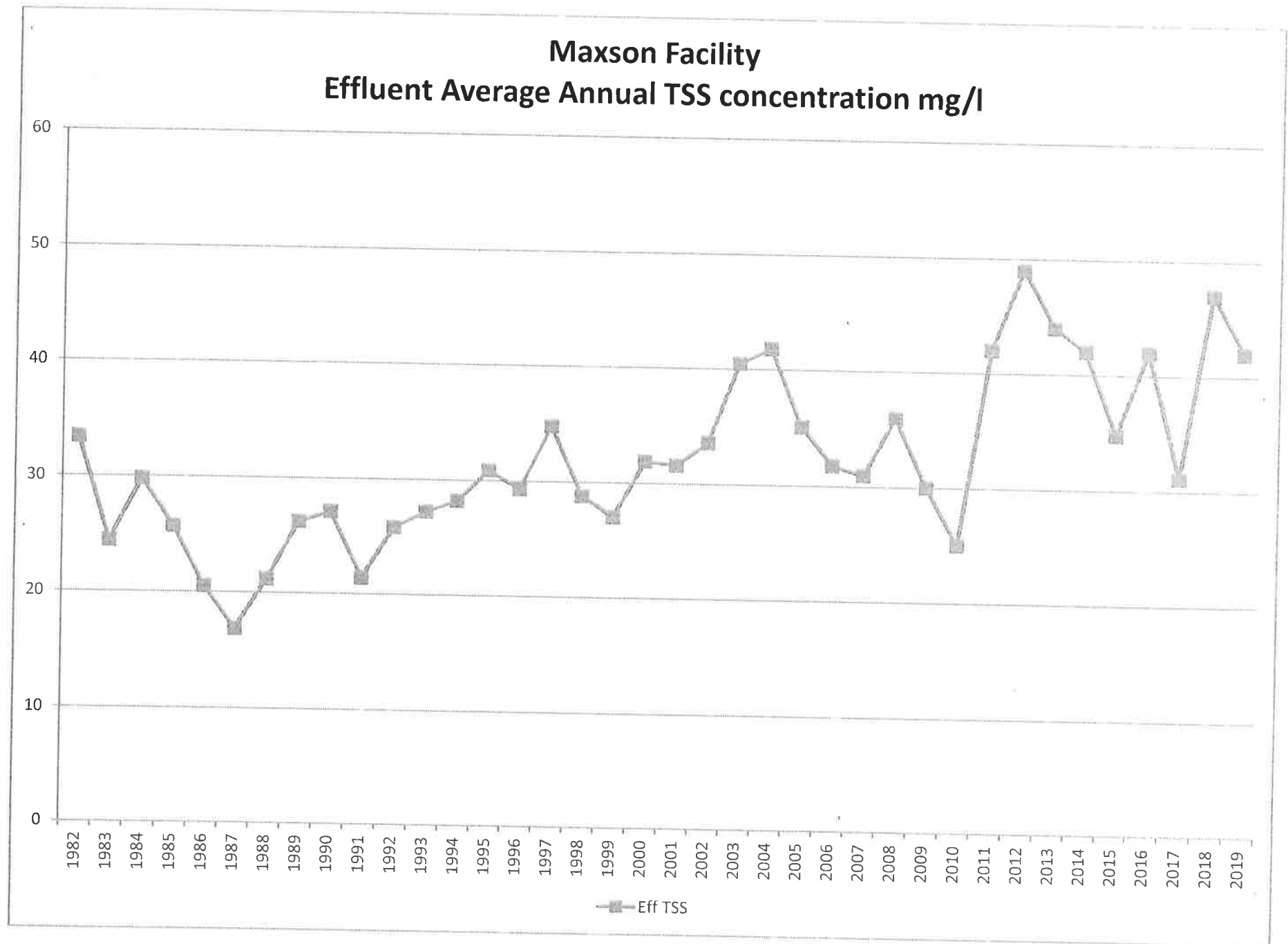


Figure 25

Maxson Facility Primary and Secondary Wasting ppd

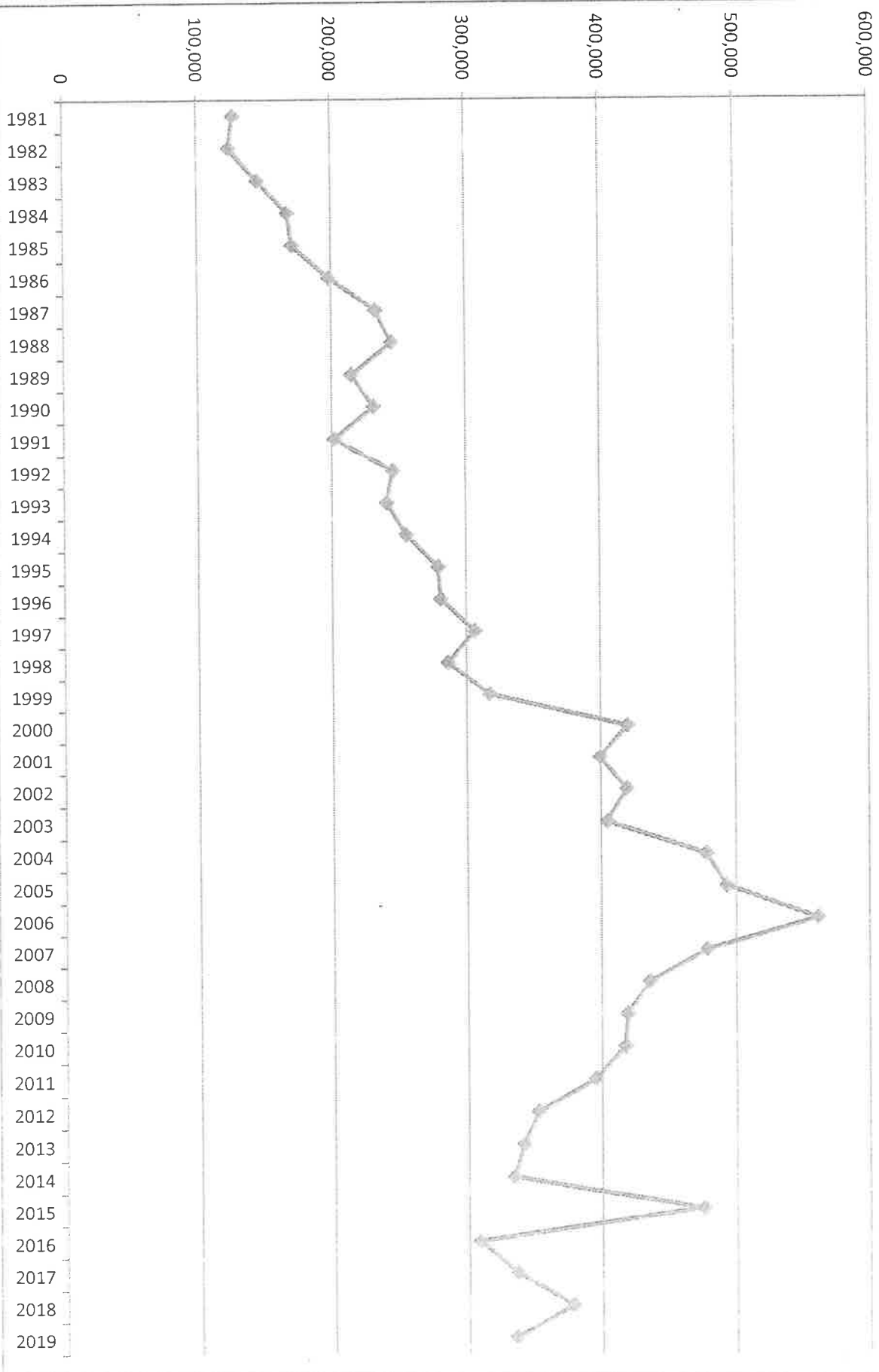


Figure 26

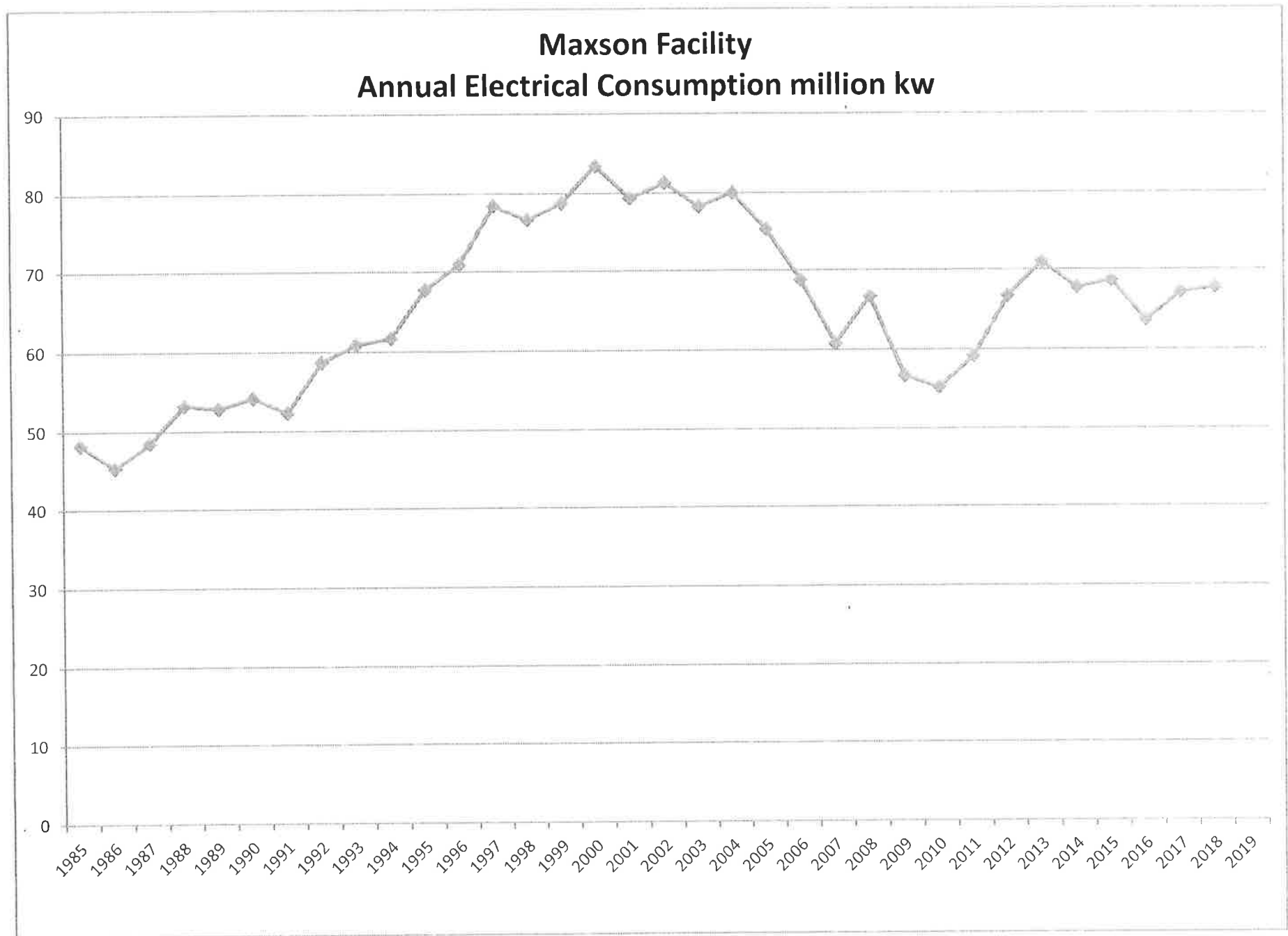


Figure 27

Stiles Facility BOD Percent Removal

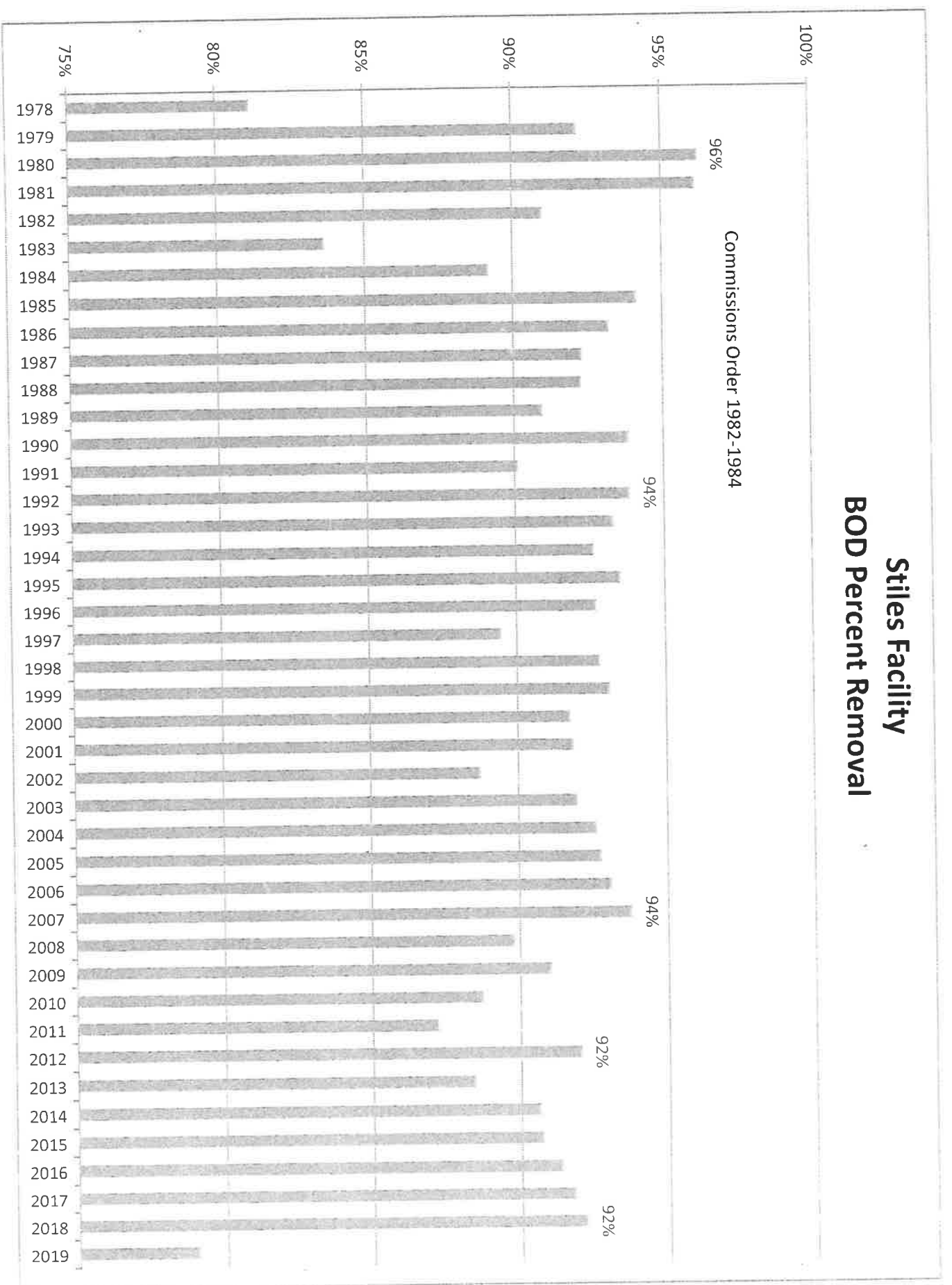


Figure 28

Stiles Facility TSS % Removal

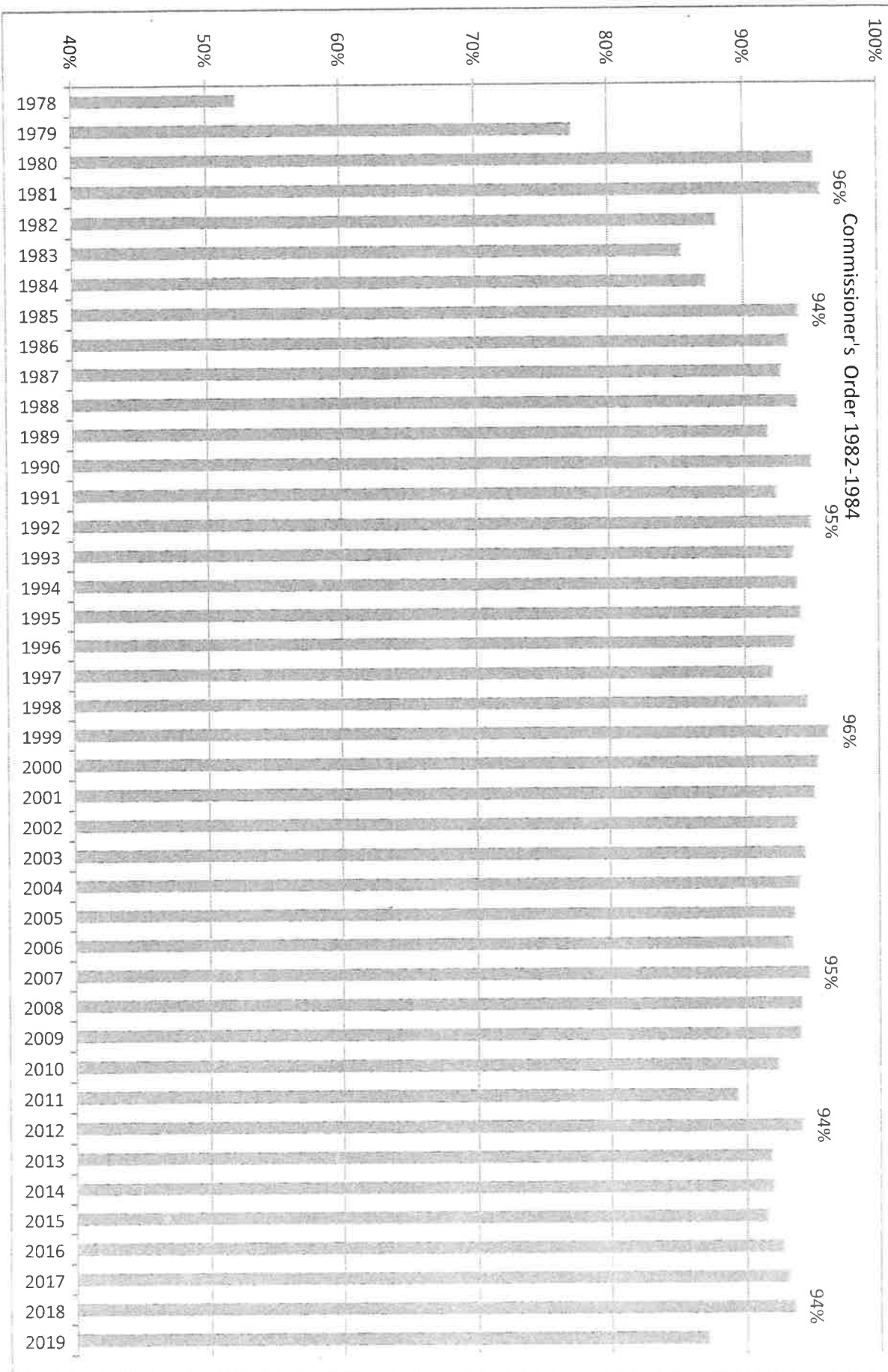


Figure 29

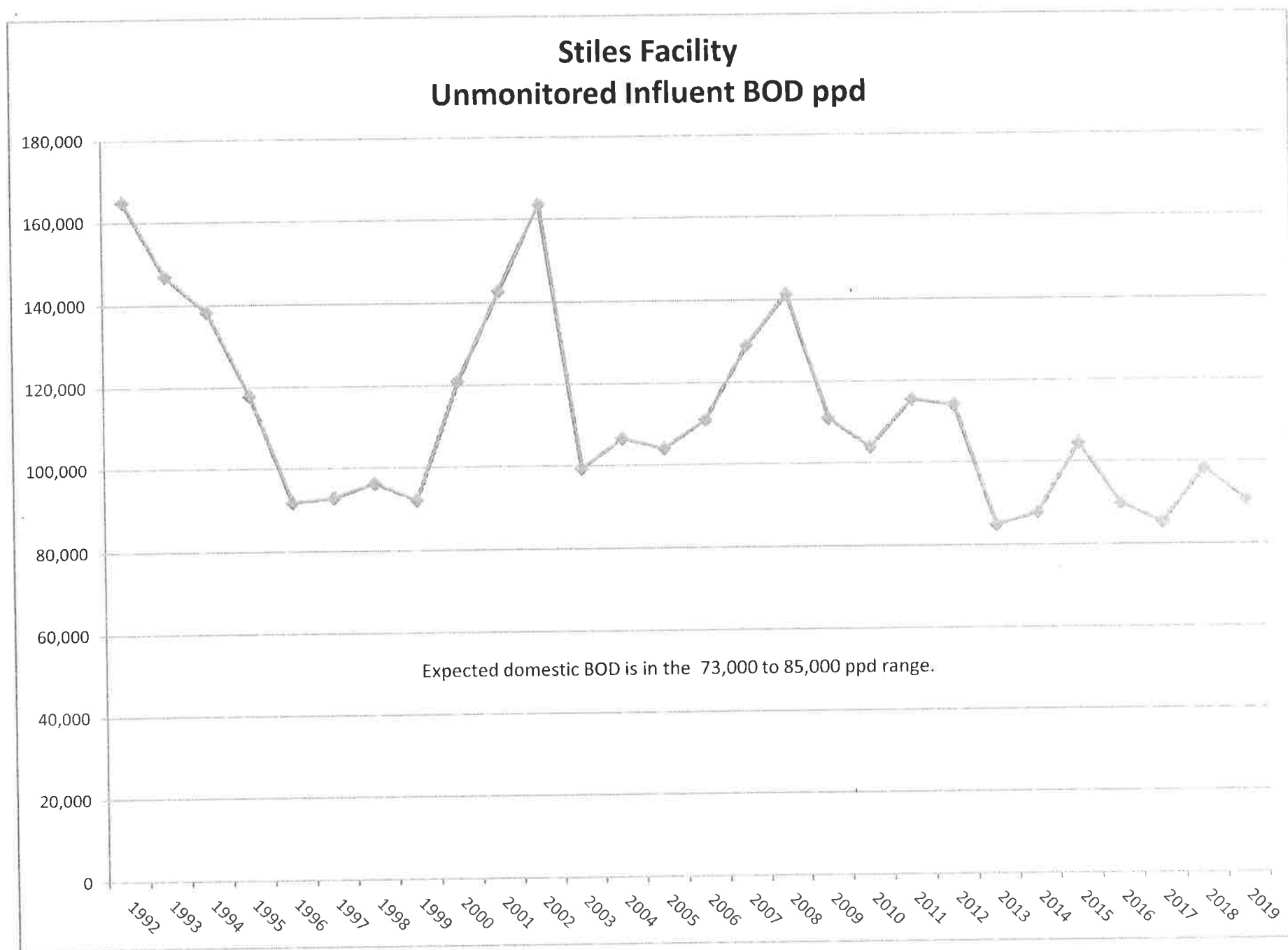


Figure 30

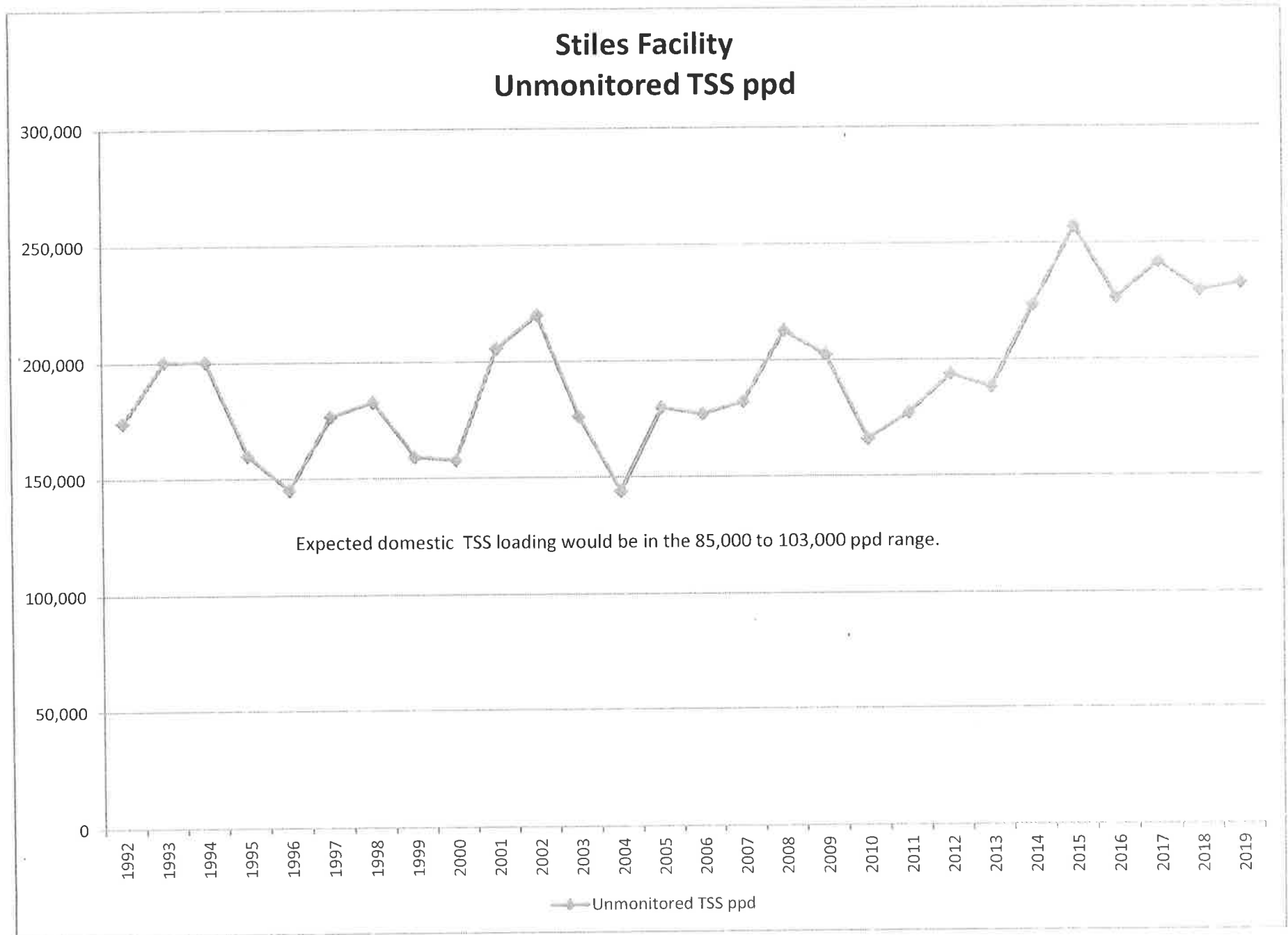


Figure 31

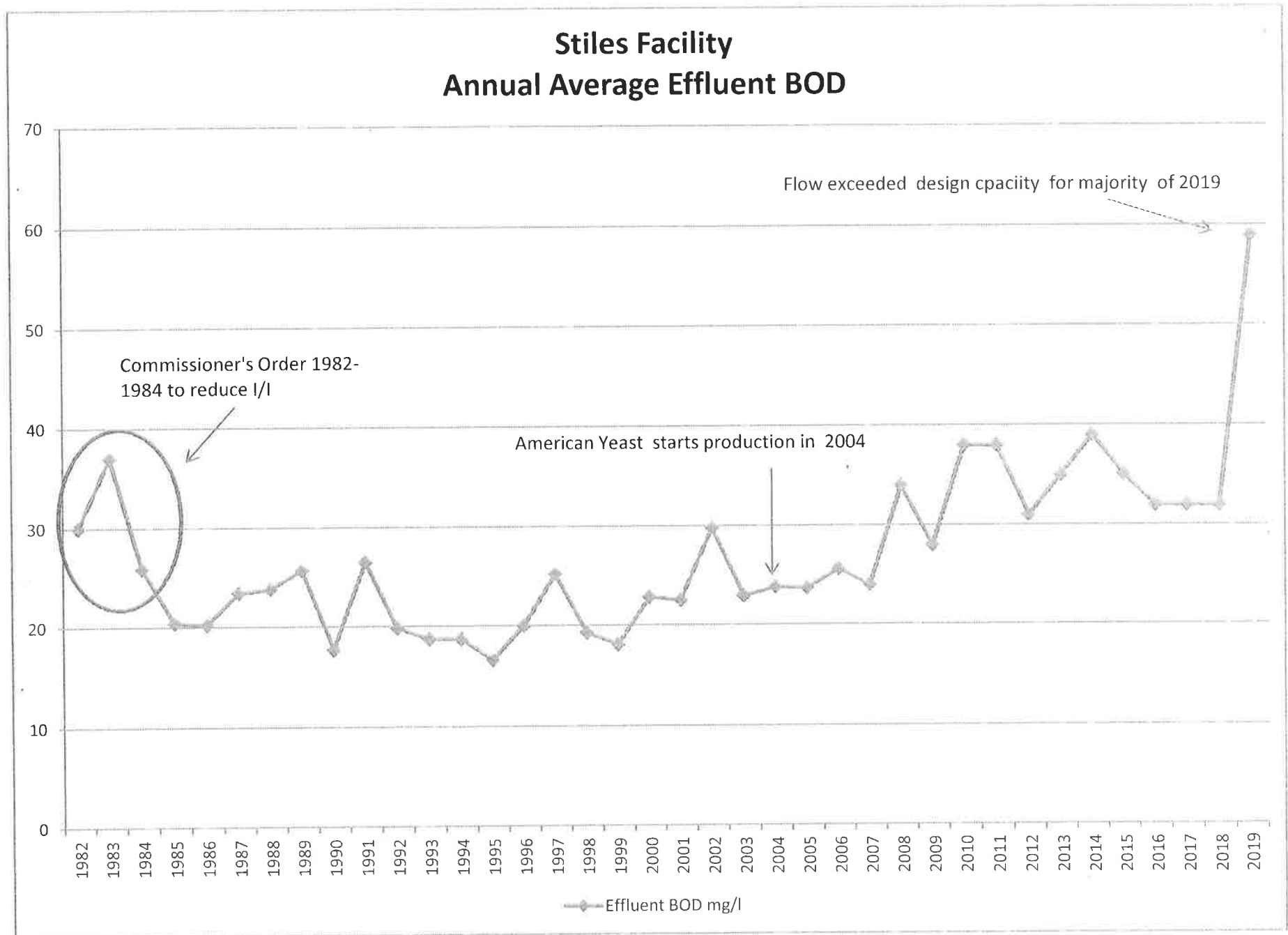


Figure 32

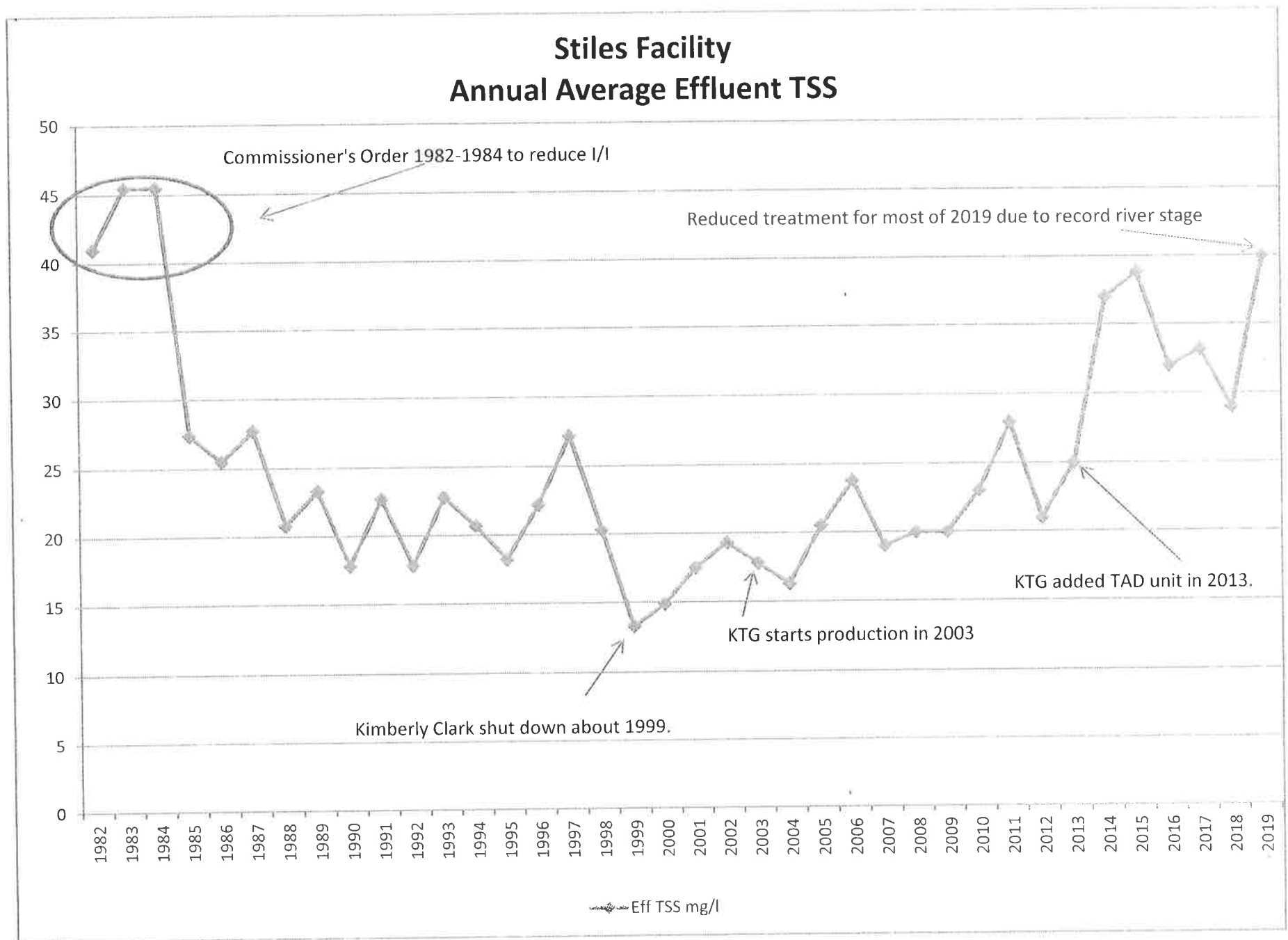


Figure 33

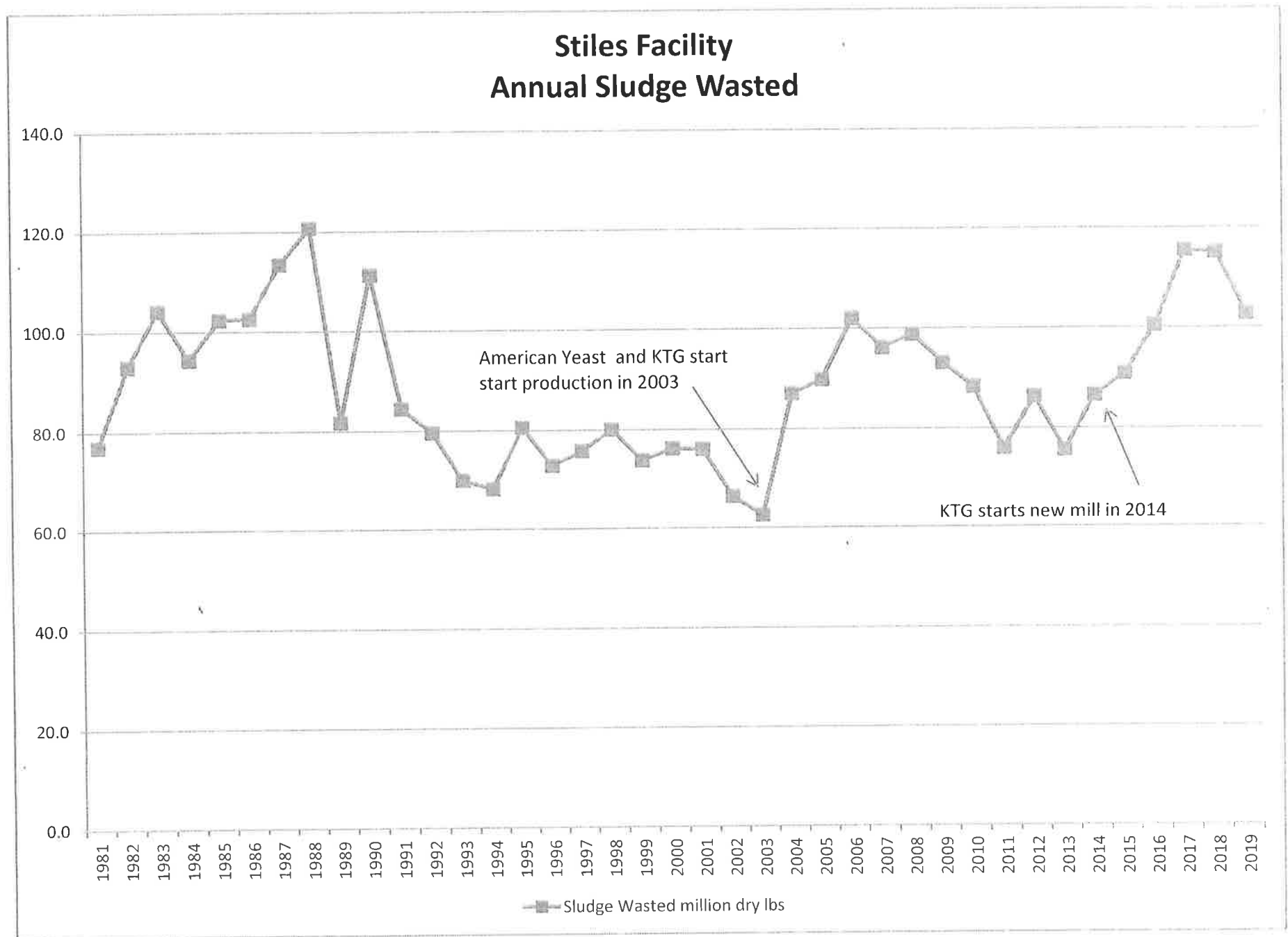


Figure 34

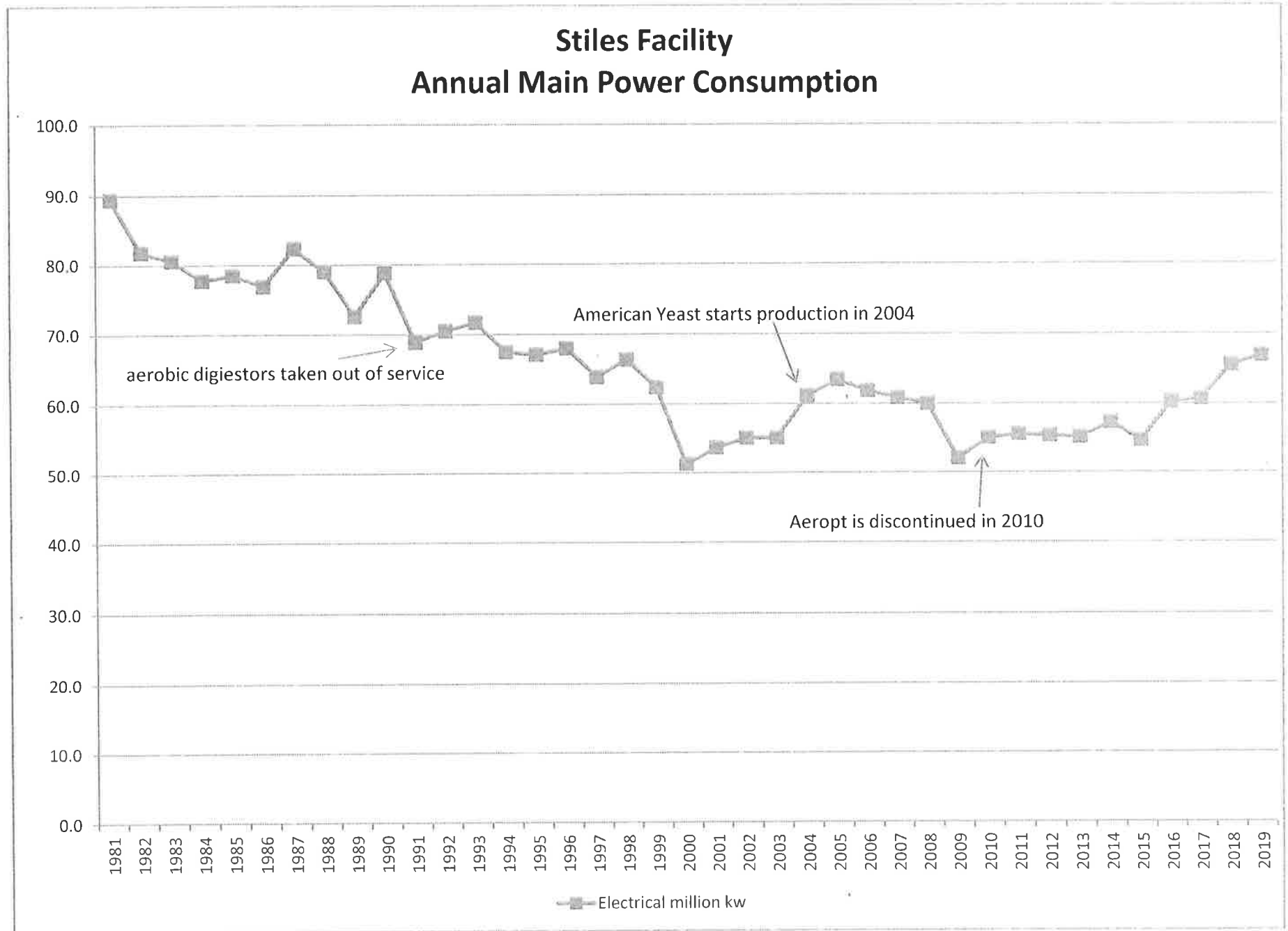


Figure 35

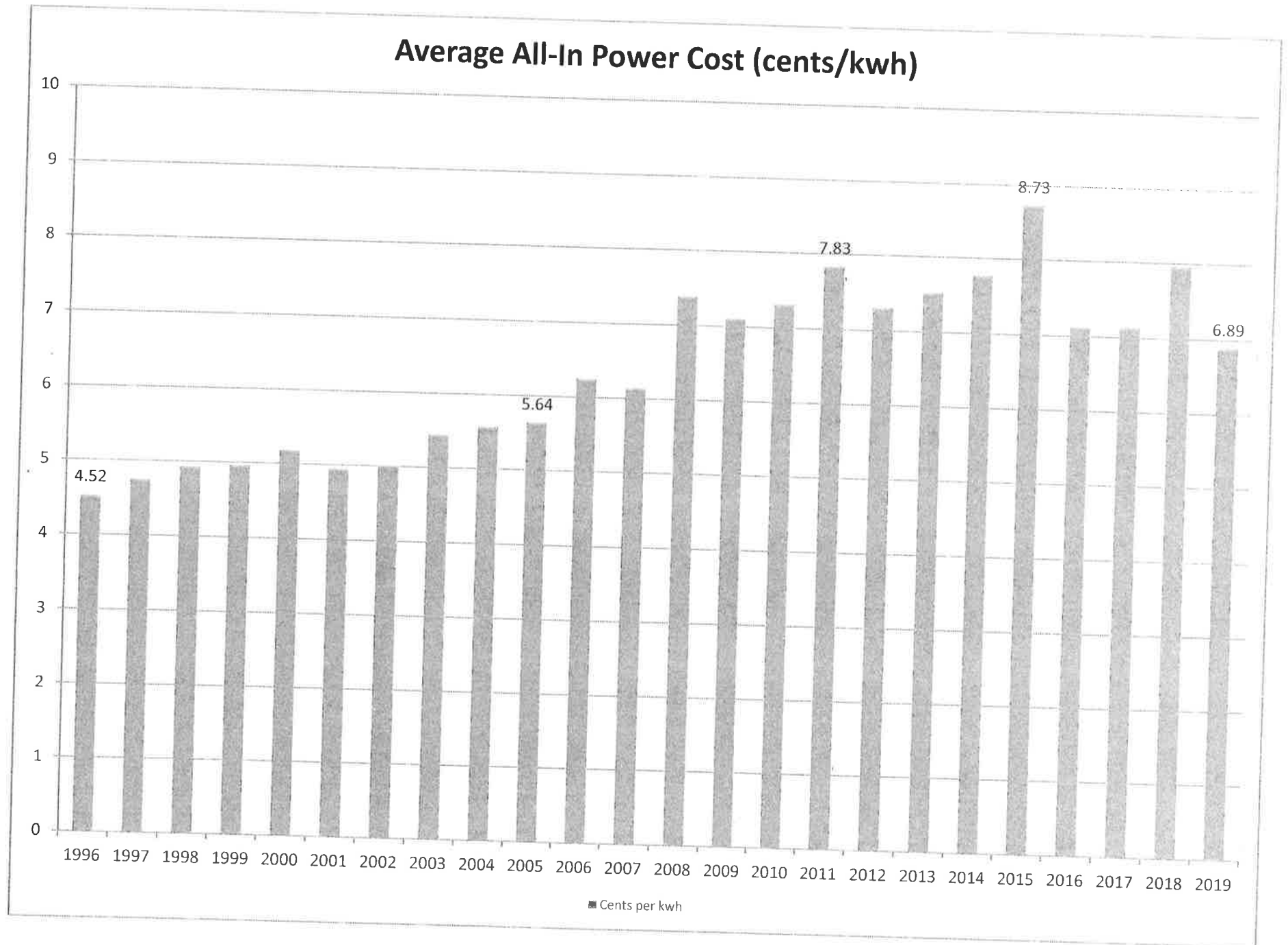


Figure 36

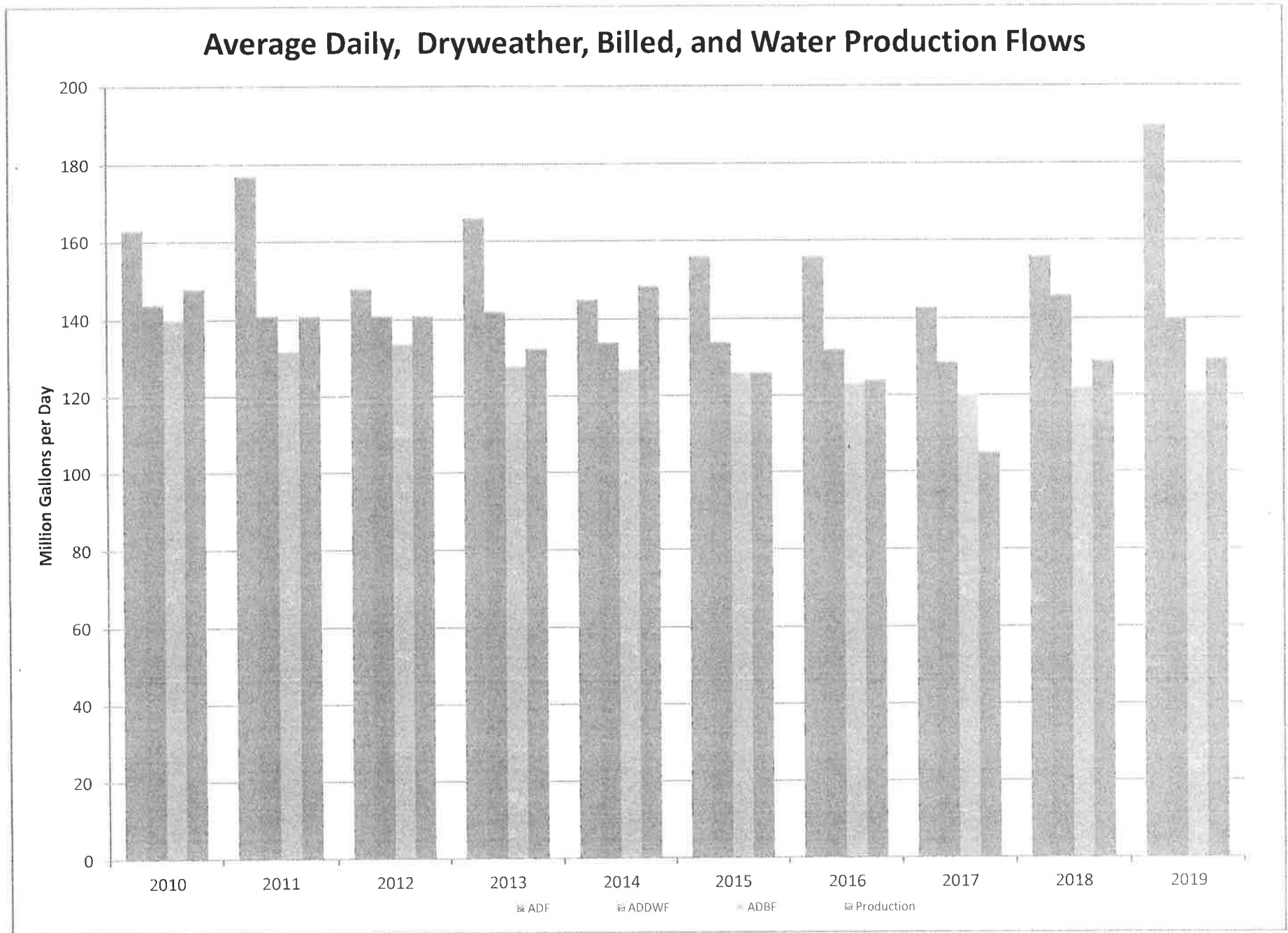


Figure 37

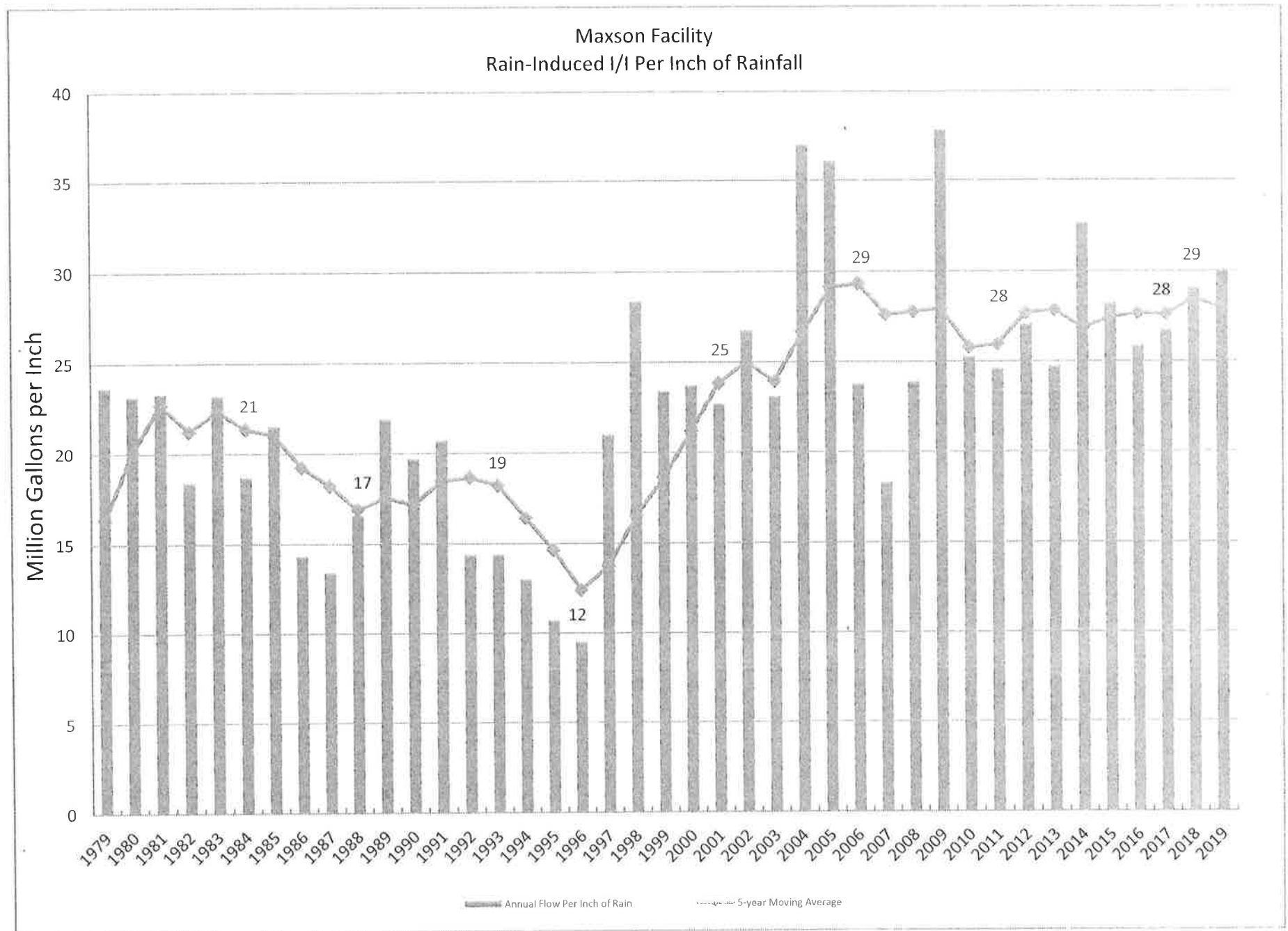


Figure 38

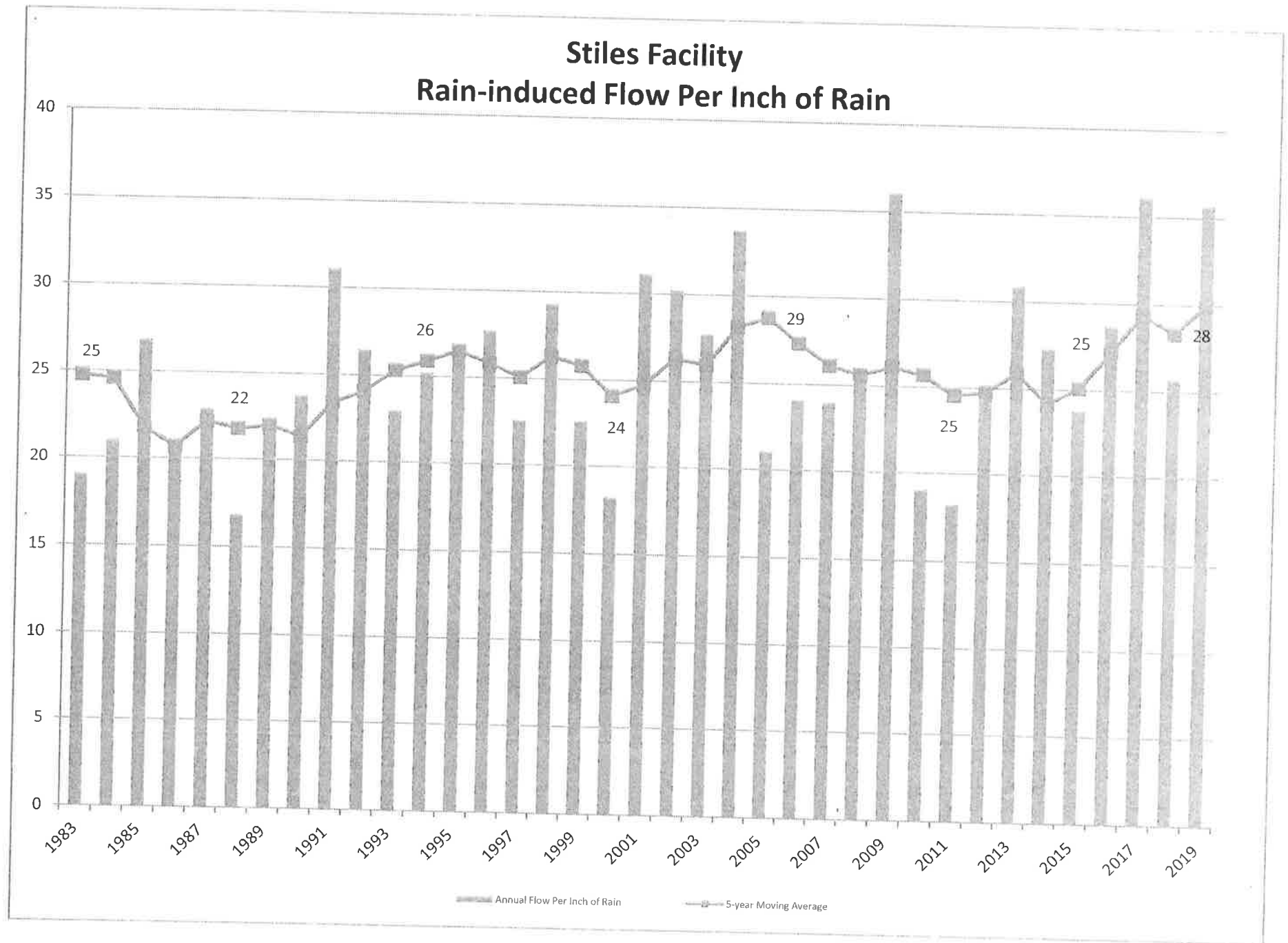


Figure 39

Combined Average Daily Flows

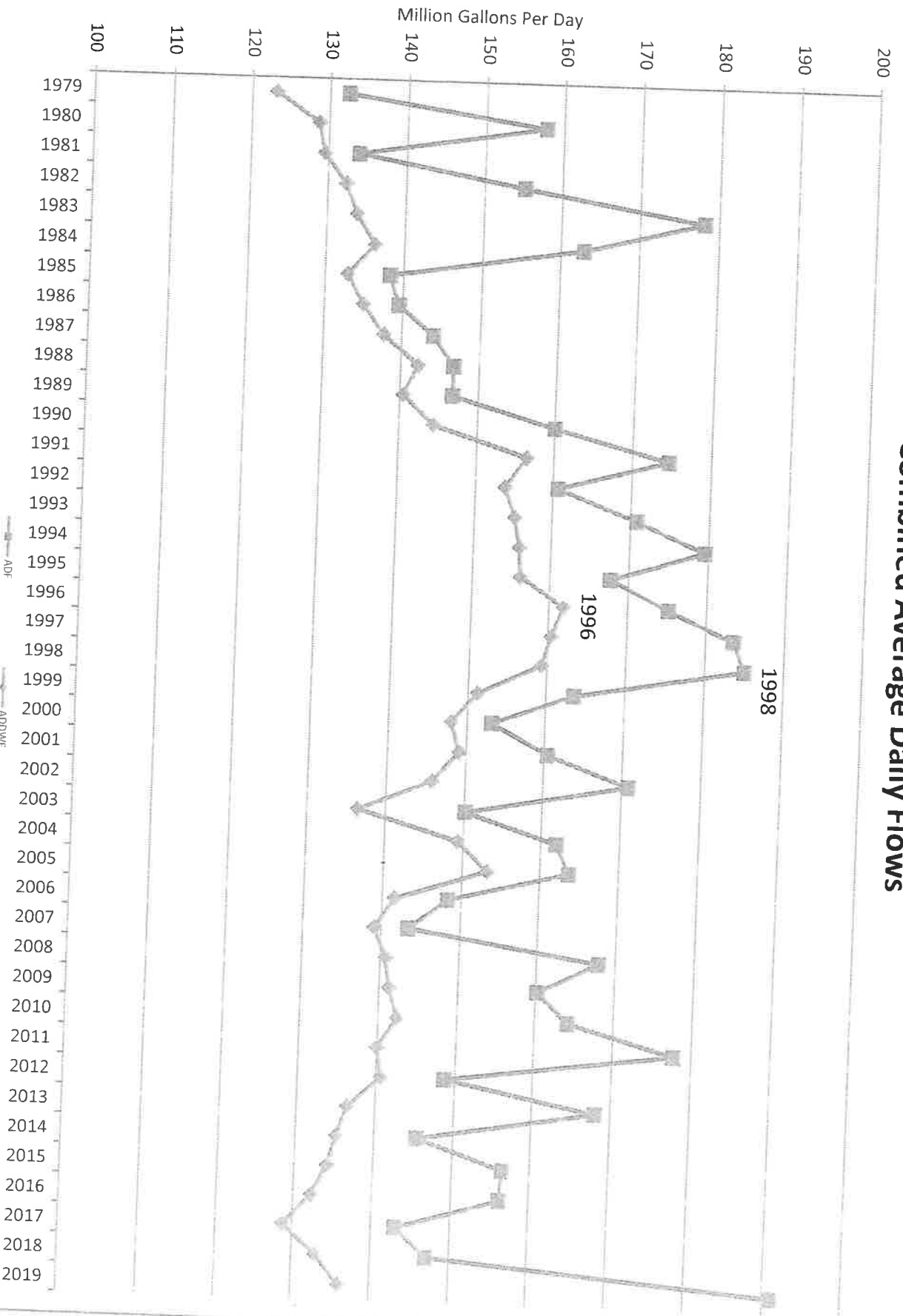
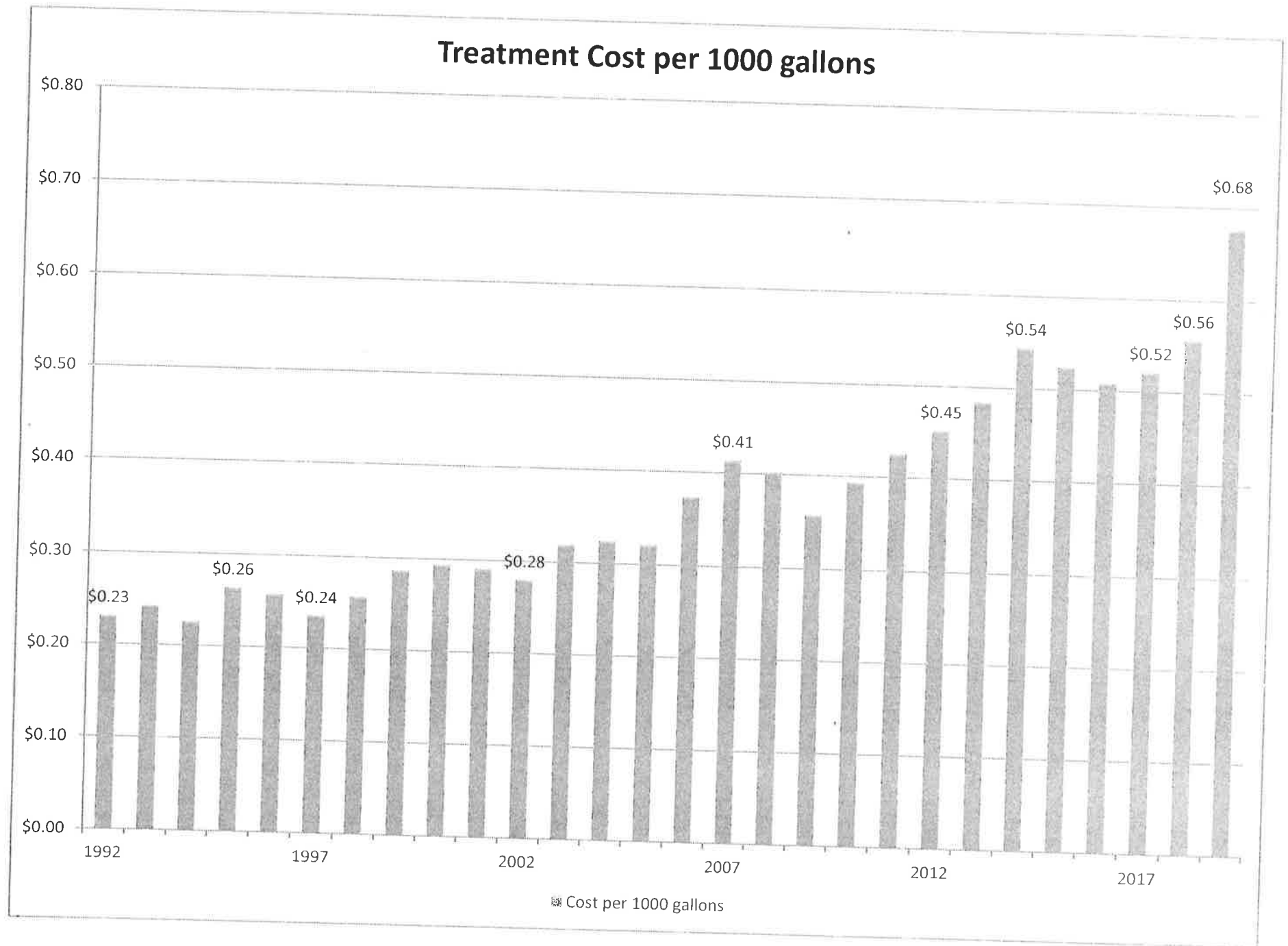


Figure 40



TABLES

Table 1
2019 Calendar Year Report
T.E. Maxson Facility

	Bio-Chemical Oxygen Demand (BOD)				Chemical Oxygen Demand (COD)			Total Suspended Solids (TSS)		
	Flow	Influent		Effluent	Influent		Effluent	Influent		Effluent
	<u>MGD</u>	<u>lbs/day</u>	<u>mg/l</u>	<u>mg/l</u>	<u>lbs/day</u>	<u>mg/l</u>	<u>mg/l</u>	<u>lbs/day</u>	<u>mg/l</u>	<u>mg/l</u>
January	85	489,850	691	34	1,088,870	1536	204	438,100	618	41
February	94	630,304	804	46	1,204,163	1536	211	601,297	767	58
March	102	546,987	643	60	962,119	1131	242	421,087	495	65
April	86	435,365	607	28	846,343	1180	157	306,261	427	33
May	77	480,993	749	16	804,009	1252	118	342,924	534	19
June	77	513,102	799	35	846,393	1318	109	375,675	585	24
July	73	456,006	749	37	882,789	1450	139	460,268	756	34
August	61	438,025	861	29	806,353	1585	141	353,574	695	29
September	60	474,379	948	33	895,716	1790	181	373,799	747	32
October	73	501,059	823	43	1,019,165	1674	193	423,130	695	43
November	72	520,016	866	50	979,383	1631	229	401,121	668	58
December	72	547,037	911	68	1,255,003	2090	283	458,166	763	68
2019 Average	78	502,760	788	40	965,859	1514	184	412,950	646	42
2018 Average	78	535,000	829	40	1,033,357	1602	227	526,650	807	47
2017 Average	68	450,747	802	28	936,909	1663	143	427,315	759	31
2016 Average	77	456,184	721	40	958,908	1556	144	378,486	622	42
2015 Average	72	532,083	908	31				524,421	903	36
2014 Average	71	540,103	920	34				364,476	621	42
2013 Average	70	508,695	869	36				360,547	615	44
2012 Average	68	450,168	794	39				324,884	573	49
2011 Average	75	389,421	642	31				330,378	541	42
2010 Average	75	404,628	648	20				410,627	650	25
2009 Average	71	364,049	625	24				299,750	511	30

Table 2
2019 Calendar Year Report
M.C. Stiles Facility

	Bio-Chemical Oxygen Demand (BOD)					Chemical Oxygen Demand (COD)			Total Suspended Solids (TSS)		
	Flow MGD	Influent		Effluent mg/l	Effluent Adj for PAA mg/l	Influent		Effluent mg/l	Influent		Effluent mg/l
		lbs/day	mg/l			lbs/day	mg/l		lbs/day	mg/l	
January	107	256,693	289	37	34	669,710	754	223	275,345	310	30
February	128	240,506	226	45	38	634,254	596	201	312,870	294	38
March	176	231,260	158	85	83	594,250	406	252	228,333	156	61
April	148	292,139	237	83	76	594,138	482	259	331,583	269	53
May	153	276,534	217	93	97	616,786	484	268	322,411	253	63
June	152	255,139	201	78	75	599,132	472	258	232,291	183	66
July	121	365,712	363	105	83	867,433	861	322	560,154	556	56
August	73	259,966	427	57	47	526,629	865	273	217,958	358	44
September	70	290,806	496	30	23	705,321	1203	211	322,466	550	21
October	79	269,815	409	30	24	740,836	1123	193	295,543	448	15
November	71	213,771	360	34	25	587,277	989	207	187,643	316	18
December	75	231,435	370	35	26	580,464	928	191	261,459	418	17
2019 Average	113	265,315	313	59	53	643,019	764	238	295,671	343	40
2018 Average	79	268,114	417	32		612,655	949	200	294,417	456	29
2017 Average	75	247,351	411	32		621,586	1030	176	304,940	504	33
2016 Average	79	244,332	383	32		658,837	1041	172	289,269	460	32
2015 Average	85	268,505	395	35					326,334	479	39
2014 Average	75	259,864	418	39					288,417	461	37
2013 Average	98	246,641	314	35					250,482	317	25
2012 Average	81	263,148	392	31					240,627	355	21
2011 Average	102	253,235	333	38					225,537	297	28
2010 Average	89	250,252	361	38					223,553	314	23
2009 Average	89	232,766	328	28					251,045	344	20

Table 3
2019 Calendar Year Report
Total Influent Flow, Average Daily Dryweather Flow, and Average Daily Billed Flow at Both Treatment Plants

Month	ADF (mgd)	ADDWF (mgd)	Average Daily Billed Flow (mgd)	MLGW Production Wells
Jan-17	148	138	111	111
Feb-17	133	128	120	107
Mar-17	140	134	108	106
Apr-17	140	129	115	112
May-17	206	135	111	125
Jun-17	148	132	125	105
Jul-17	133	128	130	110
Aug-17	138	127	133	106
Sep-17	137	130	135	105
Oct-17	126	120	121	100
Nov-17	125	118	120	87
Dec-17	141	123	112	88
Average 17	143	129	120	105
Jan-18	145	134	120	134
Feb-18	187	155	127	122
Mar-18	197	185	107	111
Apr-18	171	151	113	110
May-18	151	146	108	130
Jun-18	143	139	126	147
Jul-18	140	136	153	135
Aug-18	141	136	127	147
Sep-18	149	143	130	146
Oct-18	146	141	120	133
Nov-18	150	141	118	119
Dec-18	160	145	114	114
Average 18	156	146	122	129
Jan-19	192	145	113	119
Feb-19	222	145	120	118
Mar-19	278	145	107	116
Apr-19	234	145	114	116
May-19	230	145	115	134
Jun-19	229	144	125	138
Jul-19	194	140	129	139
Aug-19	134	132	125	143
Sep-19	130	130	137	154
Oct-19	152	134	130	135
Nov-19	143	136	122	125
Dec-19	147	135	111	120
Average 19	190	140	121	130

Notes: ADF = Average Daily Influent Flow as directly measured at the treatment plants.
 ADDWF = Average Daily Dryweather Flow as indirectly determined by I/I analysis
 Average Daily Billed Flow based on City Sewer Billing monthly records and includes
 residential/commercial accounts, wholesale customers, and large industrial users.
 MLGW would not include private wells and other utilities.

Table 4
2019 Calendar Year Report
Rain and River-Induced Flow Annual Data Summary
Maxson Facility
Page 1 of 1

Year	RIF/inch of rain	5-year moving average	% Rain- induced Flow	ADF (mgd)	ADDWF (mgd)	Annual Rainfall (in)	% River- induced Flow
1975	4		4%	22	21	74	-1%
1976	10		3%	43	41	51	-1%
1977	26		5%	48	46	36	0%
1978	18		5%	52	48	58	2%
1979	24	16	8%	57	48	69	8%
1980	23	20	7%	51	50	54	-5%
1981	23	23	5%	55	53	42	-1%
1982	18	21	6%	58	54	71	1%
1983	23	22	6%	59	55	58	0%
1984	19	21	5%	58	56	55	0%
1985	22	21	5%	60	57	47	0%
1986	14	19	3%	61	59	41	0%
1987	13	18	3%	60	58	56	0%
1988	17	17	4%	60	58	52	0%
1989	22	18	5%	63	59	58	1%
1990	20	17	5%	65	61	60	1%
1991	21	18	5%	70	66	59	1%
1992	14	19	3%	67	64	48	2%
1993	14	18	3%	69	66	44	2%
1994	13	16	2%	71	68	50	2%
1995	11	15	2%	70	67	50	2%
1996	10	12	1%	77	74	29	3%
1997	21	14	5%	79	72	72	4%
1998	28	17	5%	86	73	53	11%
1999	23	19	4%	76	73	45	0%
2000	24	21	4%	73	70	43	0%
2001	23	24	6%	74	69	68	1%
2002	27	25	8%	73	68	75	0%
2003	23	24	5%	67	63	52	2%
2004	37	27	7%	78	70	55	2%
2005	36	29	5%	76	73	40	-1%
2006	24	29	4%	71	68	48	0%
2007	18	28	3%	68	66	38	0%
2008	24	28	6%	72	67	63	1%
2009	38	28	8%	71	65	56	1%
2010	25	26	4%	75	69	49	4%
2011	25	26	5%	75	67	59	6%
2012	27	28	4%	68	66	34	0%
2013	25	28	5%	70	65	55	2%
2014	33	27	7%	71	66	57	0%
2015	28	27	6%	72	64	52	5%
2016	26	28	5%	77	64	59	12%
2017	27	28	5%	67	64	50	-1%
2018	29	29	7%	78	70	64	3%
2019	30	28	8%	78	66	74	8%
Average	23		5%	70	62	53	2%

Table 5
2019 Calendar Year Report
Rain and River-Induced Flow Annual Data Summary
Stiles Facility
Page 1 of 1

Year	RIF/inch of rain	5 Year Moving Average	% Rain- induced Flow	ADF (mgd)	ADDWF (mgd)	Rainfall (inches)	% River- Induced Flow
1978	17		4%	70	61	59	9%
1979	22		5%	75	75	68	-5%
1980	41		6%	107	79	56	20%
1981	27		3%	79	77	38	-1%
1982	15		3%	97	79	67	16%
1983	19	25	3%	119	79	58	31%
1984	21	25	3%	105	80	55	20%
1985	27	22	4%	78	76	44	-1%
1986	21	21	3%	78	76	40	0%
1987	23	22	4%	84	80	54	1%
1988	17	22	3%	86	84	58	-1%
1989	22	22	4%	83	81	58	-2%
1990	24	21	4%	95	83	57	8%
1991	31	23	5%	105	91	59	8%
1992	26	24	4%	94	90	50	0%
1993	23	25	3%	102	89	51	9%
1994	25	26	4%	109	88	58	15%
1995	27	27	4%	98	90	56	4%
1996	28	26	5%	98	88	66	6%
1997	23	25	4%	104	88	61	12%
1998	29	26	4%	99	87	48	8%
1999	23	26	3%	88	78	46	7%
2000	18	24	3%	80	78	43	1%
2001	31	25	7%	87	80	68	1%
2002	30	26	6%	98	78	74	14%
2003	28	26	5%	83	74	50	6%
2004	34	28	6%	84	79	57	0%
2005	21	29	3%	88	80	46	5%
2006	24	27	4%	77	73	47	1%
2007	24	26	3%	75	73	38	0%
2008	25	26	5%	96	74	62	19%
2009	36	26	7%	89	76	66	7%
2010	19	26	3%	89	73	46	15%
2011	18	25	3%	102	73	62	26%
2012	25	25	3%	81	75	39	4%
2013	31	26	5%	98	71	59	22%
2014	27	24	6%	75	70	60	1%
2015	24	25	5%	85	70	61	13%
2016	29	27	5%	79	68	50	8%
2017	36	29	6%	76	64	43	9%
2018	26	28	6%	78	69	63	6%
2019	36	30	6%	113	70	72	32%
Average	26		4%	90	78	55	8%

Table 6
2019 Calendar Year Report
T.E. Maxson Facility

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Avg. Daily Flow (mgd)	75	75	68	70	71	69	77	68	78	78
Eff. BOD (mg/l)	20	31	39	36	34	30	40	32	40	40
Eff. TSS (mg/l)	25	42	49	44	42	35	42	31	47	42
Permit Violations	0	35	74	24	24	4	117	3	53	56
Inf. BOD (lbs/day)	404,628	389,421	450,168	508,695	540,103	531,426	442,780	450,747	535,000	502,760
Inf. TSS (lbs/day)	410,627	322,044	324,884	360,547	364,476	528,158	372,265	427,315	526,650	412,950
Sludge wasted to Lagoon (million pounds) (secondary)	84.8	70.0	55.2	45.6	34.8	36.6	25.5	22.2	24.8	30.4
Sludge wasted to lagoons (million lbs-primary)	67.7	74.5	73.6	79.2	87.4	136.9	87.2	101	113.5	91.7
Total	152.5	144.5	128.8	124.8	122.2	173.5	112.7	123.2	138.3	122.1
Primary Sludge Dewatered (million lbs.)	0	0	0	0	0	0	0	0	0	0
Lagoon Sludge Land Applied (millions lbs.)	56.3	41.67	24.1	31.2	23.4	27.2	29.95	24.7	23	15.8
KW Hrs. (millions)	55.22	59.2	66.73	71	67.8	68.7	63.7	67.2	67.8	70.8
Avg. Peak KW Demand	7,101	7,208	8,683	9,183	9,593	9,175	8,524	8,633	10329	9,091
Electrical Utility Cost (millions \$)	3.9	4.5	5.10	5.28	5.7	5.3	4.7	4.83	4.8	5.097
Water Cost (\$)	450,790	559,074	634,613	635,667	682,890	845,385	1,018,755	1,065,475	1,010,064	747,748
Sludge Disposal Cost (million \$)	2.809	3.221	2.974	2.969	3.061	2.915	3.004	2.745	3.047	4.34

Table 7
2019 Calendar Year Report
M.C. Stiles Facility

	<u>2010</u>	<u>2011</u>	<u>2012</u>	<u>2013</u>	<u>2014</u>	<u>2015</u>	<u>2016</u>	<u>2017</u>	<u>2018</u>	2019
Avg. Daily Flow (mgd)	89	102	81	98	75	85	79	75	79	113
Eff. BOD (mg/l)	38	38	31	35	39	35	32	32	32	59
Eff. TSS (mg/l)	23	28	21	25	37	39	32	33	29	40
Permit Violations	8	26	5	12	14	56	46	5	12	181
Inf. BOD (lbs./day)	250,252	253,235	263,148	246,641	259,864	268,505	244,332	247,351	268,114	265,315
Inf. TSS (lbs./day)	223,553	225,537	240,627	250,482	288,417	326,334	289,269	304,940	294,417	295,671
Sludge Wasted (million lbs.)	88	76	86	76	87	91	101	115.5	115.1	101.4
Sludge Dewatered (million lbs.)	31	16	33	26	26	26	40.5	28.1	38.6	39
KW Hrs. (millions)	55.1	55.6	55.4	55.2	57.3	54.7	60.3	60.8	65.6	66.902
Avg. Peak KW Demand	7,548	7,756	7,563	7,333	7,623	7,633	7,869	7871	8475	8581
Electrical Utility Cost (million \$)	3.972	4.349	4.295	4.151	4.452	4.584	4.479	4.335	4.7	4.61
Water Cost (\$)	103,188	91,650	112,897	19,401	18,089	21,928	75,891	77,519	353,480	341,066
Sludge Disposal Cost (millions\$)	2.242	2.158	2.67	2.5	2.541	3.103	3.408	3.198	3.267	3.282

Table 8
2019 Calendar Year Report
Monthly Operating Expenditures

	T.E. Maxson Facility	M.C. Stiles Facility	Total (both plants)	Gallons Treated (000)	Cost per 1000 gallons (\$)
January	\$ 723,487	\$ 1,821,096	\$ 2,544,583	5,936,500	\$0.43
February	\$ 1,645,652	\$ 2,056,205	\$ 3,701,857	6,204,800	\$0.60
March	\$ 925,851	\$ 2,737,963	\$ 3,663,814	8,602,500	\$0.43
April	\$ 766,743	\$ 2,836,078	\$ 3,602,821	7,014,000	\$0.51
May	\$ 1,565,734	\$ 3,297,507	\$ 4,863,241	7,123,800	\$0.68
June	\$ 1,926,264	\$ 6,787,543	\$ 8,713,807	6,876,000	\$1.27
July	\$ 221,529	\$ 698,254	\$ 919,782	6,007,800	\$0.15
August	\$ 1,098,576	\$ 3,414,891	\$ 4,513,467	4,154,000	\$1.09
September	\$ 1,958,750	\$ 2,278,956	\$ 4,237,706	3,909,000	\$1.08
October	\$ 908,065	\$ 1,358,715	\$ 2,266,781	4,715,100	\$0.48
November	\$ 1,696,868	\$ 2,567,655	\$ 4,264,523	4,296,000	\$0.99
December	\$ 830,814	\$ 2,835,843	\$ 3,666,657	4,557,000	\$0.80
2019 Total	\$ 14,268,334	\$ 32,690,706	\$ 46,959,040	69,465,583	\$0.68
2018 Total	\$ 15,679,711	\$ 16,037,020	\$ 31,716,731	57,031,250	\$0.56
2017 Total	\$ 14,390,204	\$ 12,804,033	\$ 27,194,237	52,185,875	\$0.52
2016 Total	\$ 14,857,582	\$ 14,091,216	\$ 28,948,798	56,909,583	\$0.51
2015 Total	\$ 15,185,101	\$ 14,720,084	\$ 29,905,185	57,036,254	\$0.52
2014 Total	\$ 15,323,120	\$ 13,558,501	\$ 28,881,620	53,077,083	\$0.54
2013 Total	\$ 17,682,760	\$ 12,005,497	\$ 29,688,257	61,307,833	\$0.48
2012 Total	\$ 11,864,318	\$ 12,690,471	\$ 24,554,789	54,409,333	\$0.45
2011 Total	\$ 14,516,778	\$ 12,957,042	\$ 27,473,819	64,632,375	\$0.43
2010 Total	\$ 11,478,192	\$ 12,116,869	\$ 23,595,061	59,957,333	\$0.39

Table 9
2019 Calendar Year Report
Treatment Facility Expenditures by Major Category

								Total Plant	Total	Total Plant
								Capital	Expenditures	Generated
Maxson	Overtime	Part Time	Personnel	Chemicals	Supplies	Pro. Svcs	Utilities			Revenue
FY11	\$413,023	\$175,309	\$4,293,611	\$492,971	\$732,421	\$458,720	\$5,298,945	\$793,376	\$13,605,672	\$16,078
FY12	\$256,745	\$166,453	\$3,981,808	\$391,361	\$664,295	\$630,646	\$5,749,536	\$812,928	\$13,892,818	\$49,007
FY13	\$240,127	\$157,989	\$4,034,189	\$181,358	\$764,939	\$827,176	\$6,153,680	\$1,761,286	\$16,620,295	\$34,331
FY14	\$598,376	\$114,305	\$3,461,901	\$426,797	\$714,223	\$678,573	\$5,748,753	\$2,032,762	\$15,264,457	\$379,215
FY15	\$495,697	\$87,882	\$3,920,331	\$294,424	\$975,954	\$158,058	\$5,827,819	\$1,395,695	\$14,997,000	\$185,640
FY16	\$563,384	\$33,006	\$4,394,037	\$439,828	\$1,658,892	\$77,644	\$5,360,999	\$0	\$14,637,391	\$57,241
FY17	\$470,518	\$15,624	\$4,288,252	\$225,676	\$780,606	\$32,680	\$6,886,790	\$1,508,299	\$14,788,376	\$37,251
FY18	\$702,636	\$28,556	\$4,273,772	\$72,023	\$839,597	\$124,294	\$5,885,832	\$339,678	\$14,858,659	\$81,354
FY19	\$1,106,994	\$2,016	\$4,466,218	\$248,140	\$1,047,728	\$378,513	\$4,550,749	\$1,577,445	\$15,102,714	\$56,843
FY20	\$1,420,436	\$2,544	\$4,314,455	\$411,298	\$1,009,581	\$197,800	\$3,738,899	\$2,628,777	\$14,775,265	\$66,828
Stiles	Overtime	Part Time	Personnel	Chemicals	Supplies	Pro. Svcs	Utilities	Total Plant	Total	Total Plant
Facility								Capital	Expenditures	Generated
FY11	\$305,568	\$163,767	\$4,957,254	\$551,139	\$621,089	\$169,108	\$3,810,994	\$858,102	\$11,836,389	\$1,508,335
FY12	\$246,825	\$219,126	\$4,646,026	\$805,648	\$482,264	\$652,080	\$5,885,923	\$819,571	\$14,413,717	\$1,254,657
FY13	\$295,294	\$292,356	\$4,431,072	\$788,167	\$578,878	\$1,029,366	\$4,401,258	\$1,223,929	\$13,085,886	\$1,341,042
FY14	\$246,192	\$259,244	\$3,975,514	\$422,749	\$575,288	\$696,395	\$3,877,525	\$503,400	\$11,848,841	\$1,239,253
FY15	\$359,009	\$166,993	\$3,932,825	\$437,418	\$888,940	\$1,567,527	\$4,656,836	\$1,360,057	\$14,992,000	\$1,624,432
FY16	\$602,935	\$345,687	\$4,224,469	\$532,847	\$726,353	\$1,006,784	\$4,259,085	\$39,291	\$13,531,589	\$1,445,196
FY17	\$328,998	\$149,310	\$4,043,883	\$554,603	\$704,138	\$1,003,681	\$5,020,073	\$639,248	\$13,554,099	\$159,582
FY18	\$462,865	\$101,010	\$4,256,607	\$447,266	\$986,159	\$915,253	\$4,914,027	\$391,707	\$14,107,167	\$123,762
FY19	\$570,595	\$48,642	\$4,560,915	\$9,604,092	\$1,071,798	\$1,729,812	\$5,134,789	\$3,376,563	\$27,200,877	\$136,099
FY20	\$606,892	\$35,684	\$4,247,763	\$12,141,974	\$973,548	\$836,289	\$5,266,303	\$1,270,589	\$27,165,192	\$126,133

Table 10
2019 Calendar Year Report
T.E. Maxson Nine Largest Industrial Sewer Fees

	Blues City Brewery	Cargill Corn	Con Agra	Fleischmann's	Kellogg's	Riviana Foods	Solae	Valley Products	Valero	TOTAL
January	\$120,083	\$32,194	\$6,861	\$59,001	\$80,516	\$143,538	\$648,877	\$23,271	\$214,859	\$ 1,329,200
February	\$119,938	\$31,047	\$3,638	\$52,310	\$68,461	\$139,458	\$593,568	\$9,645	\$214,098	\$ 1,232,163
March	\$130,808	\$31,127	\$45,250	\$59,715	\$75,650	\$131,028	\$833,493	\$10,518	\$228,462	\$ 1,546,051
April	\$120,668	\$23,579	\$41,539	\$59,152	\$67,510	\$104,781	\$584,557	\$5,090	\$220,911	\$ 1,227,788
May	\$115,385	\$19,709	\$23,336	\$67,408	\$87,272	\$110,492	\$716,522	\$7,089	\$174,723	\$ 1,321,935
June	\$123,527	\$22,329	\$30,754	\$68,167	\$78,065	\$114,312	\$717,821	\$6,323	\$158,556	\$ 1,319,856
July	\$118,587	\$28,618	\$67,280	\$73,257	\$84,884	\$90,092	\$770,938	\$5,134	\$195,152	\$ 1,433,942
August	\$113,705	\$25,953	\$36,655	\$77,758	\$88,132	\$133,848	\$849,040	\$5,762	\$194,535	\$ 1,525,388
September	\$112,799	\$22,689	\$32,962	\$74,795	\$82,005	\$95,370	\$782,899	\$7,664	\$188,723	\$ 1,399,907
October	\$121,706	\$39,234	\$39,137	\$79,475	\$77,427	\$118,616	\$810,469	\$15,919	\$224,061	\$ 1,526,044
November	\$124,178	\$19,807	\$35,451	\$74,597	\$74,832	\$140,294	\$789,092	\$45,399	\$199,362	\$ 1,503,011
December	\$106,609	\$24,298	\$33,643	\$74,943	\$76,370	\$122,075	\$771,053	\$67,717	\$206,383	\$ 1,483,088
2019 Total	\$1,427,992	\$320,585	\$396,507	\$820,578	\$941,124	\$1,443,904	\$8,868,328	\$209,530	\$2,419,825	\$ 16,848,373
2018 Total	\$1,243,058	\$376,447	\$603,732	\$803,249	\$914,020	\$1,215,861	\$9,917,474	\$121,924	\$2,296,962	\$ 17,492,727
2017 Total	\$1,110,597	\$201,739	\$507,879	\$793,988	\$700,543	\$990,482	\$8,331,347	\$68,616	\$1,575,376	\$ 14,280,567
2016 Total	\$627,115	\$239,396	\$503,276	\$1,096,210	\$653,107	\$1,044,440	\$8,292,176	\$60,521	\$1,567,558	\$ 14,083,799
2015 Total	\$289,259	\$1,456,615	\$758,724	\$1,213,428	\$752,548	\$913,269	\$6,770,531	\$66,880	\$1,477,919	\$ 13,699,172
2014 Total	\$213,457	\$5,551,496	\$885,377	\$958,306	\$663,008	\$810,965	\$7,476,364	\$69,236	\$1,721,756	\$ 18,349,965
2013 Total	\$197,996	\$5,614,787	\$714,170	\$1,046,400	\$555,219	\$949,354	\$6,661,731	\$77,971	\$1,829,815	\$ 17,647,444
2012 Total	\$202,264	\$5,352,464	\$855,509	\$867,952	\$1,108,860	\$747,278	\$6,475,093	\$101,071	\$1,199,057	\$ 16,909,548
2011 Total	\$48,598	\$4,801,320	\$677,725	\$759,159	\$974,767	\$809,560	\$6,194,325	\$108,178	\$1,355,157	\$ 15,728,789
2010 Total	\$22,100	\$3,574,202	\$564,493	\$671,727	\$440,166	\$293,429	\$5,530,540	\$52,653	\$1,017,918	\$ 11,443,902

Table 11
2019 Calendar Year Report
M.C. Stiles Facility Ten Largest Industrial Customer's Sewer Fees Plus the Hauled Waste Fees

	American Yeast	Memphis Cellulose	Cascade Tissue	Dupont	KTG Corp.	Penn Specialty Chemicals/ PENN A KEM	PMC BIOGENIX (formerly Chemtura)	Turner Holding Co./Forrest Dairy	Runyon Industries	Velsicol	Hauled Waste	TOTAL
January	\$155,486	\$441,917	\$46,798	\$447,304	\$241,844	\$37,331	\$186,893	\$13,359	\$32,828	\$7,001	\$51,557	\$1,662,318
February	\$142,252	\$363,981	\$43,534	\$383,310	\$213,577	\$48,979	\$175,204	\$14,802	\$26,534	\$8,051	\$31,245	\$1,451,469
March	\$145,220	\$436,949	\$51,380	\$333,949	\$197,121	\$46,622	\$203,510	\$14,476	\$32,888	\$5,535	\$20,955	\$1,488,605
April	\$137,804	\$415,346	\$53,022	\$326,992	\$218,218	\$49,203	\$154,072	\$14,466	\$40,256	\$6,518	\$19,090	\$1,434,986
May	\$136,146	\$409,107	\$53,409	\$422,933	\$229,596	\$51,132	\$159,133	\$11,918	\$55,765	\$7,978	\$18,505	\$1,555,623
June	\$154,161	\$358,597	\$50,002	\$397,953	\$218,399	\$51,521	\$129,312	\$11,800	\$44,437	\$8,277	\$26,239	\$1,450,697
July	\$151,617	\$329,757	\$49,425	\$386,728	\$218,077	\$49,511	\$60,299	\$42,854	\$16,587	\$8,738	\$10,153	\$1,323,746
August	\$167,964	\$305,154	\$48,753	\$414,792	\$241,860	\$52,713	\$151,040	\$28,911	\$12,876	\$8,401	\$27,676	\$1,460,139
September	\$122,366	\$331,435	\$44,751	\$386,366	\$247,126	\$40,142	\$151,203	\$21,169	\$6,886	\$7,583	\$40,153	\$1,399,179
October	\$140,357	\$293,572	\$43,005	\$391,609	\$210,976	\$43,326	\$167,640	\$34,424	\$15,209	\$9,277	\$43,393	\$1,392,786
November	\$130,700	\$321,225	\$38,930	\$411,702	\$248,121	\$51,197	\$125,230	\$18,702	\$12,288	\$6,563	\$37,668	\$1,402,328
December	\$128,691	\$244,372	\$43,125	\$432,346	\$293,313	\$50,364	\$182,908	\$21,714	\$17,277	\$5,625	\$38,588	\$1,458,322
2019 Total	\$1,712,764	\$4,251,410	\$566,134	\$4,735,984	\$2,778,225	\$572,041	\$1,846,444	\$248,594	\$313,831	\$89,549	\$365,222	\$ 17,480,198
2018 Total	\$1,687,432	\$4,937,006	\$633,473	\$4,957,178	\$2,931,079	\$415,515	\$2,058,936	\$151,018	\$454,289	\$63,966	\$626,357	\$ 18,916,248
2017 Total	\$1,430,416	\$3,524,571	\$461,503	\$4,087,697	\$2,443,475	\$358,765	\$1,471,641	\$88,162	\$488,079	\$56,657	\$604,553	\$ 15,015,519
2016 Total	\$1,458,928	\$3,762,434	\$433,382	\$3,569,602	\$3,236,602	\$391,818	\$1,623,679	\$53,253	\$432,827	\$58,778	\$651,251	\$ 15,672,554
2015 Total	\$1,406,898	\$3,830,828	\$448,421	\$3,552,664	\$3,451,372	\$411,315	\$1,471,952	\$54,599	\$540,672	\$51,263	\$1,111,607	\$ 16,331,590
2014 Total	\$1,419,425	\$3,623,823	\$461,792	\$3,852,549	\$3,323,278	\$381,409	\$1,608,485	\$80,844	\$489,817	\$50,512	\$1,211,507	\$ 16,503,442
2013 Total	\$2,565,426	\$3,345,514	\$472,309	\$3,870,106	\$3,737,672	\$328,491	\$1,562,415	\$83,478	\$318,344	\$61,458	\$831,482	\$ 17,176,694
2012 Total	\$1,718,561	\$2,923,832	\$580,438	\$3,403,949	\$2,375,317	\$311,960	\$1,406,544	\$87,036	\$223,920	\$84,797	\$1,139,951	\$ 14,256,307
2011 Total	\$1,243,105	\$2,730,099	\$511,634	\$2,384,798	\$2,070,288	\$260,276	\$1,321,708	\$92,144	\$153,366	\$220,216	\$1,336,280	\$ 12,323,914
2010 Total	\$1,194,987	\$2,000,830	\$400,292	\$1,878,144	\$1,613,101	\$171,137	\$1,303,698	\$104,868	\$160,623	\$222,638	\$1,073,240	\$ 10,123,559

Table 12
2019 Calendar Year Report
Sewer Fees for Other Municipal Customers

	<u>Germantown</u>	<u>Bartlett</u>	<u>Horn Lake</u>	<u>Totals</u>
January	\$96,028	\$73,731	\$106,832	\$276,591
February	\$80,629	\$74,909	\$131,748	\$287,285
March	\$88,687	\$65,964	\$109,105	\$263,756
April	\$95,314	\$71,918	\$140,976	\$308,208
May	\$105,721	\$75,079	\$136,459	\$317,259
June	\$128,984	\$91,323	\$122,861	\$343,168
July	\$130,526	\$88,432	\$146,141	\$365,100
August	\$154,355	\$88,563	\$135,907	\$378,824
September	\$141,926	\$104,039	\$117,943	\$363,908
October	\$166,724	\$97,151	\$85,552	\$349,427
November	\$139,546	\$84,197	\$90,955	\$314,697
December	\$87,243	\$66,542	\$91,650	\$245,435
Total of 2019	\$1,415,683	\$981,848	\$1,416,128	\$3,813,659
Total of 2018	\$1,455,355	\$1,026,024	\$1,252,848	\$3,734,227
Total of 2017	\$1,452,582	\$999,012	\$1,173,778	\$3,625,372
Total of 2016	\$1,319,969	\$894,179	\$1,269,080	\$3,483,228
Total of 2015	\$1,299,562	\$893,284	\$1,308,650	\$3,501,495
Total of 2014	\$1,382,360	\$879,295	\$1,232,793	\$3,494,447
Total of 2013	\$1,229,471	\$905,908	\$1,067,626	\$3,203,005
Total for 2012	\$1,167,104	\$833,714	\$1,050,337	\$3,051,155
Total for 2011	\$1,133,659	\$954,622	\$910,979	\$2,999,260
Total for 2010	\$1,012,253	\$795,413	\$1,172,324	\$2,979,990
Total for 2009	\$894,010	\$709,631	\$1,117,055	\$2,720,696
Total for 2008	\$764,380	\$671,256	\$1,065,788	\$2,501,425
Total for 2007	\$961,845	\$743,008	\$954,325	\$2,659,177
Total for 2006	\$1,023,640	\$750,598	\$889,511	\$2,663,749
Total for 2005	\$1,008,405	\$820,178	\$868,607	\$2,697,190
Total for 2004	\$710,303	\$567,394	\$885,396	\$2,163,093
Total for 2003	\$684,344	\$563,144	\$915,688	\$2,163,176
Total for 2002	\$652,688	\$571,784	\$739,082	\$1,963,555
Total for 2001	\$682,639	\$577,566	\$651,397	\$1,911,601
Total for 2000	\$785,287	\$611,900	\$693,492	\$2,090,678
Total for 1999	\$730,413	\$592,640	\$695,907	\$2,018,960

Table 13
2019 Calendar Year Report
Annual Hauled Waste Volume and Revenue

Year	Rate/gallon	Volume	Revenue
2019	13 cents	2,416,765	\$365,222
2018	13 cents	4,818,131	\$626,357
2017	13 cents	4,649,995	\$604,553
2016	13 cents	5,009,616	\$651,160
2015	13 cents	8,550,823	\$1,111,607
2014	13 cents	9,319,286	\$1,211,507
2013	13 cents	6,007,191	\$831,482
2012	13 cents	9,804,557	\$1,139,951
2011	11 cents	14,025,493	\$1,336,278
2010	8 cents	15,017,219	\$1,046,637
2009	6 cents	14,481,480	\$868,889
2008	6 cents	14,431,159	\$865,870
2007	6 cents	8,750,070	\$525,004
2006	6 cents	8,011,537	\$480,692
2005	6 cents	7,136,268	\$428,147
2004	6 cents	8,044,020	\$420,593
2003	5 cents	8,229,569	\$411,478
2002	5 cents	6,877,473	\$343,874

Table 14
2019 Calendar Year Report
Monthly Peak Demand, Main Power and Other Utility Consumption

Monthly Peak Demand, Main Power and Other Utility Consumption														
	Main Power Cost Factors				T.E. Maxson Facility				M.C. Stiles Facility					TOTAL COST
	Demand Charge (\$/kw)	Energy Cost Rate (\$/kwh)	Fuel Cost Adjustment (\$/kwh)	Peak Demand (kw)	Main Power Usage (kwh)	Main Power Cost	Other Utilities Total Cost	Total Utility Cost	Peak Demand (kw)	Main Power Usage (kwh)	Main Power Cost	Other Utilities Total Cost	Total Utility Cost	
Jan-18	\$15.20	0.027	0.01798	8807	5,680,193	\$407,784	\$192,132	\$599,916	8814	5,125,716	\$394,042	\$15,397	\$409,439	\$1,009,355
Feb-18	\$15.15	\$0.0286	\$0.01847	8549	5,255,950	\$391,375	\$81,206	\$472,581	8561	5,181,586	\$390,687	\$14,140	\$404,827	\$877,408
Mar-18	\$15.20	\$0.0272	\$0.01849	8680	5,980,300	\$420,172		\$420,172	8483	4,844,898	\$378,631	\$27,530	\$406,161	\$826,333
Apr-18	\$15.14	\$0.0248	\$0.01785	8743	5,636,900	\$387,648	\$107,059	\$494,708	8625	5,754,096	\$387,516	\$56,316	\$443,832	\$938,540
May-18	\$15.15	\$0.0239	\$0.0182	8144	5,688,046	\$378,245	\$260,572	\$638,817	8168	5,490,693	\$374,057	\$62,857	\$436,914	\$1,075,732
Jun-18	\$16.14	\$0.0288	\$0.01754	8042	5,396,978	\$394,202	\$157,435	\$551,638	8252	5,222,567	\$389,365	\$65,215	\$454,580	\$1,006,218
Jul-18	\$16.42	\$0.0285	\$0.01760	8402	5,682,580	\$411,169		\$411,169	8296	5,478,154	\$406,487	\$57,930	\$464,417	\$875,586
Aug-18	\$16.55	\$0.0268	\$0.01785	8063	5,624,796	\$415,047		\$415,047	8439	5,533,281	\$415,094	\$56,641	\$471,735	\$886,782
Sep-18	\$15.99	\$0.0286	\$0.01572	10329	5,557,502	\$410,825	\$152,035	\$562,860	8300	5,387,892	\$389,848	\$56,838	\$446,686	\$1,009,546
Oct-18	\$15.47	\$0.0206	\$0.01695	8580	5,789,618	\$366,112	\$65,927	\$432,039	8628	5,880,610	\$375,575	\$43,750	\$419,325	\$851,364
Nov-18	\$15.92	\$0.0212	\$0.02001	9048	5,639,391	\$389,190		\$389,190	8583	5,753,735	\$384,797	\$36,238	\$421,035	\$810,225
Dec-18	\$15.85	\$0.0237	\$0.02145	9344	5,903,511	\$428,434	\$148,527	\$576,961	8551	5,972,104	\$414,471	\$29,226	\$443,697	\$1,020,658
Total Monthly Average					67,835,765	\$4,800,205	\$1,164,893	\$5,965,098		65,625,332	\$4,700,570	\$522,078	\$5,222,648	\$11,187,746
	\$15.68	\$0.0258	\$0.0182	8728	5,652,980	\$400,017	\$97,074	\$497,092	8475	5,468,778	\$391,714	\$43,507	\$435,221	\$932,312
Jan-19	\$15.86	0.02507	\$0.02200	9458	6,129,896	\$450,398		\$450,398	8766	6,038,418	\$433,437		\$433,437	\$883,835
Feb-19	\$15.95	\$0.0266	\$0.01890	9373	5,603,443	\$424,464	\$63,874	\$488,338	8724	5,135,170	\$385,760	\$31,688	\$417,448	\$905,786
Mar-19	\$15.62	\$0.0250	\$0.01771	10027	6,190,366	\$443,362	\$92,117	\$535,479	6981	4,949,650	\$329,560	\$32,939	\$362,499	\$897,978
Apr-19	\$15.92	\$0.0224	\$0.01715	9123	5,868,176	\$392,553	\$72,802	\$465,356	8836	5,149,749	\$362,657	\$60,356	\$423,013	\$888,368
May-19	\$15.96	\$0.0221	\$0.0166	9740	6,418,055	\$418,852	\$155,923	\$574,775	7765	4,918,711	\$325,340	\$72,534	\$397,874	\$972,648
Jun-19	\$16.88	\$0.0262	\$0.01656	9540	6,177,763	\$438,071	\$70,045	\$508,116	8921	5,498,320	\$401,431	\$42,869	\$444,300	\$952,416
Jul-19	\$16.92	\$0.0265	\$0.01654	9100	6,113,687	\$430,540		\$430,540	8528	5,788,080	\$404,540		\$404,540	\$835,080
Aug-19	\$16.95	\$0.0266	\$0.01550	8474	5,745,177	\$400,375	\$88,582	\$488,958	9072	6,122,232	\$422,531	\$51,046	\$473,577	\$962,535
Sep-19	\$16.96	\$0.0266	\$0.01426	8311	5,688,298	\$424,827	\$193,606	\$618,433	8663	5,861,996	\$392,384	\$94,187	\$486,571	\$1,105,004
Oct-19	\$15.96	\$0.0258	\$0.01531	8843	5,983,992	\$424,827	\$100,309	\$525,136	9072	5,958,950	\$380,737	\$54,297	\$435,034	\$960,169
Nov-19	\$15.96	\$0.0225	\$0.01748	8829	5,653,375	\$424,827		\$424,827	9007	5,678,493	\$377,388		\$377,388	\$802,215
Dec-19	\$15.96	\$0.0218	\$0.01777	8275	5,228,366	\$424,827	\$78,306	\$503,133	8632	5,802,612	\$395,057	\$47,109	\$442,166	\$945,299
Total Monthly Average					70,800,594	\$5,097,924	\$915,563	\$6,013,487		66,902,381	\$4,610,822	\$487,025	\$5,097,847	\$11,111,334
	\$16.24	\$0.0248	\$0.0171	9091	5,900,050	\$424,827	\$76,297	\$501,124	8581	5,575,198	\$384,235	\$101,729	\$424,821	\$925,944

Table 15
2019 Calendar Year Report
TSS (ppd) Analysis for Maxson Facility

Year	Maxson Facility				Lagoon Supernatant					
	Influent	7 Largest Industries	% of Influent	Difference	TSS (mg/l)	volume (mgd)	TSS	% of Influent	Domestic & Unexplained	Permit Violations
2019	412,950	151,398	37%	261,552	815	2.2	14,953	4%	246,599	56
2018	534,900	171,316	32%	363,584	1947	3.2	51,692	10%	311,892	53
2017	427,315	151,199	35%	276,116	4,312	2.4	86,309	20%	189,807	3
2016	377,286	145,235	38%	232,051	1,739	2.4	37,407	10%	194,644	117
2015	524,421	118,267	23%	406,154	2222	3.9	72,273	14%	333,881	4
2014	364,476	126,836	35%	237,641	910	2.57	19,500	5%	218,141	24
2013	360,547	111,171	31%	249,376	1700	3.14	44,519	12%	204,857	24
2012	324,884	117,553	36%	207,331	1013	3.93	33,202	10%	174,129	74
2011	330,372	150,552	46%	179,820	1,441	3.41	40,981	12%	138,839	35
2010	402,342	109,120	27%	293,222	2,835	3.78	89,374	22%	203,848	0
2009	299,750	102,468	34%	197,282	2,661	2.66	61,191	20%	136,091	1
2008	391,264	122,648	31%	268,616	2,823	3.59	84,408	22%	184,208	15
2007	340,465	115,852	34%	224,613	2,172	3.63	65,769	19%	158,844	2
2006	488,953	129,316	26%	359,637	4,682	5.07	197,972	40%	161,665	4
2005	405,361	122,088	30%	283,273	3,157	4.23	110,583	27%	172,690	1
2004	385,592	102,290	27%	283,302	1,677	5.36	74,686	19%	208,616	3
2003	296,072	126,200	43%	169,872	1,554	5.7	73,874	25%	95,998	10
2002	352,012	118,653	34%	233,359	2,532	5.35	112,975	32%	120,384	4
2001	394,078	137,062	35%	257,016	4,483	4.1	153,292	39%	103,724	5
2000	434,462	143,254	33%	291,208	5,413	4.0	180,578	42%	110,630	3

Table 16
2019 Calendar Year Report
T.E. Maxson Influent and Waste Sludge Comparison
(Influent TSS in 1000 lbs/day, waste sludge values in million lbs/month, lagoon supernatant in mg/l)

	2015			2016			2017			2018			2019		
	Influent	Waste	Lagoon Super- natant	Influent	Waste	Lagoon Super- natant	Influent	Waste	Lagoon Super- natant	Influent	Waste	Lagoon Super- natant	Influent	Waste	Lagoon Super- natant
January	429	10.7	2682	756	6.6	1513	587	5.2	555	497	12.2	8664	420	10.8	871
February	531	11.3	2083	462	3.7	1847	696	6.6	3221	410	13.1	9772	534	10.7	847
March	448	9.9	3602	419	6.6	739	1001	14.4	5394	773	17.3	8382	397	10.3	652
April	559	11.7	3016	340	8.0	1290	663	9.7	3401	697	16.9	5362	299	8.2	1300
May	546	12.4	2991	400	9.8	1777	788	12.5	4590	538	14.2	791	379	10.5	877
June	387	9.3	1729	466	12.9	2421	812	12.4	3918	400	9.2	1855	370	9.4	643
July	332	7.0	886	410	13.3	949	744	11.0	3804	378	9.2	399	458	9.1	596
August	432	9.2	1295	486	14.9	1785	948	13.8	6215	458	9.4	708	355	10.1	779
September	670	14.9	3014	414	11.1	2662	632	9.5	6058	448	10.5	488	375	11.3	1165
October	815	17.9	1955	397	10.0	2255	658	8.9	5238	459	10.8	703	425	10.7	1039
November	660	13.2	2318	386	11.7	2504	701	8.6	4262	606	15.4	695	395	11.6	373
December	483	6.4	1091	414	4.1	1101	871	10.7	5087	540	13.7	2989	455	9.1	684
Influent and WAS TSS Monthly															
Avg	524	11.2		446	9.4		758	10.3		517	12.7		405	10.2	
Lagoon Supernatant TSS Avg (mg/l)			2,222			1,737			4,312			3,401			819

Table 17
2019 Calendar Year Report
T.E. Maxson Facility BOD Evaluation (lbs/day)

	T. E. Maxson Facility	Blues City Brewery	Cargill Corn	ConAgra	Riviana Foods	Fleisch- mann's Yeast	Kellogg USA	Solae	Valley Products	Valero	Total of of 9 Companies	Difference
2019												
January	489,950	35566	1168	3546	34084	13756	4588	158440	5703	3,690	270,920	219,030
February	630,304	37,366	1,668	3,141	38,130	13,696	3,906	162,352	3,251	7,410	279,646	350,658
March	546,987	33,836	1,558	3,498	30,436	13,197	5,062	188,149	3,910	2,731	261,169	285,818
April	420,146	33,065	2,019	4,699	26,740	13,958	3,887	173,612	3,189	2,948	249,801	170,345
May	480,993	29,739	940	3,208	26,066	14,000	7,409	165,007	3,432	5,293	271,651	209,342
June	513,102	39,107	2,335	9,284	29,521	14,682	8,268	165,992	2,462	4,195	261,099	252,003
July	456,006	28,693	3,820	8,794	19,252	14,198	10,739	173,755	1,848	3,980	275,430	180,576
August	438,025	25,894	1,821	5,689	32,665	15,132	6,082	185,863	2,284	2,510	256,867	181,158
September	474,349	25,247	2,822	7,522	27,716	15,018	6,229	169,959	2,354	2,601	291,453	182,896
October	501,059	32,431	11,551	5,238	25,444	17,631	4,619	186,180	8,359	4,819	304,324	196,735
November	401,121	35,212	1,411	5,671	36,076	15,506	4,532	194,334	11,582	3,641	288,429	112,692
December	547,037	27,821	960	4,562	31,965	16,883	4,955	186,446	14,837	3,707	263,355	283,682
2019 - Avg	491,590	31,674	2,810	5,573	29,456	14,900	5,972	177,423	5,228	3,985	272,845	218,745
2018 - Avg	535,000	22,175	1,425	4,364	24,554	17,307	5,017	188,733	4,594	4,034	265,305	269,695
2017 - Avg	450,747	23,962	1,192	6,408	24,091	17,976	4,761	164,094	3,722	2,779	246,575	204,172
2016 - Avg	456,184	13,909	2,687	3,827	28,062	30,318	3,939	166,128	3,075		251,114	205,070
2015 - Avg	532,083	4,555	26,302	5,892	26,707	34,444	4,784	137,757	6,237		246,679	285,404
2014 - Avg	540,103	3,580	85,138	7,833	23,775	22,755	6,499	168,442	4,485		327,604	217,598
2013 - Avg	508,695	2,768	81,069	8,409	4,318	27,238	25,716	144,648	5,219		296,181	212,515
2012 - Avg	450,168	2,559	78,951	7,814	29,425	17,785	15,219	150,312	5,501		307,566	142,602
2011 - Avg	397,452	NA	78,126	5,152	23,261	14,417	14,177	154,531	5,996		294,477	102,975
2010 - Avg	404,628	8,987	79,262	4,076		15,395	12,229	133,575	6,303		259,827	144,801
2009 - Avg	364,049	0	84,764	2,294		15,491	12,142	114,260	7,773		236,725	127,325
2008 - Avg	452,971	0	73,125	3,266		34,080	16,132	142,214	8,873		277,689	175,282
2007 - Avg	433,161	0	80,406	3,726		31,714	18,530	144,461	7,749		286,587	146,574
2006 - Avg	427,916	7,589	82,678	5,539		21,029	11,009	130,258	7,182		265,283	162,632
2005 - Avg	468,336	16,932	76,243	3,371		33,964	11,009	112,628	4,570		258,717	209,619
2004 - Avg	450,183	17,736	64,840	3,935		44,748	11,009	155,218	8,125		305,610	144,573
2003 - Avg	378,410	14,650	67,100	7,400		37,397	11,009	175,630	11,135		324,321	54,090

Note: *Revised-adjusted T. E. Maxson flow data from September 2005 to March 2006 with corresponding changes in #/day and resulting difference

*Coor Brewing Closed, August 2006, currently run by Blues City Brewery.

Table 18
2019 Calendar Year Report
T.E. Maxson Facility COD Evaluation (pounds per day)

	T. E. Maxson Facility	Blues City Brewery	Cargill	ConAgra	Riviana Foods	Fleisch- mann's Yeast	Kellogg USA	Solae	Valley Products	Valero	Total of 9 Companies	Difference
2019												
January	1,088,870	60,865	1,898	2,194	64,353	31,047	8,332	295,433	10,280	9,709	484,111	604,759
February	1,204,163	70,875	2,635	1,898	74,374	34,973	7,110	331,796	6,075	20,657	550,393	653,770
March	962,119	59,486	2,971	1,543	63,132	35,947	9,467	399,395	7,213	7,724	586,878	375,241
April	846,343	54,428	3,258	3,665	50,320	35,650	7,128	336,155	5,553	6,239	502,396	343,947
May	804,009	54,386	1,579	5,511	50,164	35,652	12,544	346,294	4,808	13,313	524,251	279,758
June	846,393	48,743	3,781	15,679	55,698	39,546	10,429	350,351	3,870	7,459	535,556	310,837
July	882,789	47,908	6,286	16,181	38,043	29,581	14,106	339,961	3,335	8,615	504,016	378,773
August	806,353	43,555	3,263	10,537	60,560	41,023	9,490	363,480	3,716	5,841	541,465	264,888
September	895,716	47,158	5,595	16,861	40,425	41,770	11,144	340,723	3,634	6,723	514,033	381,683
October	1,019,165	53,435	18,652	8,933	51,448	45,370	13,577	351,283	12,579	8,728	564,005	455,160
November	979,383	58,863	2,822	9,676	70,873	47,712	7,772	407,467	17,977	9,777	632,939	346,444
December	1,255,003	47,727	1,569	8,316	63,765	49,255	8,487	392,317	22,508	5,313	599,257	655,746
2019-Avg	965,859	53,952	4,526	8,416	56,930	38,961	9,966	354,555	8,462	9,175	544,942	420,917
2018-Avg	1,025,318	43,351	3,003	6,017	47,717	44,908	8,577	393,011	7,796	12,601	566,481	458,838

Table 19
2019 Calendar Year Report
T.E. Maxson Facility TSS Evaluation (lbs/day)

2019	T. E. Maxson Facility	Blues City Brewery	Cargill	ConAgra	Riviana Foods	Fleisch- mann's Yeast	Kellogg USA	Solae	Valley Products	Valero	Total of 9 Companies	Difference
January	438,100	3,107	132	1,014	24,545	4,782	1,030	86,483	136	1,805	122,898	315,202
February	601,297	3,417	205	1,068	26,488	4,195	1,052	69,590	145	1,314	107,474	330,626
March	421,087	3,678	188	979	21,258	5,485	1,072	135,492	153	1,042	169,347	431,950
April	306,261	2,850	167	1,283	16,535	4,766	1,192	119,115	71	1,859	147,838	273,249
May	342,924	3,702	212	1,158	17,611	4,891	4,247	105,630	101	2,163	139,715	166,546
June	375,675	3,552	236	2,126	23,033	5,631	3,090	101,761	99	1,315	140,843	202,081
July	460,268	2,534	181	2,566	13,611	6,321	2,915	120,769	102	730	149,729	225,946
August	353,574	4,516	238	1,937	21,051	6,766	2,909	161,959	173	735	200,285	259,983
September	373,799	2,577	148	3,566	18,859	5,639	4,395	127,734	109	274	163,301	190,273
October	423,130	3,537	207	2,370	16,934	4,849	1,429	121,271	437	695	151,729	222,070
November	520,016	2,900	174	2,292	23,991	6,346	1,162	118,885	387	1,042	157,179	265,951
December	458,166	2,200	127	1,543	15,903	4,540	803	111,706	513	605	137,940	382,076
2019-Avg	422,858	3,224	189	1,899	19,570	5,403	2,206	117,628	208	1,132	151,398	268,250
2018-Avg	526,650	2,975	411	1,570	17,215	4,770	1,668	140,541	223	1,926	171,316	355,234
2017-Avg	427,315	4,573	259	2,730	12,156	5,202	1,911	123,145	154	1,044	151,199	271,306
2016-Avg	378,486	3,017	465	1,801	13,025	7,936	1,495	115,524	108		143,353	235,132
2015-Avg	524,421	1,213	12,190	2,377	11,016	9,029	2,137	80,141	163		118,267	406,155
2014-Avg	364,476	334	22,323	2,876	10,027	7,930	3,972	79,190	184		126,836	237,641
2013-Avg	360,547	305	24,865	3,324	11,351	8,723	1,379	61,089	159		110,877	249,669
2012-Avg	324,884	914	25,754	4,236	14,692	7,600	5,413	58,756	188		117,553	207,331
2011 Avg	330,550	0	22,775	2,962	12,299	7,848	4,434	100,114	489		150,552	179,998
2010-Avg	402,342	0	23,366	2,367	6,822	7,922	3,109	100,605	580		144,771	257,571
2009-Avg	299,750	0	24,174	1,512		6,410	3,233	66,555	585		102,468	197,282
2008-Avg	391,264	0	18,852	1,934		9,019	4,133	87,913	797		122,648	268,616
2007-Avg	340,465	0	21,105	1,442		8,315	4,589	79,903	499		115,852	224,612
2006-Avg	488,812	2,179	27,036	2,663		7,223	2,962	86,627	625		129,316	359,496
2005-Avg	395,178	5,533	21,067	1,406		9,314	2,962	81,139	668		122,088	273,097
2004-Avg	385,592	5,627	14,604	1,770		8,805	2,962	66,800	1,730		102,298	283,294

Note: *Revised - adjusted T. E. Maxson flow data from September 2005 to March 2006 with corresponding changes in #/day and resulting difference
*Coor Brewing Closed, August 2006; currently Blues City Brewery is operating at this location.

Table 20
2019 Calendar Year Report
T.E. Maxson Facility Water Usage Costs

2019	Secondary Dewatering		Main Plant		Total of Both	
	Bill \$	Usage CCF	Bill \$	Usage CCF	Bill \$	Usage CCF
January	\$489	0	\$41,594	40,416	\$42,082	40,416
February	\$489	0	\$43,200	43,310	\$43,688	43,310
March	\$489	0	\$54,797	53,335	\$55,285	53,335
April	\$489	0	\$52,063	50,660	\$52,552	50,660
May	\$1,781	1,460	\$51,956	50,555	\$53,736	52,015
June	\$12,689	11,880	\$55,710	54,228	\$68,398	66,108
July	\$32,133	30,120	\$49,005	47,668	\$81,138	77,788
August	\$27,442	25,720	\$42,576	41,377	\$70,018	67,097
September	\$28,146	26,380	\$49,700	48,348	\$77,846	74,728
October	\$15,865	14,860	\$43,503	42,284	\$59,368	57,144
November	\$17,283	16,190	\$43,722	42,498	\$61,005	58,688
December	\$38,166	35,780	\$44,465	43,225	\$82,631	79,005
2019 Total	\$175,458	162,390	\$572,289	557,904	\$747,748	720,294
2019 Avg	\$24,532	22,990	\$47,691	46,492	\$62,312	60,025
2018 Avg	\$35,803	34,632	\$48,369	48,526	\$86,735	85,385
2017 Avg	\$36,014	35,792	\$48,363	48,611	\$84,206	79,161
2016 Avg	\$35,147	33,895	\$50,338	51,030	\$84,499	84,896
2015 Avg	\$29,988	35,758	\$39,991	49,458	\$70,449	85,870
2014 Avg	\$18,299	21,810	\$38,620	47,753	\$56,907	70,408
2013 Avg	\$19,306	23,489	\$33,666	42,496	\$52,972	67,365
2012 Avg	\$22,436	29,232	\$30,448	41,144	\$52,884	70,376
2011 Avg	\$22,099	29,046	\$24,491	33,164	\$46,589	62,210
2010 Avg	\$20,084	28,083	\$19,793	27,054	\$39,877	55,137
2009 Avg	\$22,482	31,854	\$18,084	24,731	\$40,565	56,586
2008 Avg	\$21,378	33,073	\$19,973	29,068	\$41,351	62,141
2007 Avg	\$14,177	22,408	\$28,803	46,331	\$42,980	68,739
2006 Avg	\$8,761	13,715	\$23,812	38,324	\$32,573	52,040
2005 Avg	\$9,668	15,244	\$24,820	38,751	\$34,487	53,995
2004 Avg	\$10,467	16,500	\$24,450	39,319	\$34,917	55,819
2003 Avg	\$8,863	17,630	\$22,822	46,694	\$31,686	64,324
2002 Avg	\$9,188	18,501	\$25,250	51,914	\$34,438	70,415
2001 Avg	\$14,296	28,719	\$31,016	63,591	\$45,312	92,310
2000 Avg	\$10,399	20,889	\$24,042	49,291	\$34,441	70,180
1999 Avg	\$11,347	22,367	\$27,857	56,797	\$39,204	79,164
1998 Avg	\$10,385	20,867	\$25,830	53,058	\$36,215	73,925
1997 Avg	\$11,668	24,929	\$23,918	48,048	\$35,585	72,977

Table 21
2019 Calendar Year Report
T.E. Maxson Facility Biogas System Summary

Date 2019	Electrical Usage KWH	Electric Cost Cost \$	Biogas Production			Propane Usage Gallons	Propane Cost Cost \$	Biogas Avg % O ₂
			Cubic Ft./Month	TVA usage	Amount Flared			
			Million ft ³	Million ft ³	Million ft ³			
January	36,939	\$ 4,720	50.9	0	50.9	4419	\$4,618	5.8
February	33,659	\$ 4,715	54.6	0	54.6	5754	\$7,780	5.7
March	25,439	\$ 4,792	51.5	0	51.5	5190	\$7,556	5.7
April	32,739	\$ 4,314	47.0	0	47.0	3032	\$6,366	5.2
May	25,839	\$ 3,785	47.6	0	47.6	2220	\$4,643	4.1
June	26,139	\$ 4,174	45.4	0	45.4	2211	\$4,355	5.5
July	24,339	\$ 3,564	43.6	0	43.6	1060	\$1,318	5.9
August	2,739	\$ 3,784	48.8	0	48.8	1350	\$3,409	5.1
September	30,639	\$ 4,108	44.5	0	44.5	1060	\$2,928	6.4
October	30,039	\$ 3,618	41.3	0	41.3	1350	\$3,103	5.9
November	39,339	\$ 4,560	40.4	0	40.4	1860	\$2,744	6.6
December	35,139	\$ 4,331	36.3	0	36.3	1762	\$3,893	7.1
2019 Total	342,988	\$50,464	552	0	552	31,267	\$52,713	
2019 Avg	28,582	\$4,205	46.0	0	46.0	2,606	\$4,393	5.8
2018 Avg	21,345	\$ 2,250	47.1	0.0	47.1	2223	\$ 5,797	7.0
2017 Avg	17,925	\$ 2,532	62.0	0.0	62.0	673	\$ 1,458	6.6
2016 Avg	15,415	\$ 2,389	97.3	0.0	97.3	449	\$ 1,003	4.2
2015 Avg	23,858	\$ 2,437	104.5	0.0	104.5	449	\$ 1,003	4.2
2014 Avg	NA	NA	96.2	5.0	91.2	850	\$ 2,818	5.0
2013 Avg	16,779	\$ 2,571	85.6	13.0	72.6	NA	NA	3.6
2012 Avg	15,672	\$ 1,196	90.4	33.1	57.3	432	\$ 1,040	7.8
2011 Avg	14,576	\$ 1,186	84.6	34.5	47.9	930	\$ 2,677	4.8
2010 Avg	13,518	\$ 1,303	71.0	8.8	62.2	730	\$ 1,838	7.9
2009 Avg	15,061	\$ 1,433	97.2	28.9	68.2	521	\$ 1,254	8.3

NOTE: * TVA did not take gas from May 24, 2007 to September 5, 2007 and during Sept-Dec 2010.

Table 22
2019 Calendar Year Report
T. E. Maxson Facility
Electrical Services-Other Electrical Meters

Location Information	Blowers		Pumps		Pumps		Pumps						Lagoons #3 Dredge		Blowers	
	2C		L2A		L1-2B		2C		Lagoon Shop		Dewatering				1, 2A, 2B #1 Lagoon City Blowers & Biorem	
	Address															
	Meter ID#															
MLGW Acct #	1450181		1450178		1450179		1450180		1450173		1450174		1450177		1452329	
	KWH		KWH		KWH		KWH		KWH		KWH		KWH		KWH	
	Cost	Usage	Cost	Usage	Cost	Usage	Cost	Usage	Cost	Usage	Cost	Usage	Cost	Usage	Cost	Usage
January	\$1,251	11,900	\$1,947	16,950	\$3,523	31,500	\$3,540	35,260	\$2,339	20,719	\$8,865	92,850	\$932	7,197	\$3,469	36,900
February	\$1,459	12,000	\$2,558	22,050	\$2,491	19,800	\$2,941	27,300	\$2,444	24,217	\$8,719	60,750	\$725	5,485	\$3,257	33,620
March	\$1,426	11,900	\$1,915	17,250	\$2,668	23,400	\$3,099	29,100	\$2,458	24,672	\$4,570	46,500			\$3,366	25,400
April	\$1,336	11,600	\$1,184	10,200	\$2,137	17,400	\$2,178	20,550	\$1,314	11,397	\$4,614	48,600	\$752	6,215	\$2,978	32,700
May	\$1,464	12,800	\$532	4,200	\$2,769	22,200	\$1,312	11,400	\$578	4,623	\$3,322	27,300	\$802	5,551	\$2,321	25,800
June	\$1,689	14,100	\$454	3,300	\$2,851	20,700	\$1,398	11,550	\$841	6,684	\$2,618	21,450	\$548	3,648	\$2,484	26,100
July	\$1,165	11,100	\$406	2,850	\$2,797	19,500	\$1,358	11,100	\$835	6,564	\$3,985	42,150	\$712	5,499	\$2,398	24,300
August	\$1,266	12,300	\$366	2,550	\$3,278	27,900	\$1,474	12,300	\$809	6,449	\$3,958	39,750	\$633	4,897	\$2,519	2,700
September	\$1,422	14,100	\$395	2,850	\$3,540	32,400	\$1,418	12,000	\$860	7,007	\$2,498	41,550	\$601	4,687	\$2,686	30,600
October	\$1,097	11,600	\$503	4,040	\$31	34,800	\$1,801	17,700	\$560	4,590	\$3,843	20,700	\$337	2,475	\$2,520	30,000
November	\$1,201	12,400	\$1,561	13,800	\$3,421	34,500	\$2,614	24,000	\$2,184	22,162	\$7,462	39,000	\$457	3,534	\$3,359	39,300
December	\$1,243	12,600	\$1,856	17,400	\$3,527	34,500	\$3,298	37,350	\$2,536	26,253	\$8,268	88,800	\$760	6,008	\$3,088	35,100
2019 Total	\$16,019	148,400	\$13,678	117,440	\$97	318,600	\$26,430	249,610	\$17,758	165,337	\$62,721	569,400	\$7,259	55,196	\$34,445	342,520
2019 Avg	\$2,464	12,367	\$1,140	9,787	\$2,753	26,550	\$2,202	20,801	\$1,480	13,778	\$5,227	47,450	\$660	9,199	\$2,870	28,543
2018 Avg	\$1,969	11,533	\$2,860	29,182	\$2,488	21,925	\$2,256	22,925	\$1,439	12,996	\$5,106	49,560	\$1,118	10,880		
2017 Avg	\$3,012	8,708	\$1,153	11,201	\$2,475	22,750	\$2,358	23,380	\$1,028	7,618	\$4,242	40,649	\$625	9,697		
2016 Avg	\$1,284	\$6,427	\$1,009	\$9,301	\$2,668	\$23,475	\$2,140	\$21,525	\$894	\$7,342	\$10,582	\$55,863	\$633	\$10,767		
2015 Avg	\$1,824	\$9,675	\$1,744	\$18,613	\$2,989	\$28,950	\$2,598	\$29,013	\$1,115	\$11,013	NA	NA	\$615	\$9,894		
2014 Avg	\$775	\$7,475	\$2,107	\$23,375	\$3,003	\$26,525	\$2,438	\$25,925	\$1,089	\$10,373	NA	NA	\$567	\$4,584		
2013 Avg	\$595	5933	\$2,963	35,100	\$2,864	27,575	\$2,589	29,050	\$1,096	12,026	\$3,799	42,188	\$687	6111		
2012 Avg	\$535	6,950	\$2,007	37,245	\$1,715	28,530	\$1,766	30,045	\$747	9,325	\$2,051	38,700	\$612	6,934		
2011 Avg	\$413	5,067	\$2,048	32,488	\$1,931	29,825	\$2,057	33,400	\$950	11,290	\$2,538	45,063	\$746	7,805		
2010 Avg	\$564	5,863	\$3,025	33,870	\$2,976	27,600	\$2,580	29,387	\$1,178	13,066	\$6,199	48,870	\$1,010	9,393		
2009 Avg	\$631	6,559	\$3,397	45,450	\$2,070	19,967	\$2,199	25,967	\$1,124	11,122	\$5,030	42,548	\$1,114	10,661		

Table 23
2019 Calendar Year Report
Trends in Maxson Facility Sludge Production, Influent Loading, Lagoon Return, and Biogas Production

Year	WAS (ppd)	Primary Wasting (ppd)	Combined Wasting (ppd)	Influent BOD (pounds per day)	Monitored BOD (pounds per day)	Difference in Influent BOD vs Monitored BOD (pounds per day)	BOD Removed by Primary Clarifiers (pounds per day)	Lagoon Supernatant TSS (pounds per day)	Biogas (million cubic feet)	Volatilized Sludge (million pounds)	Yield (CF biogas per pound destroyed)	Ratio of PS to WAS
1995	283,000		283,000	342,498	214,949	138,215		53,605	385	41	14.0	0.00
1996	284,667		284,667	355,436	249,770	111,739		86,856	491	54	9.1	0.00
1997	311,333		311,333	368,708	276,400	99,929		77,789	541	57	9.4	0.00
1998	283,000		283,000	371,740	275,327	102,957		74,227	504	56	9.1	0.00
1999	317,534		317,534	389,827	280,830	108,997		102,374	580	48	12.1	0.00
2000	420,274		420,274	434,462	347,728	86,765	10,687	180,578	578	56	10.3	0.00
2001	399,726		399,726	394,078	315,070	78,020	11,041	153,292	563	48	11.8	0.00
2002	419,452		419,452	391,573	302,408	89,178	5,968	112,975	647	60	10.7	0.00
2003	404,932		404,932	378,406	324,321	54,090	7,418	73,874	581	78	7.5	0.00
2004	477,808		477,808	450,183	305,610	144,573	37,512	74,686	724	87	8.3	0.00
2005	195,342	297,534	492,877	488,888	258,717	209,619	116,353	110,583	1,046	74	14.2	1.52
2006	295,068	266,575	561,644	431,115	265,283	162,632	68,898	197,972	1,120	80	14.0	0.90
2007	227,123	251,233	478,356	433,161	286,587	146,574	78,019	65,769	1,163	83	14.0	1.11
2008	223,836	212,055	435,890	452,971	277,689	175,282	71,169	84,408	1,007	67	15.1	0.95
2009	164,110	255,342	419,452	364,049	236,725	127,325	69,779	61,191	1,160	67	17.4	1.56
2010	232,329	185,479	417,808	404,628	259,827	144,801	76,586	89,374	851	58	14.7	0.80
2011	191,781	204,110	395,890	389,421	294,477	102,975	53,046	40,981	1,015	68	14.9	1.06
2012	151,233	201,644	352,877	450,168	307,566	142,602	62,997	33,202	1,085	64	17.0	1.33
2013	124,932	216,986	341,918	508,695	296,181	212,515	84,433	44,519	1,027	65	15.8	1.74
2014	95,342	239,452	334,795	540,103	322,506	217,598	124,094	19,500	1,155	66	17.5	2.51
2015	101,370	375,342	476,712	532,083	246,679	285,404	374,961	72,273	1,254	59	21.4	3.70
2016	69,608	238,425	308,033	456,184	251,114	185,509	104,114	37,407	1,167	49	23.8	3.43
2017	60,822	276,712	384,764	450,747	246,575	204,172	137,160	86,309	744	60	12.4	4.55
2018	67,831	310,907	378,738	535,000	265,305	269,199	232,506	51,962	565	68	8.3	4.58
2019	83,308	251,151	334,459	491,590	275,646	215,944	104,139	14,960	552	62	8.9	3.01

Table 24
2019 Calendar Year Report
M.C. Stiles Facility
Biogas Recovery System

	Electrical Usage (kwh)					Electrical Costs					Biogas Production (million CF)					% Oxygen				
	2015	2016	2017	2018	2019	2015	2016	2017	2018	2019	2015	2016	2017	2018	2019	2015	2016	2017	2018	2019
Jan.	NA	3,055	1,488	2,138	6,427	NA	905	166	225	688.13	19.7	7.0	5.8	6.1	12.4	2.8	0.7	0.2	0.6	0.6
Feb.	NA	1,969	1,492	3,848	8,570	NA	797	165	386	898.40	8.1	7.1	7.5	10.0	11.0	0.5	0.1	0.2	0.3	0.4
March	NA	1,922	7,511	6,008	3,661	NA	793	722	609	393.97	12.1	6.4	16.4	12.6	7.0	0.7	0.3	0.4	0.3	0.9
April	NA	1,216	9,841	6,279	7,150	NA	726	931	607	710.53	16.7	6.9	17.1	11.0	10.2	0.7	-	0.5	6.1	0.3
May	NA	6,936	5,922	2,307	8,549	NA	942	874	237	842.57	18.4	11.9	11.2	9.8	11.5	5.3	0.1	2.1	0.2	3.9
June	NA	7,470	1,245	6,021	3,447	NA	753	147	606	375.85	16.2	13.1	9.1	10.4	7.6	2.1	0.3	0.3	0.3	3.6
July	NA	7,992	7,223	7,376	1,911	NA	1,435	744	748	221.83	14.1	11.5	12.8	10.5	7.2	0.6	2.0	0.5	3.1	0.5
Aug.	NA	6,078	5,377	3,234	2,188	NA	640	548	343	246.14	14.5	9.8	10.2	9.5	6.1	0.8	0.1	0.3	0.5	0.4
Sept.	NA	4,738	1,865	7,359	2,006	NA	1,099	201	730	224.24	10.7	8.9	8.5	10.1	11.5	0.6	0.0	0.2	1.0	0.8
Oct.	NA	1,591	1,268	2,555	2,839	NA	766	139	261	287.96	9.2	6.3	8.0	7.8	19.9	2.1	1.1	0.4	0.4	0.7
Nov.	NA	1,817	1,230	1,405	2,375	NA	1,099	137	159	250.78	8.5	8.9	7.5	5.9	16.5	0.6	1.6	0.2	0.3	0.6
Dec.	NA	1,917	1,670	1,834	2,447	NA	1,566	181	203	262.03	10.4	8.0	7.8	8.2	14.6	0.2	3.4	1.9	0.4	1.8
Total	NA	46,701	46,132	50,364	51,570	NA	13,537	6,972	7,132	7,421	159	106	122	112	136					
Average	NA	3,892	3,844	4,197	4,298	NA	960	413	426	450	13.2	8.8	10.1	9.3	11.3	1.4	0.8	0.6	1.1	1.2

NOTE: No biogas was recovered beginning 9/2005 through 9/2007 due to construction of new lagoon cover system.
Biogas production resumed on 10-25-2007.

2019 Avg		4,298													11					1.2
2018 Avg			4,197												9					1.1
2017 Avg			3,844												10					0.6
2016 Avg			3,892												9					0.8
2015 Avg			NA												13					1.4
2014 Avg			NA												23					1.2
2013 Avg			4,903												22					7.3
2012 Avg			6,617												28					6.1
2011 Avg			4,622												23					2.3
2010 Avg			6,672												29					5.7

Table 25
2019 Calendar Year Report
M.C. Stiles Facility Other Utilities Usage and Costs

	Water		GAS		OLD DEWATERING		NEW DEWATERING		GAS FOR FLARE	
	CCF's	COST	CCF'S	COST	Kwh	COST	Kwh	COST	CCF'S	COST
2019										
January	24,230	\$24,292	0	\$113.96	10,650	\$869			1470	\$1,008
February	46,700	\$46,559	0	\$113.56	0	\$0	85,800	\$9,262	1261	\$783
March	22,230	\$23,008	0	\$113.56	10,800	\$1,260	28200	\$5,100	1362	\$800
April	27950	\$28,853	0	\$113.96		\$0	39600	\$5,670	1393	\$793
May	30520	\$31,480	2280.96	\$1,281.00		\$0	53100	\$6,274	1318	\$784
June	37130	\$38,235	1634.16	\$853.88		\$0	54300	\$7,136	1471	\$779
July	34330	\$35,374	922.68	\$450.94		\$0	47400	\$6,639	1374	\$622
August	34850	\$35,905	900.24	\$590.10		\$0	51600	\$6,819	1284	\$599
September	40540	\$41,720	184	\$200.19		\$0	67500	\$7,553	1477	\$0
October	34590	\$35,639	1524	\$929.00		\$0	50700	\$6,974	0	\$0
November	28720	\$29,640	2867	\$1,656.00		\$0	34800	\$6,797	0	\$0
December	36790	\$37,888	1462	\$782.00		\$0	35700	\$5,142	0	\$0
2019 Total	333,070	\$341,066	11,775	\$7,198	21,450	\$2,129	548,700	\$73,366	12,410	\$6,168
2019 Avg	33,307	\$34,107	981	\$600	10,725	\$1,064	49,882	\$6,670	1,379	\$514
2018 Avg	35,348	\$35,273	781	\$543	9,570	\$1,088	63,600	\$6,687	1,361	\$778
2017 Avg	7,620	\$7,752	1,108	\$670	8,650	\$845	51,600	\$5,458	1,357	\$662
2016 Avg	7,454	\$7,589	8,028	\$834	15,850	\$738	51,775	\$5,312	1,392	\$596
2015 Avg	2,445	\$2,193	1,016	\$646	10,125	\$1,804	31,550	\$3,905		
2014 Avg	1,603	\$1,507	1,330	\$1,016	11,088	\$2,156	37,375	\$4,724		
2013 Avg	1,750	\$1,617	756	\$520	10,813	\$1,067	36,975	\$4,138		
2012 Avg	15,078	\$11,290	1,259	\$1,026	6,120	\$473	71,610	\$3,158		
2011 Avg	10,114	\$7,637	1,635	\$1,982	5,025	\$433	42,050	\$3,637		
2010 Avg	12,002	\$8,599	714	\$622	5,595	\$501	74,550	\$4,165		
2009 Avg	10,581	\$7,604	827	\$859	5,236	\$430	72,000	\$4,358		

Table 26
2019 Calendar Year Report
Chapel Hill Utility Costs

2019	KWH	Cost \$
January	4,868	\$527
February	5,493	\$585
March	4,697	\$498
April	5,785	\$580
May	5,725	\$573
June	7,048	\$742
July	7,609	\$807
August	6,146	\$645
September	6,534	\$673
October	6,456	\$622
November	6,393	\$631
December	8,161	\$814
2019 Total	74,915	\$7,697
2019 Avg	6,243	\$641
2018 Avg	6,881	\$910
2017 Avg	8,385	\$1,009
2016 Avg	8,021	\$984
2015 Avg	7,700	\$787
2014 Avg	8,141	\$839
2013 Avg	8,433	\$837
2012 Avg	8,656	\$659
2011 Avg	8,176	\$667
2010 Avg	8,841	\$840
2009 Avg	6,977	\$662
2008 Avg	7,548	\$696

Table 27
2019 Calendar Year Report
Chapel Hill Summary

The influent flowrate for 2019 was		14,600 gpd	
Year	Effluent CBOD (mg/l)	Effluent TSS (mg/l)	Permit Violations
2010	4	8	0
2011	3	4	0
2012	2	8	0
2013	3	11	1
2014	5	15	0
2015	5	21	0
2016	6	21	0
2017	5	20	5
2018	5	13	0
2019	9	13	12

Table 28
2019 Calendar Year Report
M.C. Stiles Facility BOD Evaluation (lbs/day)

Year/Month	M. C. Stiles Plant	American Yeast	Buckeye Technologies	Cascades Tissue G.	Coke	PMC Biogenix Pope	K.T.G. (USA) LP	Penn Specialty Chemicals, Inc.	Runyon Industries	Total of 8 Companies	Difference
2019											
January	256,693	57,617	39,826	5,441	1,034	37,639	3,882	1,753	30,753	177,945	78,748
February	240,506	55,677	44,265	5,077	589	32,250	3,979	1,752	25,023	168,612	71,894
March	231,260	52,789	45,462	5,202	1,418	45,033	3,519	2,287	32,772	188,482	42,778
April	292,139	51,749	45,086	5,658	1,600	39,900	4,774	2,835	38,047	189,649	102,490
May	276,534	50,517	34,900	5,692	1,418	42,200	5,069	4,265	50,774	194,835	81,699
June	255,139	65,368	33,329	6,020	1,771	29,970	3,410	4,318	44,623	188,809	66,330
July	365,712	59,331	27,808	4,795	1,156	39,252	3,064	3,317	35,508	174,231	191,481
August	259,966	68,510	24,946	6,382	2,859	30,391	5,194	3,471	29,904	171,657	88,309
September	290,806	48,063	26,518	6,239	1,123	41,870	4,759	1,584	24,607	154,763	136,043
October	269,815	55,222	25,937	4,866	1,008	34,328	5,898	2,242	37,848	167,349	102,466
November	231,435	50,164	25,162	5,037	1,096	20,707	5,356	3,162	34,594	145,278	86,157
December	213,771	49,835	29,264	4,995	1,311	39,864	7,490	3,382	37,233	173,374	40,397
2019 Avg	265,315	55,404	33,542	5,450	1,365	36,117	4,700	2,864	35,141	174,582	90,733
2018 Avg	268,114	50,397	40,135	5,482	2,727	28,591	3,457	1,953	37,037	169,779	98,335
2017 Avg	247,351	48,012	38,238	5,394	1,634	21,264	4,780	1,942	40,016	161,903	85,448
2016 Avg	244,332	44,343	37,996	5,393		25,115	6,606	1,864	33,041	154,358	89,974
2015 Avg	268,505	41,986	38,041	5,344		27,519	8,798	1,726	40,562	163,977	104,528
2014 Avg	259,864	41,053	41,081	10,915		31,746	9,138	1,542	36,580	172,055	87,810
2013 Avg	246,641	47,518	48,682	4,591		29,895	6,259	1,135	25,747	161,682	84,959
2012 Avg	263,148	54,192	40,249	5,605		23,146	1,548	1,120	23,127	148,988	114,160
2011 Avg	253,235	38,039	41,116	7,251		33,989	2,475	1,389	13,443	137,701	115,535
2010 Avg	250,252	41,086	44,072	5,296		37,878	2,567	1,474	13,917	146,289	103,963
2009 Avg	233,850	34,622	35,247	3,977		29,521	3,447	1,009	13,948	121,771	112,078
2008 Avg	266,730	39,813	38,508	6,587		20,808	3,879	1,073	14,585	125,253	141,477
2007 Avg	243,021	34,950	38,899	2,583		14,608	3,335	1,761	17,878	114,014	129,007
Note:											
*Name changed: Enenco - ceased production November 15, 2004, now known as Chemtura/Bell; Crompton is now known as Chemtura Corporation											
Added American Yeast Corporation - Feb. 2003											
Added K.T.G. (USA) LP - March 2003; previous data belongs to Super American Tissue Mills of Tennessee sold to K.T.G. (USA) LP.											
Added Runyon Industries - 2003; Cascades Tissue Group - Tennessee Inc. - January 2005											

Table 29
2019 Calendar Year Report
M.C. Stiles Facility COD Evaluation (lbs/day)

Year/Month	M. C. Stiles Plant	American Yeast	Buckeye Technologi	Cascades Tissue G.	Coke	PMC Biogenix Pope	K.T.G. (USA) LP	Penn Specialty Chemicals, Inc.	Runyon Industries	Total of 8 Companies	Difference
2019											
January	669,710	134,933	131,332	9,963	1,585	60,256	24,445	4,202	42,923	409,639	260,071
February	634,254	137,209	141,118	9,748	929	55,809	29,324	4,246	35,588	413,971	220,283
March	594,250	117,016	135,659	10,196	2,419	72,602	24,420	5,987	45,782	414,081	180,169
April	594,138	117,825	122,212	10,917	2,500	64,289	23,772	6,209	51,453	399,177	194,961
May	616,786	111,144	109,229	11,820	2,099	60,947	29,597	8,541	64,779	398,156	218,630
June	599,132	140,601	105,403	11,634	2,659	47,458	30,205	8,823	65,908	412,691	186,441
July	867,433	125,473	87,717	9,216	1,764	66,911	28,926	7,741	42,781	370,529	496,904
August	526,629	131,116	78,148	12,053	4,620	55,784	32,329	7,663	41,951	363,664	162,965
September	705,321	110,791	86,950	12,481	1,869	71,702	33,798	3,888	58,800	380,279	325,042
October	740,836	108,569	73,111	9,251	1,636	51,781	41,313	4,698	50,682	341,041	399,795
November	618,620	102,887	77,988	9,706	1,785	35,458	34,727	7,363	46,000	315,914	302,706
December	551,054	74,423	96,236	9,615	2,108	68,626	60,961	6,989	48,735	367,693	183,361
2019 Average	643,180	117,666	103,759	10,550	2,164	59,302	32,818	6,363	49,615	382,236	260,944
2018 Average	611,458	132,137	132,226	9,317	4,680	50,654	24,744	4,440	51,058	409,255	202,203

Table 30
2019 Calendar Year Report
M.C. Stiles Facility TSS Evaluation (lbs/day)

Year/Month	M. C. Stiles Plant	American Yeast	Buckeye	Cascades	*Chemtura Corp/ PMC Biogenix	Coke	KTG	Penn Specialty Chemical	Runyon	Total of 8 Companies	Difference
2019											
January	275,345	12,379	28,804	764	4,039	21	20,743	50	1384.00	68,184	244,686
February	312,870	12,356	30,721	358	4,163	10	21,813	47	426	69,894	158,439
March	228,333	8,836	30,433	355	6,482	27	18,007	67	476	64,683	266,855
April	331,538	9,974	30,465	710	3,870	84	18,028	81	532	63,744	258,667
May	322,411	10,165	22,959	1,937	3,948	83	20,953	108	767	60,920	171,371
June	232,291	9,734	19,026	997	3,706	51	22,915	61	786	57,276	502,878
July	560,154	10,914	16,674	1,571	5,437	125	19,859	77	445	55,102	162,856
August	217,958	11,769	14,874	751	4,893	75	25,982	128	491	58,963	263,503
September	322,466	7,519	18,523	799	4,979	105	26,089	179	776	58,969	236,574
October	295,543	6,158	13,972	940	3,983	168	30,322	99	1,330	56,972	204,487
November	261,459	6,683	19,639	707	3,329	111	28,754	114	1,299	60,636	127,007
December	187,643	6,332	20,173	1,545	5,750	59	43,860	149	829	78,697	108,946
2019 Avg	295,668	9,402	22,189	953	4,548	77	24,777	97	795	62,837	225,522
2018 Avg	294,512	8,242	29,675	2,842	4,234	152	18,729	100	722	64,697	233,977
2017 Avg	304,940	6,752	26,784	1,511	3,964	210	23,064	68	631	63,052	245,437
2016 Avg	289,269	6,804	25,806	1,204	4,432		23,975	294	287	62,801	231,284
2015 Avg	326,334	7,102	30,196	684	4,280		26,710	77	296	69,345	255,079
2014 Avg	288,417	6,988	27,692	3,399	4,785		21,669	72	215	64,819	222,287
2013 Avg	250,482	5,442	33,026	813	6,151		16,456	84	210	62,165	177,645
2012 Avg	240,627	7,681	28,503	1,865	5,209		3,253	72	155	46,737	194,326
2011 Avg	225,537	6,901	28,141	2,375	5,799		4,854	68	77	48,215	177,322
2010 Avg	223,553	10,666	30,338	3,601	7,910		4,768	76	208	57,567	165,986
2009 Avg	251,045	7,827	18,655	5,766	6,067		10,106	46	96	48,563	202,482
2008 Avg	274,035	9,872	25,655	10,794	4,626		9,833	66	431	61,277	212,758
2007 Avg	226,842	9,028	18,357	5,931	3,137		7,174	273	251	44,151	182,691
2006 Avg	234,454	11,430	24,969	6,030	3,436		5,491	656	4,971	56,982	177,472
2005 Avg	236,041	7,794	23,527	5,435	3,140		9,529	593	6,298	56,323	179,718
Note:											
*Name changed: Enenco - ceased production November 15, 2004, now known as Chemtura/Bell; Crompton is now known as Chemtura Corporation											
Added American Yeast Corporation - Feb. 2003											
Added K.T.G. (USA) LP - March 2003; previous data belongs to Super American Tissue Mills of Tennessee sold to K.T.G. (USA) LP.											
Added Runyon Industries - 2003; Cascades Tissue Group - Tennessee Inc. - January 2005											

Table 31
2019 Calendar Year Report
Biogas Purchased by American Yeast

Month	Total Biogas Generated by Stiles	Biogas Used by Am Yeast (million CF)	% Methane	BTU/CF	1/2 Henry Hub Price	Invoice
Jan-18	6.06	3.39	70	710	\$1.79	\$3,692
Feb-18	10.04	6.30	66	670	\$1.61	\$7,640
Mar-18	12.57	11.07	70	710	\$1.31	\$9,775
Apr-18	10.95	8.79	46	470	\$1.38	\$7,691
May-18	9.78	7.73	67	680	\$1.34	\$4,770
Jun-18	10.35	6.69	63	640	\$1.45	\$6,613
Jul-18	10.47	6.00	56	570	\$1.45	\$3,844
Aug-18	9.50	5.32	64	650	\$1.41	\$3,489
Sep-18	10.09	3.99	70	720	\$1.47	\$1,907
Oct-18	7.77	5.36	64	650	\$1.50	\$7,270
Nov-18	5.89	5.77	72	730	\$2.29	\$4,556
Dec-18	8.24	1.46	67	680	\$1.65	\$7,506
2018 Totals	111.7	71.88				\$68,752
2018 Average	9.3	5.99	65	657	\$1.55	\$5,729
Jan-19	12.44	0.00	66.2	660	\$1.62	\$0
Feb-19	11.02	1.57	62.8	625	\$1.42	\$1,384
Mar-19	7.01	3.31	60.6	656	\$1.47	\$3,177
Apr-19	10.22	2.30	63.6	593	\$1.35	\$1,839
May-19	11.54	4.14	55.1	514	\$1.30	\$2,767
Jun-19	7.55	2.79	57.6	563	\$1.24	\$1,941
Jul-19	7.20	4.20	70.4	373	\$1.18	\$1,841
Aug-19	6.07	4.49	68.3	635	\$1.12	\$3,181
Sep-19	11.55	5.37	68.4	644	\$1.15	\$3,962
Oct-19	19.87	5.58	60.9	596	\$1.17	\$3,872
Nov-19	16.55	6.71	71.8	594	\$1.34	\$5,345
Dec-19	14.59	8.41	68.5	594	\$1.17	\$7,506
2019 Totals	135.6	48.87				\$36,815
2019 Average	11.3	4.07	65	587	\$1.29	\$3,068

NOTES

- American Yeast did not start taking biogas until Feb 2009.
- Henry Hub price is per 1 million BTUs.
- CF = Cubic Feet

Table 32
2019 Calendar Year Report
Annual Electrical Usage and Cost
Combined Totals for Both Treatment Plants

<u>With ESP or VPI or ENERNOC</u>	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Maxson Facility	NA	NA	NA	\$5,240,763	\$5,593,494	\$5,236,611	\$4,692,045	\$4,750,044	\$4,741,403	\$5,040,423
Stiles Facility	NA	NA	NA	\$4,111,087	\$4,332,448	\$4,737,732	\$4,273,480	\$4,303,847	\$5,190,271	\$4,570,326
Total	NA	NA	NA	\$9,351,850	\$9,925,942	\$9,974,343	\$8,965,525	\$9,053,891	\$9,931,674	\$4,610,822
<u>Savings</u>	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Maxson Facility	NA	NA	NA	\$ 39,987	\$ 103,829	\$ 96,933	\$71,106	\$79,718	\$58,803	\$57,500
Stiles Facility	NA	NA	NA	\$ 39,987	\$ 119,904	\$ 41,478	\$21,534	\$31,562	\$32,377	\$40,496
Total	NA	NA	NA	\$ 79,973	\$ 223,733	\$ 138,411	\$92,640	\$111,280	\$91,179	\$97,996
<u>Without ESP or VPI or ENERNOC</u>	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Maxson Facility	\$3,897,724	\$4,497,064	\$5,101,364	\$5,280,750	\$5,697,323	\$5,333,543	\$4,763,151	\$4,829,762	\$4,800,205	\$5,097,924
Stiles Facility	\$3,972,467	\$4,349,363	\$4,295,247	\$4,151,074	\$4,452,352	\$4,779,211	\$4,295,014	\$4,335,409	\$5,222,648	\$4,610,822
Total	\$7,870,191	\$8,846,427	\$9,396,610	\$9,431,824	\$10,149,675	\$10,112,754	\$9,060,181	\$9,165,171	\$10,022,853	\$9,708,746
<u>Annual Usage (million kwh)</u>	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Maxson Facility	55.22	58.85	66.73	71.00	73.77	68.69	63.74	67.195	67.835	70.8
Stiles Facility	54.39	55.58	58.85	55.23	57.32	54.73	60.27	60.763	65.625	66.902
Total	109.61	114.43	125.58	126.23	131.09	123.42	124.01	127.958	133.46	137.702
<u>Average Cost per kwh</u>	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019
Mx with ESP/VPI//ENERNOC	NA	NA	NA	7.39	7.58	7.62	7.36	7.07	6.99	7.12
Mx w/o ESP/VPI/ENERNOC	7.06	7.64	7.64	7.44	7.73	7.76	7.47	7.19	7.08	7.20
Stiles with ESP/VPI/ENERNOC	NA	NA	NA	7.41	7.56	8.66	7.09	7.08	7.91	6.83
Stiles w/o ESP/VPI/ENERNOC	7.31	7.83	7.3	7.52	7.77	8.73	7.13	7.13	7.96	6.89

Note: program was from 1991 to 2000. From 2000 to 2006, the plants participated in the Variable Price Interruptible (VPI) program. Since 2006, both treatment plants have been in the Firm Power program. Began participating in ENERNOC in 2013.

Table 33
2019 Calendar Year Report
T.E. Maxson Facility Influent Flow Study (mgd)

	Monthly Average Treatment Plant Flow	Monthly Average Industrial Flow	Monthly Average Difference	Monthly Total Rainfall (inches)
2017				
January	77	9	69	4.26
February	68	8	60	2.04
March	72	9	63	2.94
April	66	9	58	5.36
May	67	9	58	3.05
June	66	10	56	4.61
July	63	9	54	3.91
August	65	10	55	9.29
September	70	9	61	1.59
October	64	9	55	4.13
November	61	9	52	1.81
December	71	9	62	7.26
2018				
January	75	10	65	4.02
February	92	11	81	13.43
March	93	10	83	4.44
April	82	9	73	7.86
May	75	9	66	3.32
June	70	9	61	2.06
July	70	10	60	5
August	71	9	62	2.07
September	76	9	67	5.27
October	76	10	66	3.59
November	74	10	64	4.25
December	79	9	70	8.84
2019				
January	85	9	76	4.75
February	94	10	84	8.78
March	102	10	92	4.12
April	86	9	77	7.89
May	77	9	68	5.17
June	77	9	68	9.51
July	73	9	64	7.53
August	61	11	50	4.48
September	60	9	51	0.12
October	73	10	63	12.95
November	72	10	62	4.31
December	72	10	62	4.04

Table 34
2019 Calendar Year Report
T.E. Maxson Facility Influent Flow Study (mgd)

	Monthly Average Treatment Plant Flow	Monthly Average Industrial Flow	Monthly Average Difference	Monthly Total Rainfall (inches)
2017				
January	77	9	69	4.26
February	68	8	60	2.04
March	72	9	63	2.94
April	66	9	58	5.36
May	67	9	58	3.05
June	66	10	56	4.61
July	63	9	54	3.91
August	65	10	55	9.29
September	70	9	61	1.59
October	64	9	55	4.13
November	61	9	52	1.81
December	71	9	62	7.26
2018				
January	75	10	65	4.02
February	92	11	81	13.43
March	93	10	83	4.44
April	82	9	73	7.86
May	75	9	66	3.32
June	70	9	61	2.06
July	70	10	60	5
August	71	9	62	2.07
September	76	9	67	5.27
October	76	10	66	3.59
November	74	10	64	4.25
December	79	9	70	8.84
2019				
January	85	9	76	4.75
February	94	10	84	8.78
March	102	10	92	4.12
April	86	9	77	7.89
May	77	9	68	5.17
June	77	9	68	9.51
July	73	9	64	7.53
August	61	11	50	4.48
September	60	9	51	0.12
October	73	10	63	12.95
November	72	10	62	4.31
December	72	10	62	4.04

Table 35
2019 Calendar Year Report
Purchase Order Summary
Page 1 of 1

Requisition			Purchase Order		
Item	Number	Description	Requisition Amount	Order Number	Facility
1	25434	Janitorial Services	\$28,400	36460	Stiles
2	28291	Flender Gear Unit	\$61,300	28291	Maxson
3	28265	Concrete Sealing and Repair	\$49,000	36768	Stiles
4	28287	350 HP motor for biopumps	\$79,500	36947	Maxson
5	28627	Lab Supplies	\$96,099	37072	Stiles
6	28715	12-inch Hydraulic Pump	\$163,598	36948	Maxson
7	28716	Pump and Motor for Lagoon Dredge	\$49,877	36945	Maxson
8	28955	Vactor Services	\$28,955	NA	Maxson
9	28018	3 dump trucks	\$365,250	36941	Maxson
10	29019	Lagoon Cover Repairs	\$150,000	36949	Maxson
Total Requisitions			\$1,071,979		

Table 36
2019 Calendar Year Report
Secondary Treatment Permit Limits Over The Years

Maxson Facility	2/22/1982	9/4/1984	3/31/1987	1/29/1993	4/1/2000	1/1/2011
Monthly Avg BOD conc	46	46	52	35	42	42
Monthly Avg BOD lbs	30,691	30,691	34,970	23,352	31,525	31,525
Weekly Avg BOD conc	65	65	73	53	63	63
Weekly Avg BOD lbs	43,368	43,368	48,380	35,362	47,288	47,288
Daily Max BOD conc	84	84	93	75	84	84
Daily Max BOD lbs	56,045	56,045	61,790	None	None	None
Monthly Avg TSS conc	44	44	35	40	48	48
Monthly Avg TSS lbs	29,357	29,357	23,350	26,688	36,029	36,029
Weekly Avg TSS conc	66	66	44	60	72	72
Weekly Avg TSS lbs	44,035	44,035	29,550	40,032	54,043	54,043
Daily Max TSS conc	88	88	54	70	96	96
Daily Max TSS lbs	58,714	58,714	35,760	None	None	None
Stiles Facility	2/22/1982	9/4/1984	3/31/1987	1/29/1993	4/1/2000	1/1/2011
Monthly Avg BOD conc	51	51	40	35	45	43.1
Monthly Avg BOD lbs	57,881	57,881	44,960	39,250	50,666	None
Weekly Avg BOD conc	64	64	52	51	68	64.7
Weekly Avg BOD lbs	72,058	72,058	58,130	57,309	76,561	None
Daily Max BOD conc	77	77	63	60	90	86.1
Daily Max BOD lbs	86,820	86,820	71,290	None	None	None
Monthly Avg TSS conc	46	46	38	36	52	52
Monthly Avg TSS lbs	51,982	51,982	43,130	41,558	58,547	None
Weekly Avg TSS conc	58	58	49	49	78	78
Weekly Avg TSS lbs	64,739	64,739	54,770	55,061	87,820	None
Daily Max TSS conc	69	69	59	70	104	104
Daily Max TSS lbs	77,972	77,972	66,400	None	None	None

Table 37
2019 Calendar Year Report
503 Sludge Limits and Sludge Values
Page 2 of 2

PLANT METALS CONCENTRATION IN SLUDGE (mg/kg)			
(Metals included are total of Cd, Cr, Cu, Pb, Ni & Zn)			
Time Period	Maxson Facility Primary Sludge	Maxson Plant Secondary Sludge	Stiles Facility Secondary Sludge
1985-1986			1969
1986-1987			1834
1987-1988			1852
1988-1989			1548
1989-1990	1430	3432	1351
1990-1991	1573	3461	1525
1991-1992	1756	2808	1577
1993	1065	2307	1478
1994	728	1475	1344
1995	799	1619	1534
1996	1225	1727	1471
1997	699	1100	1447
1998	1139	2076	1609
1999	788	1728	1915
2000	1256	2585	2020
2001	905	1248	2057
2002	818	1011	1925
2003	530	1470	1620
2004	**	1686	2009
2005	**	1655	1692
2006	**	1680	2384
2007	**	1701	2173
2008	**	1688	1895
2009	**	1856	1544
2010	**	1546	1483
2011	**	1663	1659
2012	**	1648	1294
2013	**	2013	1275
2014	**	2014	1224
2015	**	1688	1396
2016	**	1708	1418
2017	**	1681	1540
2018	**	1598	1431
2019		1411	1369

Note: **operation closed after April 2004

Table 38
Employees of the Month
2019 Calendar Year Report
Page 1 of 2

Month	Maxson	Stiles
Jan-10	Derek Tipton	Eugene Kaszas
Feb-10	Chris Luss	Alvin Clark, Jr.
Mar-10	Jeff Haggans	Baldwin Eng
Apr-10	Jack Wilkinson	Fred Moore
May-10	Yvonne Holmes	Tammy Franklin
Jun-10	Ken Frost	Gene Horton
Jul-10	Creola Samuels	Essie Washington
Aug-10	Henry Clark	Kim Curan
Sep-10	Jeff Alloway	Louvena Boone
Oct-10	Charles Hargrove	Walter Delaney
Nov-10	Paul Hollowell	Edward Ransom
Dec-10	Rick Watkins	Daryl Watson
Jan-11	Terry Blair	Ricky Bramlett
Feb-11	Jackie Sisco	Kim Griswold
Mar-11	Rodney Williams	Nickey Lamb
Apr-11	Jerome Kimball	Jimmy Howell
May-11	Jessie Williams	David Keen
Jun-11	Kim Crawford	Kim Curan
Jul-11	Nathaniel Partee	Darryl Brady
Aug-11	Paul Hollowell	Jennifer Kisner-Johnson
Sep-11	Michael Jones	David South
Oct-11	Beth Thomas	Johnny Stroud
Nov-11	Jim Mottley	Fred Moore
Dec-11	Ricky Martin	Tammy Westbrook
Jan-12	James Laster	Fred Moore
Feb-12	Ronnie Blakney	Jimmy Howell
Mar-12	Ronald Hood	David Dorris
Apr-12	Paul Mask	Charles Taylor
May-12	Johnny Garner	James Murray
Jun-12	Rickey Watkins	Chris Morrical
Jul-12	Terrence Hampton	Daryl Watson
Aug-12	Rodney Williams	Randy Fowler
Sep-12	Jim Mottley	Baldwin Eng
Oct-12	Carlton Mull	Robert Humphrey
Nov-12	Chris Luss	Mike Herron
Dec-12	Thurman Denton	Rickey Bramlett
Jan-13	Terence Hampton	L.J. Frazier
Feb-13	Keith Presson	Lannie McBride
Mar-13	James Laster	James Murray
Apr-13	Charles Hargrove	Brandon Griffin
May-13	Tim Barnes	Walter Delaney
Jun-13	Ken Frost	Nickey Lamb
Jul-13	Paul Mask	Christopher Pate
Aug-13	Terrell Maxwell	Ricky Bramlett
Sep-13	Tim Arnold	Jennifer Kisner-Johnson
Oct-13	Gladys Nakayama	Daryl Brady
Nov-13	Creola Samuels	Edward Ransom
Dec-13	Charles Hargrove	Derek McElroy
Jan-14	Lenora LaFlora	Charles Taylor
Feb-14	Ronald Hood	Bret Stanton
Mar-14	Yvonne Holmes	Christopher Rider
Apr-14	Michael Jones	Kristen Stephens

Table 38
Employees of the Month
2019 Calendar Year Report
Page 2 of 2

Month	Maxson	Stiles
May-14	Rick Williams	Chrisopher McCaster
Jun-14	Kim Crawford	Chrisopher Pate
Jul-14	Yvonne Holmes	Steve Weakley
Aug-14	Marvin Wilson	Rickey Bramlett
Sep-14	Terence Hampton	David Dorris
Oct-14	Derek Tipton	Louvena Boone
Nov-14	Carlton Mull	Robert Dorris
Dec-14	Creola Samuels	Edward Ransom
Jan-15	Jeff Alloway	Derek Porter
Feb-15	Ronnie Blakney	Kristen Stephens
Mar-15	Gladys Nakayama	Jimmy Howell
Apr-15	Ricky Craig	L.J. Frazier
May-15	Gary Young	Rickey Bramlett
Jun-15	Kevin Stephenson	Kim Curan-Butler
Jul-15	Michael Johnson	Steve Weakley
Aug-15	Harvey Woodall	John Vasquez
Sep-15	Rodney Williams	Daryl Brady
Oct-15	Jeffery Haggans	Jimmy Howell
Nov-15	Stephanie Dockersy	L.J. Frazier
Dec-15	Jeffery Haggans	Brandon Griffin

* Employee of the Month Program discontinued in 2016 when ABC Program started.

Table 39
Number of Full Time Positions by Year

Year	Maxson	Stiles
1977		
1978		
1979		87
1980		84
1981		84
1982		82
1983		80
1984		85
1985	73	85
1986		85
1987	74	87
1988	74	
1989	74	
1990		
1991		
1992		
1993	75	90
1994	89	89
1995	80	84
1996	75	79
1997	68	81
1998		79
1999	70	75
2000	66	74
2001	66	74
2002	66	74
2003		73
2004	70	72
2005	69	74
2006	69	75
2007	69	75
2008	69	75
2009	69	75
2010	69	75
2011	67	75
2012	67	75
2013	73	79
2014	73	78
2015	73	72
2016	74	72
2017	75	71
2018	75	71
2019	78	71
2020	78	77

Table 37
2019 Calendar Year Report
503 Sludge Limits and Sludge Values
Page 1 of 2

Metals [mg/kg (dry)]*	503 Limits	Maxson Facility Secondary Sludge	Stiles Facility Dewatering Sludge
Arsenic	73	12.6	7.2
Cadmium	39	1.3	1.1
Chromium	600	34.2	41.1
Copper	1500	330	389
Lead	300	26.7	26
Mercury	17	0.5	0.61
Molybdenum	75	30.5	6.7
Nickel	420	42.7	74.9
Selenium	36	8.7	3.4
Zinc	2800	963	830

* average values

PLANT METAL INFLUENT LOADINGS

(Metals included are total of Cd, Cr, Cu, Pb, Ni & Zn)

Time Period	Maxson Facility (lbs/day)	Stiles Facility (lbs/day)	Both Plants Combined (lbs/day)
1980-1984	428	742	1170
1984-1985	501	496	997
1985-1986	374	400	773
1986-1987	441	390	831
1987-1988	384	319	703
1988-1989	400	304	704
1989-1990	376	335	712
1990-1991	268	344	612
1991-1992	403	321	724
1993	361	361	722
1994	227	284	511
1995	208	212	420
1996	373	236	610
1997	360	262	623
1998	461	256	717
1999	348	286	634
2000	689	333	1022
2001	441	283	724
2002	390	425	816
2003	327	320	648
2004	370	337	707
2005	378	433	811
2006	464	359	823
2007	320	363	682
2008	388	265	653
2009	327	232	559
2010	339	346	685
2011	239	296	535
2012	207	176	383
2013	255	175	430
2014	359	120	479
2015	422	221	643
2016	206	224	430
2017	225	188	413
2018	105	114	219
2019	200	137	337